

Fundamentals of Spatial Random Processes

You are here. What happens north, south, east, or west of you is very likely to be (positively) dependent on what is happening here. This expresses a “law of Geography” that says nearby things tend to be more alike than those far apart. Why? Because there is a flux of causal relationships in the space–time continuum that, when integrated out over time or captured in a micro-instant of time, shows neighboring values to be more highly correlated. A caveat that should be mentioned is that physical barriers like rivers and mountains can affect what is meant by “nearby” or “neighboring.” Also, the presence of competition can lead to correlations that are negative; to put a biological face on the point, under big trees only small trees can grow.

A spatial process can be a temporal snapshot, a temporal aggregation, or a temporally frozen state of a space–time process. The snapshot may actually be the result of aggregation over a very short time scale, resulting in “instantaneous” spatial dependence (e.g., contiguous scenes from a series of remotely sensed images). Aggregation over longer time scales occurs in studies of climate; climate is sometimes characterized as a long-term average of daily weather (i.e., climate is what you expect, weather is what you get!), although we prefer to think of it as a long-term statistical distribution. Finally, an ore deposit deep underground represents a temporally “frozen” state in the evolution of its mineralization.

In this chapter, we shall review the ways that spatial dependence has been incorporated into spatial statistical models, the eventual goal being to incorporate them into hierarchical statistical models. Spatial statistics is a relatively new development within Statistics; ironically, Fisher (1935) was well aware of potential complications from the spatial component in agricultural experiments but, as explained in Chapter 1, he used *randomization* to neutralize its effect (and to control for bias in the assignment of treatments to experimental units).

In 1938, Fairfield Smith wrote an article (Fairfield Smith, 1938) showing empirically that, as plot size increased, the variance of the average yield computed from his crop data decreased, but beyond a particular plot size the decrease was negligible. Clearly, this cannot happen if the data are uncorrelated, so that Fairfield Smith's study indirectly recognized the presence of spatial correlation; it has only been recently that a spatial statistical model has been developed that is consistent with Fairfield Smith's observations (McCullagh and Clifford, 2006).

Papadakis (1937) and Bartlett (1938) gave spatial analyses for data from agricultural field trials, but *spatial statistical models* for such phenomena did not begin to appear until much later (e.g., Whittle, 1954; Besag, 1974). It has become clear in the last two decades that spatial statistical analyses do not usurp the standard designs and analyses of variance proposed by Fisher, but they can augment them to yield more efficient inferences and hence shorter less costly experiments (e.g., Gotway and Cressie, 1990; Grondona and Cressie 1991, 1993; Brownie and Gumpertz, 1997; Federer, Newton, and Altman, 1997; Legendre et al., 2004).

Like Chapter 3, this chapter does not attempt to review the whole field. Our goal is to establish enough spatial statistical methodology so that our subsequent expositions on spatio-temporal statistical methodology can be adequately explained. In some sections, we have added a subsection of "Other Topics" that represent areas of special interest. By no means do these topics cover the explosion of spatial statistical research seen in the last twenty years. An excellent reference for that is the *Handbook of Spatial Statistics*, edited by Gelfand, Diggle, Fuentes, and Guttorp (2010).

Spatial Hierarchical Statistical Models

In this chapter, we continue with our emphasis on process modeling within hierarchical statistical models (HMs), here for *spatial* data. To provide context, the data model is considered along with the process model, but we discuss in less depth the treatment of the unknown parameters. Recall that a Bayesian hierarchical model (BHM) results from modeling uncertainty of parameters at a parameter-model level, and an empirical hierarchical model (EHM) results from estimating the parameters directly from the data and "plugging" them back into the model. Statistical inference in this chapter, for both the EHM and the BHM, is primarily concerned with prediction of the spatial random process, given the data. (In the case of an EHM, the variability of the estimated parameter values is typically, but not always, ignored.) Inference on the parameters themselves can also be of scientific interest.

Accounting for Measurement Error

A lot of the earlier literature in spatial statistics did not explicitly specify, nor did it disentangle, the variability from two important sources. One of the sources is due to the *measurement* of the phenomenon of interest, and the other is due to *incomplete knowledge* of the phenomenon. In a time-series context, Berliner (1996) called these the "data model" and the "process model," respectively; in this book, we shall also use this terminology. This is most

straightforwardly seen in the geostatistics literature, where the nugget effect is defined as being equal to the discontinuity of the variogram at the origin (e.g., Journel and Huijbregts, 1978, pp. 151–152). Although it is sometimes recognized by geostatisticians that this discontinuity can be made up of *both* measurement error *and* spatial dependence at scales smaller than the available distances between observations, subsequent kriging (i.e., optimal spatial prediction) has often ignored the difference between these two sources of variability. Cressie (1988) derived optimal spatial prediction (kriging) in the presence of measurement error. When using geostatistical software, caution is advised, since many kriging algorithms fail to filter out the variability due to measurement error. In Section 4.1, we put measurement error, spatial prediction, and related issues into an HM context.

In Chapter 3, HMs are temporal; in this chapter, HMs are spatial; and in Chapters 6 and 7, HMs are spatio-temporal. The articles by Besag, York, and Mollié (1991) and by Diggle, Tawn, and Moyeed (1998) show the power of spatial HMs in diverse settings, albeit with considerable computational demands, and the book by Banerjee, Carlin, and Gelfand (2004) gives a modern perspective on spatial HMs.

Consider the basic *geostatistical* problem of *kriging* (i.e., spatial prediction) of, say, an unobserved climate variable such as temperature, at a known location, from observed temperatures at known locations throughout the region of interest $D_s \subset \mathbb{R}^d$. Let $Y(\cdot) \equiv \{Y(s) : s \in D_s\}$ be the true temperature. The observations are “noisy” versions of the true temperatures at known locations $\{s_1, \dots, s_m\}$; assuming additive measurement error, write the observations as,

$$Z(s_i) = Y(s_i) + \varepsilon(s_i), \quad i = 1, \dots, m,$$

where, independently of $Y(\cdot)$, $\varepsilon(\cdot) \equiv \{\varepsilon(s) : s \in D_s\}$ is a white-noise process with mean zero and variance $\sigma_\varepsilon^2 \geq 0$. Figure 4.1 shows $D_s \subset \mathbb{R}^2$ and data location(s) of temperature changes from the 1980s to the 1990s, produced by a model developed by the National Center for Atmospheric Research (NCAR); the temperature-change data are considered in Section 4.1.1.

From Section 2.2.1, an optimal spatial predictor of $Y(s_0)$ at known location s_0 , based on squared error loss and data,

$$\mathbf{Z} \equiv (Z(s_1), \dots, Z(s_m))',$$

is

$$E(Y(s_0)|\mathbf{Z}).$$

Different loss functions imply different optimal summaries of the posterior distribution. For example, to make inference on exceedances and exceedance regions of $Y(\cdot)$, Zhang, Craigmile, and Cressie (2008) use a *weighted ranks* squared error loss function [adapted from Wright, Stern, and Cressie (2003)] and Baddeley’s distance between binary images.

Notice that $E(Y(s_0)|\mathbf{Z})$ is different from the quantity $E(Z(s_0)|\mathbf{Z})$, which is the optimal predictor of the noisy (scientifically less relevant quantity) $Z(s_0)$. There *are* instances where one might be interested in predicting the data process, $Z(\cdot)$, from observations (see Section 2.2.2). In cross-validation,

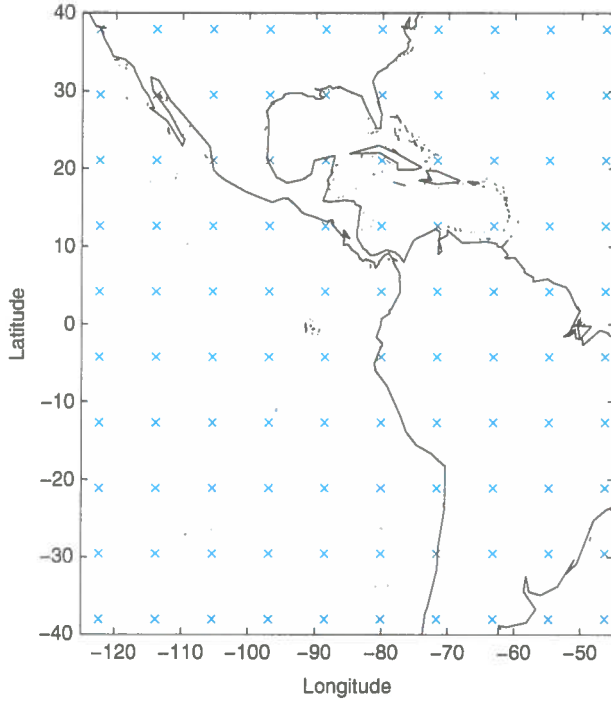


Figure 4.1 Data locations of temperature change for the Americas; “x” represents a sampled grid cell whose resolution is $2.8^\circ \times 2.8^\circ$. The whole map is made up of 28×28 grid cells and the 10×10 sampled grid cells are shown.

$Z(s_i)$ is deleted, so that logically it should be compared to the optimal spatial predictor of $Z(s_i)$ (not $Y(s_i)$), given

$$\mathbf{Z}^{(-i)} \equiv (Z(s_1), \dots, Z(s_{i-1}), Z(s_{i+1}), \dots, Z(s_m))'.$$

That is, $E(Z(s_i)|\mathbf{Z}^{(-i)})$ is computed and compared to $Z(s_i)$; $i = 1, \dots, m$. In a similar vein, suppose we generate a sample from the posterior distribution, $Y(\cdot)|\mathbf{Z}$, and then use the resulting $\mathbf{Y} \equiv (Y(s_1), \dots, Y(s_n))'$ to generate a sample, $\mathbf{Z}_{rep}$, from the conditional distribution of $\mathbf{Z}|\mathbf{Y}$ (see Section 2.2.2). If this is repeated many times, one obtains a sample $\mathbf{Z}_{rep}^{(1)}, \mathbf{Z}_{rep}^{(2)}, \dots$ from the *posterior predictive distribution*, $[\mathbf{Z}_{rep}|\mathbf{Z}]$ (e.g., Gelman et al., 2003, Chapter 6), which, like cross-validation, can be used to check the fit of the HM.

A Spatial Hierarchical Model to Explain the Nugget Effect

For simplicity, we assume here that $Y(\cdot)$ and $\varepsilon(\cdot)$ in $Z(\cdot) = Y(\cdot) + \varepsilon(\cdot)$, are independent Gaussian processes, and hence only the first two moments are needed to characterize completely the probability distributions. A further assumption of second-order stationarity (see Section 4.1.6) allows us to define

$$C_Y(\mathbf{h}) \equiv \text{cov}(Y(\mathbf{s}), Y(\mathbf{s} + \mathbf{h})), \quad \text{for all } \mathbf{s}, \mathbf{s} + \mathbf{h} \in D_s.$$

Define $\sigma_Y^2 \equiv C_Y(\mathbf{0})$, and

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} (C_Y(\mathbf{0}) - C_Y(\mathbf{h})) \equiv C_Y(\mathbf{0}+) \equiv \sigma_0^2 \geq 0,$$

where $\sigma_0^2 \leq \sigma_Y^2$. Then σ_0^2 represents the variance of the microscale component of the process $Y(\cdot)$. Recall that $\text{var}(\varepsilon(\mathbf{s})) = \sigma_\varepsilon^2$. Hence,

$$\text{cov}(Z(\mathbf{s}), Z(\mathbf{s} + \mathbf{h})) \equiv C_Z(\mathbf{h}) = \begin{cases} \sigma_Y^2 + \sigma_\varepsilon^2, & \mathbf{h} = \mathbf{0}, \\ C_Y(\mathbf{h}), & \mathbf{h} \neq \mathbf{0}, \end{cases}$$

which implies that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} (C_Z(\mathbf{0}) - C_Z(\mathbf{h})) \equiv C_Z(\mathbf{0}+) = \sigma_0^2 + \sigma_\varepsilon^2 \equiv c_0 \geq 0.$$

It is this quantity, c_0 , that has been called the *nugget effect* in the geostatistics literature (e.g., Journel and Huijbregts, 1978, p. 151), but we see that it is made up of *two* nonnegative components, σ_0^2 (the microscale variance of $Y(\cdot)$) and σ_ε^2 (the measurement-error variance of $Z(\cdot)$).

Optimal spatial prediction should filter out the measurement-error variability $\varepsilon(\cdot)$ and predict $Y(\mathbf{s}_0)$, just as algorithms such as the Kalman filter (Kalman, 1960) smooth, filter, or forecast past, present, or future signals, respectively, in the time domain (Section 8.2). However, much of the geostatistics literature has formulated spatial prediction (kriging) inappropriately in terms of predicting $Z(\mathbf{s}_0)$, not $Y(\mathbf{s}_0)$. This makes scientific sense if in fact $\sigma_\varepsilon^2 = 0$ and hence $Z(\cdot) \equiv Y(\cdot)$.

It is implicit in a lot of the geostatistics literature, that measurements are made perfectly, without error, and hence the nugget effect c_0 is made up of only microscale variance: $c_0 = \sigma_0^2$. Even the etiology of the term “nugget effect”—variability due to small discrete nuggets (of gold) surrounded by dross—only refers to the microscale variability. To sum up, the correct way to formulate kriging is that of optimal (linear) spatial prediction of $Y(\mathbf{s}_0)$ based on noisy data $Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_m)$ (Section 4.1.2).

The simple geostatistical model on $D_S \subset \mathbb{R}^d$ that we have just described can be written hierarchically as follows.

Data model: Conditional on σ_ε^2 , and for $i = 1, \dots, m$,

$$Z(\mathbf{s}_i) | Y(\mathbf{s}_i), \sigma_\varepsilon^2 \sim \text{ind. Gau}(Y(\mathbf{s}_i), \sigma_\varepsilon^2).$$

Process model: Conditional on μ and C_Y , $Y(\cdot)$ is a stationary Gaussian process with mean μ and covariance function $C_Y(\mathbf{h})$; $\mathbf{h} \in \mathbb{R}^d$.

For optimal spatial prediction, Geostatistics has traditionally taken an EHT approach, estimating unknown parameters of the HM (e.g., σ_ε^2 , μ , C_Y) from the empirical variogram (Section 4.1.1). In contrast, contributions by Diggle, Tawn, and Moyeed (1998), Banerjee, Carlin, and Gelfand (2004), Le and Zidek (2006), and Diggle and Ribeiro (2007), among others, give a BHM approach to these geostatistical prediction problems.

In this chapter, we present the spatial models that we think are the most important and most relevant to the spatio-temporal models presented and discussed in Chapters 6 and 7. Other topics of special interest to us are included and broad coverage of recent developments in spatial statistics is given in the Bibliographic Notes in Section 4.5. Our presentation is divided into seven sections, presenting spatial process models based on geostatistical processes (Section 4.1), lattice processes (Section 4.2), spatial point processes (Section 4.3), and random-set processes (Section 4.4). General references to each are found in the Bibliographic Notes in Section 4.5.

4.1 GEOSTATISTICAL PROCESSES

The prefix *geo* in geostatistics originally implied Statistics pertaining to the Earth (see Matheron, 1962, 1963). However, more recently, Geostatistics has been used in a variety of disciplines from Agriculture to Zoology. [Within Meteorology, a virtually identical theory called *objective analysis* was developed by Gandin (1963).] It is known for its applicability under rather mild assumptions about the mean and covariance function of the phenomenon of interest. Less well known is its ability to incorporate the known action of scientific processes (e.g., a transport model for aerosol on continental scales might be used to define the mean function, with residual uncertainty represented by random spatial variability).

Geostatistical processes are spatial random processes defined on $D_s \subset \mathbb{R}^d$ whose parameters are defined directly through first and second moments (means, variances, and covariances). Often, although not always, D_s has positive d -dimensional volume. To see how a geostatistical process can be used as a process model in an HM, consider the following generalization of the simple HM given above.

Data model: Conditional on σ_ε^2 , and for $i = 1, \dots, m$,

$$Z(\mathbf{s}_i) | Y(\mathbf{s}_i), \sigma_\varepsilon^2 \sim \text{ind. Gau}(Y(\mathbf{s}_i), \sigma_\varepsilon^2).$$

Process model: Conditional on the p -dimensional vector β and C_Y , $Y(\cdot)$ is a Gaussian process with the following properties: For $\mathbf{s} \in D_s$, $\mathbf{h} \in \mathbb{R}^d$,

$$\begin{pmatrix} Y(\mathbf{s}) \\ Y(\mathbf{s} + \mathbf{h}) \end{pmatrix} \Big| \beta, C_Y(\cdot) \\ \sim \text{Gau} \left(\begin{pmatrix} \mathbf{x}(\mathbf{s})' \\ \mathbf{x}(\mathbf{s} + \mathbf{h})' \end{pmatrix} \beta, \Sigma_2(\mathbf{h}) \right),$$

where $\mathbf{x}(\cdot)$ is a p -dimensional vector of covariates that is potentially available at every location in D_s , $\Sigma_2(\mathbf{h})$ is a 2×2 covariance matrix whose two diagonal elements are both σ_Y^2 and whose off-diagonal element is $C_Y(\mathbf{h})$; and C_Y is a positive-definite function on $\mathbb{R}^d$ with $C_Y(\mathbf{0}) = \sigma_Y^2 < \infty$.

An equivalent way to write this process model is

$$Y(\mathbf{s}) = \mathbf{x}(\mathbf{s})'\boldsymbol{\beta} + \delta(\mathbf{s}), \quad \mathbf{s} \in D_s, \quad (4.1)$$

where $\delta(\cdot)$ is a mean-zero Gaussian process with stationary covariance function C_Y . Then, for the observation locations, $\{\mathbf{s}_1, \dots, \mathbf{s}_m\}$, we can write $\mathbf{Y} \equiv (Y(\mathbf{s}_1), \dots, Y(\mathbf{s}_m))'$ according to the linear mixed model:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}, \quad (4.2)$$

where $\boldsymbol{\beta}$ is a p -dimensional *fixed* effect, $\mathbf{X} \equiv (x_j(\mathbf{s}_i))$ is a known $m \times p$ matrix ($p < m$) of covariates, $\boldsymbol{\delta}$ is a (spatial) *random* effect, and $\text{var}(\boldsymbol{\delta}) \equiv (C_Y(\mathbf{s}_i - \mathbf{s}_j))$ is an $m \times m$ positive-definite matrix. Furthermore, the data model can be written as (conditional on $\{Y(\mathbf{s}_i)\}$):

$$Z(\mathbf{s}_i) = Y(\mathbf{s}_i) + \varepsilon(\mathbf{s}_i), \quad i = 1, \dots, m, \quad (4.3)$$

where $\{\varepsilon(\mathbf{s}_i)\}$ are *iid* $\text{Gau}(0, \sigma_\varepsilon^2)$ and independent from $Y(\cdot)$.

The geostatistical model that is equivalent to the HM given just above is obtained from interpreting (4.3) unconditionally, where $Y(\cdot)$ is defined by (4.1). Consequently,

$$\text{var}(Z(\mathbf{s})) \equiv \sigma_Z^2 = \sigma_Y^2 + \sigma_\varepsilon^2, \quad \mathbf{s} \in D_s,$$

which is referred to as the *sill* in the geostatistics literature. We have already seen in the introduction to this chapter that the *nugget effect* is,

$$c_0 = \sigma_0^2 + \sigma_\varepsilon^2,$$

where $\sigma_0^2 \equiv C_Y(\mathbf{0}+)$.

Geostatistics is mostly concerned with building models of spatial dependence and with predicting (some functional of) the spatial process optimally. The emphasis in this section will be on the spatial-dependence models (based on the variogram or the covariance function) and on a spatial-prediction method known as *kriging*, viewed in the context of hierarchical modeling.

4.1.1 The Variogram and the Covariance Function

First, we present a model-based measure of the spatial statistical dependence in a geostatistical process, called the *variogram*. In Chapter 3, we saw that time series analysts have traditionally expressed temporal dependence through the *covariance function* and the mathematically equivalent power spectrum, and in Yaglom (1962, 1987) analogous measures of spatial dependence are developed. In fact, the class of processes with a stationary variogram contains the class of processes with a stationary covariance function, and optimal linear prediction can be carried out on a wider class of processes than the second-order stationary

class traditionally used in Statistics (Matheron, 1963; Cressie, 1993, Section 2.3.2). It should be noted that early papers by Kolmogorov (1941) and Jowett (1952) defined and used the variogram in one dimension. (Kolmogorov called it the *structure function* and Jowett called it the *mean-squared difference*.)

Stationary Variogram

Let $\{Y(\mathbf{s}) : \mathbf{s} \in D_s \subset \mathbb{R}^d\}$ be a real-valued spatial process defined on a domain D_s of the d -dimensional Euclidean space $\mathbb{R}^d$, and suppose that differences of variables displaced $\mathbf{h}$ -apart vary in a way that depends only on $\mathbf{h}$. Specifically, suppose that

$$\text{var}(Y(\mathbf{s} + \mathbf{h}) - Y(\mathbf{s})) = 2\gamma_Y(\mathbf{h}), \quad \text{for all } \mathbf{s}, \mathbf{s} + \mathbf{h} \in D_s. \quad (4.4)$$

The quantity $2\gamma_Y(\cdot)$, which is a function only of the *difference* between the spatial locations $\mathbf{s}$ and $(\mathbf{s} + \mathbf{h})$, is called the *stationary variogram*. The variogram must satisfy the conditional nonpositive-definiteness condition,

$$\sum_{i=1}^k \sum_{j=1}^k \alpha_i \bar{\alpha}_j 2\gamma_Y(\mathbf{s}_i - \mathbf{s}_j) \leq 0, \quad (4.5)$$

for any positive integer k , any set of spatial locations $\{\mathbf{s}_i : i = 1, \dots, k\}$, and any set of complex numbers $\{\alpha_i : i = 1, \dots, k\}$ satisfying $\sum_{i=1}^k \alpha_i = 0$. (If α is a complex number, then recall that $\bar{\alpha}$ denotes its complex conjugate.) The condition (4.5) is extremely important, because it guarantees that all model-based variances are nonnegative (e.g., Cressie, 1993, Section 2.5.2). Any variogram that satisfies (4.5) is referred to as *valid*.

When $2\gamma_Y(\mathbf{h})$ can be written as a function of $\|\mathbf{h}\|$, for $\mathbf{h} = (h_1, \dots, h_d)' \in \mathbb{R}^d$, where $\|\mathbf{h}\| \equiv (h_1^2 + \dots + h_d^2)^{1/2}$, the variogram is said to be *isotropic*; otherwise it is said to be *anisotropic*. When clarity demands it, we write an isotropic semivariogram as, $\gamma_Y(\mathbf{h}) = \gamma_Y^o(\|\mathbf{h}\|)$, where γ_Y^o has domain $\mathbb{R}$. Figure 4.4 shows a commonly used isotropic (semi)variogram exponential model $\gamma_Y^o(\cdot)$. Note the discontinuity at the origin (which is due to the microscale variation, σ_0^2 , discussed in the introduction to this chapter).

Variogram models that depend on only a few parameters, $\boldsymbol{\theta}$, can be used as succinct summaries of the spatial dependence. Variograms are an important component of optimal spatial linear prediction (kriging), as we shall see later in Section 4.1.2. For example, the *Matérn semivariogram* is

$$\gamma_Y(\mathbf{h}; \boldsymbol{\theta}) = C_Y(\mathbf{0}; \boldsymbol{\theta}) - C_Y(\mathbf{h}; \boldsymbol{\theta}); \quad \mathbf{h} \in \mathbb{R}^d,$$

where

$$C_Y(\mathbf{h}; \boldsymbol{\theta}) = \sigma_0^2 I(\|\mathbf{h}\| = 0) + \sigma_1^2 \{2^{\theta_2-1} \Gamma(\theta_2)\}^{-1} \{\|\mathbf{h}\|/\theta_1\}^{\theta_2} K_{\theta_2}(\|\mathbf{h}\|/\theta_1) \quad (4.6)$$

is the Matérn covariance function (Stein, 1999, p. 31)). In (4.6), $C_Y(\mathbf{0}; \boldsymbol{\theta}) = \sigma_0^2 + \sigma_1^2$, K_{θ_2} is a modified Bessel function of the second

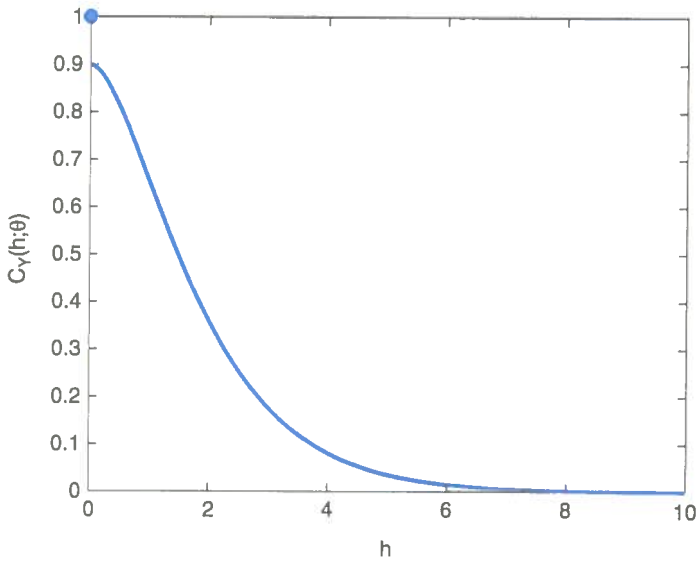


Figure 4.2 Matérn covariance function with scale parameter $\theta_1 = 1$ and smoothness parameter $\theta_2 = 1.5$; the nugget effect is 10% of the total variance.

kind of order θ_2 (e.g., Abramowitz and Stegun, 1964, pp. 374–379), and $\theta \in \{(\theta_1, \theta_2, \sigma_0^2, \sigma_1^2)': \theta_1 > 0, \theta_2 > 0, \sigma_0^2 \geq 0, \sigma_1^2 \geq 0\}$. Figure 4.2 shows a plot of (4.6) versus $\|\mathbf{h}\|$, for the particular case of $\theta = (1, 1.5, 0.1, 0.9)'$. Its corresponding semivariogram is

$$\gamma_Y(\mathbf{h}) = (0.1)I(\|\mathbf{h}\| \neq 0) + (0.9) - (0.9)e^{-\|\mathbf{h}\|}(1 + \|\mathbf{h}\|), \quad \mathbf{h} \in \mathbb{R}^d.$$

Another common isotropic semivariogram used is the *powered-exponential*,

$$\gamma_Y(\mathbf{h}; \theta) = C_Y(\mathbf{0}; \theta) - C_Y(\mathbf{h}; \theta),$$

where

$$C_Y(\mathbf{h}; \theta) = \sigma_0^2 I(\|\mathbf{h}\| = 0) + \sigma_1^2 \exp\{-(\|\mathbf{h}\|/\theta_1)^{\theta_2}\}, \quad \mathbf{h} \in \mathbb{R}^d, \quad (4.7)$$

and $\theta \in \{(\theta_1, \theta_2, \sigma_0^2, \sigma_1^2)': \theta_1 > 0, 0 < \theta_2 \leq 2, \sigma_0^2 \geq 0, \sigma_1^2 \geq 0\}$. The powered-exponential covariance function (4.7) is discussed, for example, in Diggle and Ribeiro (2007, Section 3.4.2). The most familiar member of the class is the exponential covariance function, obtained when $\theta_2 = 1$:

$$C_Y(\mathbf{h}; \theta) = \sigma_0^2 I(\|\mathbf{h}\| = 0) + \sigma_1^2 \exp\{-\|\mathbf{h}\|/\theta_1\}, \quad \mathbf{h} \in \mathbb{R}^d. \quad (4.8)$$

Figure 4.3 shows the particular case of (4.7) with $\theta = (1, 1, 0.1, 0.9)'$; that is, it is a plot of the exponential covariance function, $(0.1)I(\|\mathbf{h}\| = 0) +$

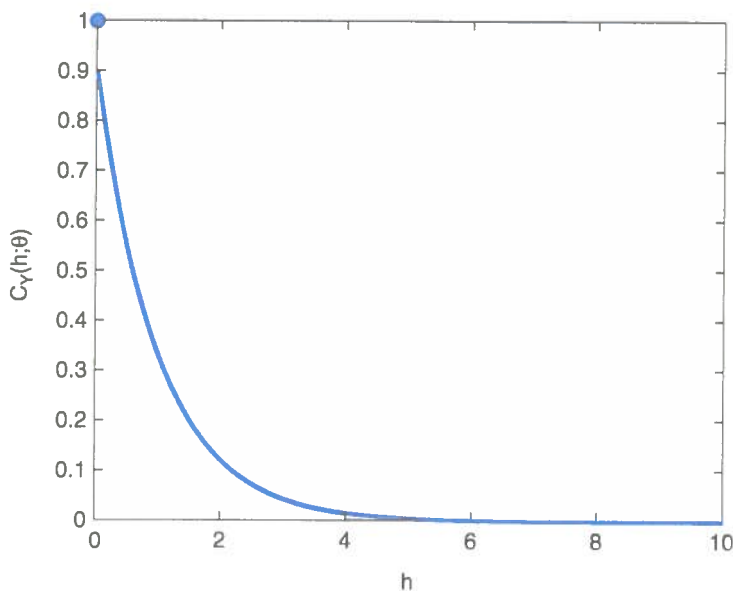


Figure 4.3 Powered exponential covariance function with scale parameter $\theta_1 = 1$ and power parameter $\theta_2 = 1$; the nugget effect is 10% of the total variance.

$(0.9)e^{-\|\mathbf{h}\|}$, versus $\|\mathbf{h}\|$. Further details on the powered-exponential class given by (4.7) can be found in Yaglom (1987, Vol. II, pp. 48–49). Figure 4.4 shows a plot of the corresponding *exponential semivariogram* given by

$$\gamma_Y(\mathbf{h}) = (0.1)I(\|\mathbf{h}\| \neq 0) + (0.9) - (0.9)e^{-\|\mathbf{h}\|}, \quad \mathbf{h} \in \mathbb{R}^d,$$

versus $\|\mathbf{h}\|$. Other models, such as the spherical, the rational-quadratic, the wave, the linear, and the power semivariograms, are discussed, for example, in Cressie (1993, pp. 61–63).

When the variogram is anisotropic, it is no longer a function purely of distance between the two spatial locations. Anisotropies are caused by the underlying physical process evolving differentially in space. That is, different (temporal) dynamics in different directions can cause the process to be anisotropic in space. Sometimes the anisotropy can be corrected by an invertible linear transformation of the lag vector $\mathbf{h}$. If $2\gamma_Y(\cdot)$ can be written as

$$2\gamma_Y(\mathbf{h}) = 2\gamma_Y^\circ(\|\mathbf{A}\mathbf{h}\|), \quad \mathbf{h} \in \mathbb{R}^d, \quad (4.9)$$

where $\mathbf{A}$ is an invertible $d \times d$ matrix and γ_Y° has domain $\mathbb{R}$, it is said to be *geometrically anisotropic*.

In the paragraphs that follow, it will be seen that stationary variogram models are more general than stationary covariance-function models.

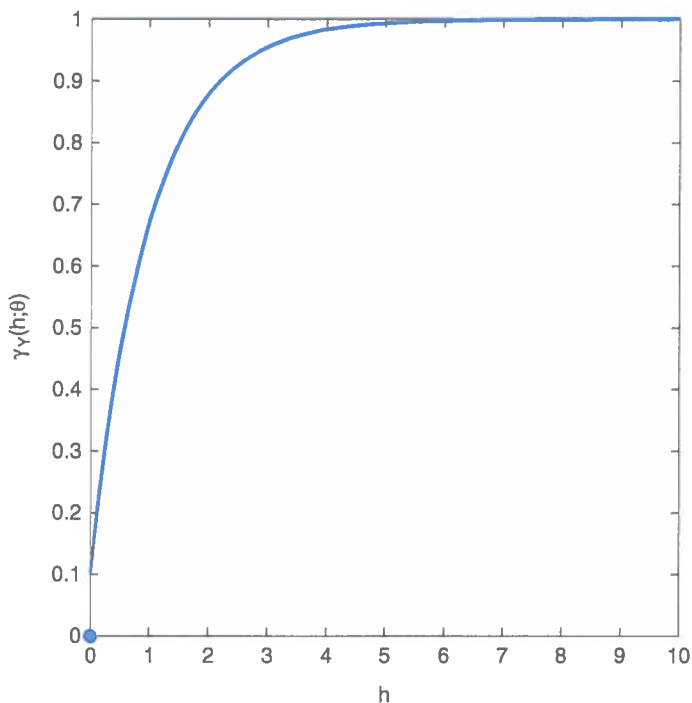


Figure 4.4 Exponential semivariogram with scale parameter $\theta_1 = 1$; the nugget effect is 10% of the total variance (sill).

Stationary Covariance Function

Replace (4.4) with the (stronger) assumption,

$$\text{cov}(Y(\mathbf{s} + \mathbf{h}), Y(\mathbf{s})) = C_Y(\mathbf{h}), \quad \text{for all } \mathbf{s}, \mathbf{s} + \mathbf{h} \in D_s, \quad (4.10)$$

and specify the mean function to be constant; that is,

$$E(Y(\mathbf{s})) = \mu, \quad \text{for all } \mathbf{s} \in D_s. \quad (4.11)$$

Then, restrictions (4.10) and (4.11) define the class of *second-order* (or weakly) *stationary* processes in D_s , with stationary *covariance function* $C_Y(\cdot)$. Time series analysts often assume the temporal analogue of (4.10) and work with the correlation function,

$$\rho_Y(\cdot) \equiv C_Y(\cdot)/C_Y(\mathbf{0}). \quad (4.12)$$

The quantity $C_Y(\cdot)$ is called the (stationary) covariance function, which must satisfy the nonnegative-definiteness condition,

$$\sum_{i=1}^k \sum_{j=1}^k \alpha_i \bar{\alpha}_j C_Y(\mathbf{s}_i - \mathbf{s}_j) \geq 0, \quad (4.13)$$

for any positive integer k , any set of spatial locations $\{s_i : i = 1, \dots, k\}$, and any set of complex numbers $\{\alpha_i : i = 1, \dots, k\}$ (not only those that sum to zero see (4.5)). Any covariance function that satisfies (4.13) is referred to as *valid*. The condition (4.13) is analogous to the conditional-nonpositive-definiteness condition (4.5) that a variogram must satisfy, and it likewise guarantees that model-based prediction variances are nonnegative (e.g., Cressie, 1993, Section 2.5.1). Notice that the nonnegative-definiteness condition (4.13) has to hold for *all* complex numbers $\{\alpha_i\}$. Hence, there will be some processes with valid stationary variograms that do not have valid stationary covariance functions (e.g., Cressie, 1993, p. 68).

When $C_Y(\mathbf{h})$ can be written as a function of $\|\mathbf{h}\|$, we say that it is *isotropic*; otherwise it is said to be *anisotropic*. When clarity demands it, we write an isotropic covariance function as, $C_Y(\mathbf{h}) = C_Y^o(\|\mathbf{h}\|)$, where C_Y^o has domain $\mathbb{R}$. Two examples of $C_Y^o(\cdot)$ are given in Figures 4.2 and 4.3. There is also a notion of geometric anisotropy, $C_Y(\mathbf{h}) = C_Y^o(\|\mathbf{A}\mathbf{h}\|)$; $\mathbf{h} \in \mathbb{R}^d$, analogous to (4.9).

Intrinsic Stationarity

Conditions (4.4) and (4.11) define the class of *intrinsically stationary* processes, which we now show contains the class of second-order stationary processes. Assuming only the existence of the stationary covariance function given by (4.10), the *semivariogram* (i.e., one half the variogram) of $Y(\cdot)$ exists and is given by

$$\gamma_Y(\mathbf{h}) = C_Y(\mathbf{0}) - C_Y(\mathbf{h}), \quad \mathbf{h} \in \mathbb{R}^d. \quad (4.14)$$

As an example, see Figure 4.3, which shows an exponential covariance function, and Figure 4.4, which shows the corresponding exponential semivariogram. Equation (4.14) is easy to prove and establishes that a second-order stationary process is always intrinsically stationary. To see that the converse is *not* true, consider a one-dimensional standard Wiener process $\{W(s) : s \geq 0\}$. Here, $2\gamma_W(h) = -|h|$ (where $-\infty < h < \infty$), but $\text{cov}(W(s), W(u)) = \min(s, u)$, which is *not* a function of $|s - u|$; an analogous result is true in $\mathbb{R}^d$ (e.g., Cressie, 1993, p. 62). Thus, since the constant-mean assumption (4.11) is common to both, the class of second-order stationary processes is *strictly* contained in the class of intrinsically stationary processes. In other words, intrinsic stationarity holds (and hence the stationary variogram exists) for a larger class of processes.

Spectral Representation

Both the variogram and the covariance function have a representation in terms of a (power) spectral density, under appropriate regularity conditions. We present the result here for the more familiar case of the covariance function; the result for the variogram can be found in, for example, Cressie (1993, Section 2.5).

Theorem 4.1 (Bochner, 1955). If $\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} |C_Y(\mathbf{h})| d\mathbf{h} < \infty$, then a valid, real-valued covariance function can be written as

$$C_Y(\mathbf{h}) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \cos(\boldsymbol{\omega}'\mathbf{h}) f_Y(\boldsymbol{\omega}) d\boldsymbol{\omega}, \quad (4.15)$$

where $f_Y(\boldsymbol{\omega}) \geq 0$ is symmetric about $\boldsymbol{\omega} = \mathbf{0}$.

The quantity $f_Y(\cdot)$ is called the *spectral density* of $C_Y(\cdot)$, and it partitions the total variation through $\sigma_Y^2 = C_Y(\mathbf{0}) = \int_{\mathbb{R}^d} f(\boldsymbol{\omega}) d\boldsymbol{\omega}$. It is the nonnegativity of the spectral density that guarantees nonnegative-definiteness of the covariance function, and vice versa. Consequently, substituting any absolutely integrable, nonnegative, symmetric function $f(\cdot)$ into (4.15) generates a valid, real-valued covariance function; further details on these constructions can be found in Cressie (1993, pp. 84–86). Sometimes the spectral density has a more succinct representation than the covariance function; such is the case with the Matérn covariance function (Stein, 1991; Fuentes, Chen, and Davis, 2008).

As we noticed above, the nonnegative-definiteness condition (4.13) for $C_Y(\cdot)$ is stronger than the *conditional* nonpositive-definiteness condition (4.5) for $\gamma_Y(\cdot)$. However, not surprisingly, $\gamma_Y(\cdot)$ has a spectral representation analogous to Theorem 4.1 (e.g., Cressie, 1993, p. 87).

The Variogram (Covariance Function) is a Parameter

There has been some confusion in the geostatistics literature whether the variogram is something computed from spatial data $\mathbf{Z} = (Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_m))'$, or whether it is a theoretical function. We would like to think of the variogram as a (infinite-dimensional) parameter in the space of all conditionally negative-definite functions, and then we refer to the *empirical variogram* as the quantity computed from the data:

$$2\hat{\gamma}_Z^o(h) \equiv \text{ave}\{(Z(\mathbf{s}_i) - Z(\mathbf{s}_j))^2 : \|\mathbf{s}_i - \mathbf{s}_j\| \in T(h); i, j = 1, \dots, m\}, \quad (4.16)$$

where $T(h)$ is a tolerance region around h (such as $h \pm \Delta$, Δ small). The *empirical semivariogram* is $\hat{\gamma}_Z^o(\cdot)$. For (4.16) to be an *appropriate estimator* of the parameter $2\gamma_Y(\cdot)$, the isotropy assumption that $2\gamma_Y(\mathbf{h}) \equiv 2\gamma_Y^o(\|\mathbf{h}\|)$ has to be made; the mean of $Z(\cdot)$ and of $Y(\cdot)$ has to be assumed constant; and the measurement-error variance component $2\sigma_\epsilon^2$ has to be subtracted from (4.16).

Initially, we wish to test the null hypothesis, $H_0: Y(\cdot)$ has *no* spatial dependence. This initial test is important to carry out, to avoid modeling spatial structure that is not actually there. Consider the test statistic

$$F \equiv \hat{\gamma}_Z^o(h_1) / \hat{\sigma}_Z^2,$$

where $\hat{\sigma}_Z^2 \equiv \sum_{i=1}^m (Z(\mathbf{s}_i) - \hat{\mu}_Z)^2 / (m - 1)$, $\hat{\mu}_Z \equiv \sum_{i=1}^m Z(\mathbf{s}_i) / m$, and h_1 is the smallest lag from all possible lags $h_1, \dots, h_L$. We reject H_0 for $|F - 1|$ large.

A permutation-based test of H_0 does not make any distributional assumptions about $\mathbf{Z}$; the data locations are permuted and F is recomputed for all possible permutations (or for a computer-generated sample of random permutations). If the observed F is above the 97.5 percentile or below the 2.5 percentile of the permutation distribution, then reject H_0 (at a 5% level of significance). If we make the additional assumption that $Y(\cdot)$ is a Gaussian process and the measurement-error process $\varepsilon(\cdot)$ is Gaussian, then H_0 can be tested based on the percentiles of the ratio of two quadratic forms (cf. Moran's I statistic discussed at the beginning of Section 4.2).

The variogram, $2\gamma_Y(\cdot)$, is generally an unknown quantity, for which statistical inference, based on spatial data $\mathbf{Z} \equiv (Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_m))'$, is needed. As we discussed above, the process $Z(\cdot)$ is not the same as the process $Y(\cdot)$, but they are related quite simply through (4.3), where the measurement-error variance is defined to be σ_ε^2 . It is straightforward to show that

$$2\gamma_Z(\mathbf{h}) = 2\gamma_Y(\mathbf{h}) + 2\sigma_\varepsilon^2 I(\mathbf{h} \neq \mathbf{0}), \quad \mathbf{h} \in \mathbb{R}^d,$$

and

$$C_Z(\mathbf{h}) = C_Y(\mathbf{h}) + \sigma_\varepsilon^2 I(\mathbf{h} = \mathbf{0}), \quad \mathbf{h} \in \mathbb{R}^d.$$

Assuming isotropy, analogous relations are available between $2\gamma_Z^o(\cdot)$, $2\gamma_Y^o(\cdot)$, $C_Z^o(\cdot)$, and $C_Y^o(\cdot)$.

From the point of view of the HM outlined in Section 2.1, the variogram (or the covariance function) could be thought of as a parameter. Thus, its estimation and subsequent use in optimal spatial prediction (kriging) could be viewed as working within the framework of an empirical hierarchical model (EHM). Specifically, the estimate is “plugged” into optimal-spatial-prediction equations (Section 4.1.2). Putter and Young (2001) show that, for Gaussian processes, the effect of this estimation is negligible under continuity conditions for the estimated and true Gaussian probability measures. Variogram-model fitting using several different criteria, including maximum likelihood, REML, and weighted least squares, is considered in Cressie (1993, Section 2.6), for a data model that assumes additive measurement error. Having more general data models (e.g., Diggle, Tawn, and Moyeed, 1998) requires rethinking some of these estimators. The composite-likelihood approach of Vecchia (1988) is one possibility, and EM estimation is another (Zhu, Gu, and Peterson, 2007).

Ecker and Gelfand (1997) consider inference on the variogram in the context of a Bayesian hierarchical model (BHM) and give the posterior (conditional on the data $\mathbf{Z}$) distribution of the variogram parameters. Daniels and Cressie (2001) give inference on a stationary covariance function in an HM setting (in one dimension). Assuming a BHM, inference on model parameters θ and process $Y(\cdot)$ is based on the posterior distribution $[Y(\cdot), \theta | \mathbf{Z}]$; see Diggle and Ribeiro (2007, Chapter 7) for a discussion of analytical and Monte Carlo methods for obtaining the posterior distribution. Omre and Tjelmeland (1997) and Diggle, Tawn, and Moyeed (1998) show how to carry out this inference, particularly

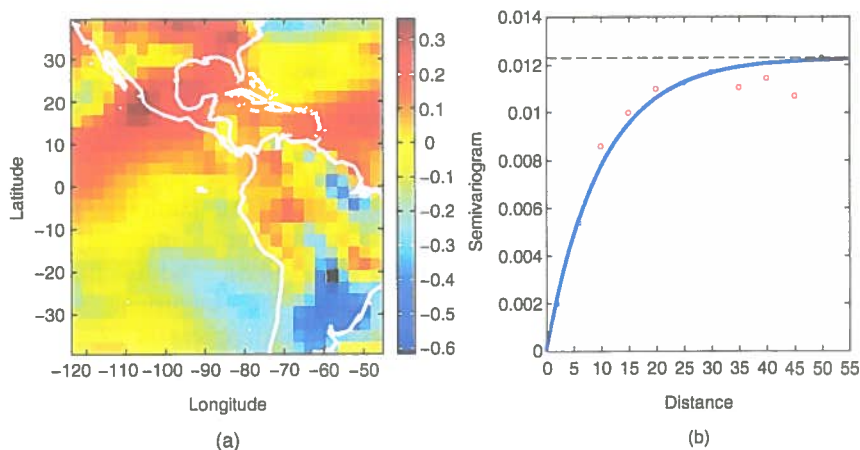


Figure 4.5 (a) Temperature change (1990s minus 1980s) in 28×28 grid cells over the Americas. The values were originally produced by an NCAR model, and they represent “the truth.” (b) Empirical and fitted (exponential model) semivariogram after the values in (a) were detrended by latitude and longitude. Units on the horizontal axis are the same as those given in (a).

on $Y(\cdot)$, based on MCMC. In a sense, inference in an EHM is simpler, in that once an estimator of θ is obtained, say $\hat{\theta}(\mathbf{Z})$, inference on $Y(\cdot)$ is based on the posterior distribution $[Y(\cdot)|\mathbf{Z}, \theta]$, with $\hat{\theta}(\mathbf{Z})$ substituted for θ in that posterior. However, inference in an EHM is often overly liberal in finite samples, since the variability in the estimator $\hat{\theta}(\mathbf{Z})$ is not accounted for.

Having considered the variogram as a parameter, we now turn to inference on the process $Y(\cdot)$; that is, we turn to spatial prediction. One commonly used Bayes estimator is $E(Y(\cdot)|\mathbf{Z})$ in a BHM, or $E(Y(\cdot)|\mathbf{Z}, \hat{\theta})$ in an EHM. In the small example that follows, we assume Gaussian distributions for $Y(\cdot)$ and $\epsilon(\cdot)$, and we motivate optimal spatial prediction through a series of figures showing the results of (simple) kriging.

Decadal Temperature Differences over the Americas

The basis of this illustrative example is output from the Climate System Model at the National Center for Atmospheric Research (NCAR) in Boulder, CO. The output is monthly temperature on longitude–latitude grid cells, each of which is roughly 2.8° longitude by 2.8° latitude. We have chosen to concentrate on a region over the Americas made up of 28×28 grid cells. Monthly temperatures for the 1980s were averaged for each grid cell and likewise for the 1990s. Consider these two averages in grid cell centered at $\mathbf{s} \in D_s \subset \mathbb{R}^2$, and define $Y(\mathbf{s}) \equiv 1990\text{s average} - 1980\text{s average}$. The values of $Y(\cdot)$ are shown in Figure 4.5a.

Although $Y(\cdot)$ was produced by a deterministic model, in this example we think of it as the “truth,” and we model it statistically as a random process defined on the 28×28 grid centers, D_s . After exploratory spatial data analysis,

Zhang, Craigmile, and Cressie (2008) found the following spatial model to provide a good fit:

$$Y(\mathbf{s}) = \mathbf{x}(\mathbf{s})'\boldsymbol{\beta} + \delta(\mathbf{s}), \quad \mathbf{s} \in D_s,$$

where for $\mathbf{s} = (s_1, s_2)'$, $\mathbf{x}(\mathbf{s}) = (1, s_1, s_2, s_2^2, s_2^3)'$, and $\boldsymbol{\beta} \equiv (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4)' = (-.1419, -.0020, .0091, -.000087, -.0000051)'$. The component process $\delta(\cdot)$ is second-order stationary with mean zero and exponential covariance function, $C_Y(\mathbf{h}; \boldsymbol{\theta})$, given by (4.8), with parameters $(\theta_1, \sigma_0^2, \sigma_1^2) = (10.0284, 0, 0.0123)$. Its fit to the empirical semivariogram [see (4.16)] of the detrended temperature differences is excellent, as can be seen in Figure 4.5b. Notice that all 28×28 values of $Y(\cdot)$ were used to fit this model; for the purposes of this example, we shall consider the fitted parameter values as *known* and proceed to introduce optimal spatial prediction (here, simple kriging), to motivate the next subsection.

We now “corrupt” $Y(\cdot)$ by subsampling only a 10×10 subset and adding random noise $\varepsilon(\cdot)$ (with known σ_ε^2) to it. By doing this, we can see how well simple kriging recovers the true process $Y(\cdot)$ at all 28×28 locations. Following Zhang, Craigmile, and Cressie (2008), we choose to sample at 10×10 spatially regular locations illustrated by the crosses in Figure 4.1. That is, the spatial dataset we shall consider has $m = 100$, and

$$Z(\mathbf{s}_i) = Y(\mathbf{s}_i) + \varepsilon(\mathbf{s}_i), \quad i = 1, \dots, 100,$$

where $\{\mathbf{s}_i : i = 1, \dots, 100\}$ are the grid cells defined by the crosses in Figure 4.1, and $\varepsilon(\cdot)$ is a white-noise process with mean zero and variance $\sigma_\varepsilon^2 = 0.00615$. Our goal is to make inference on $\{Y(\mathbf{s}) : \mathbf{s} \in D_s\}$ based on noisy and incomplete data $\mathbf{Z} \equiv (Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_{100}))'$. This includes not only predicting $Y(\mathbf{s}_0)$, but also providing a measure of uncertainty associated with that prediction.

Using the theory presented in Section 4.1.2, we use formula (4.28) to produce the optimal spatial predictor,

$$\{Y^*(\mathbf{s}_0) : \mathbf{s}_0 \in D_s\},$$

and formula (4.26) to produce the associated measure of uncertainty,

$$\{\sigma_{Y,sk}(\mathbf{s}_0) : \mathbf{s}_0 \in D_s\}.$$

The former is called the (simple) *kriging predictor*, and the latter is called the (simple) *kriging standard error*. Figure 4.6 illustrates all this through a sequence of maps: Figure 4.6a shows the smooth, complete $Y(\cdot)$ that we have called the “truth” (it is identical to the map shown in Figure 4.5a). Figure 4.6b shows the noisy, corrupted data $\mathbf{Z}$ from which we intend to recover $Y(\cdot)$.

Figure 4.6c shows the optimal (simple kriging) predictor, which should be compared directly to the map above it in Figure 4.6a. Based on a data fraction of

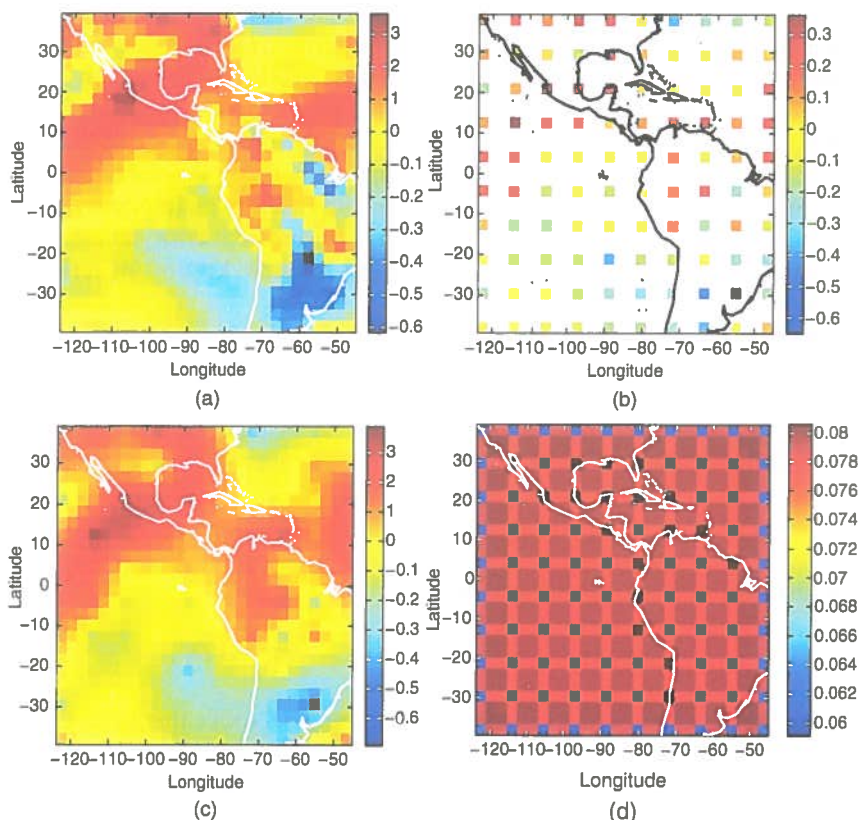


Figure 4.6 (a) Same plot as Figure 4.5a: Temperature change (1990s minus 1980s); the 28×28 values represent “the truth.” (b) The 10×10 observations are obtained by subsampling “the truth” and adding mean-zero Gaussian noise. (c) Simple kriging predictor obtained from the missing and noisy data in (b). (d) Kriging standard error corresponding to simple kriging; the pattern is expected due to regular sampling in (b).

only $(10 \times 10)/(28 \times 28) = 13\%$, the reconstruction of $Y(\cdot)$ given by $Y^*(\cdot)$ in Figure 4.6c is visually very good, albeit somewhat “smoother” than the original $Y(\cdot)$ in Figure 4.6a. The map in Figure 4.6d shows how good we expect $Y^*(\cdot)$ to be based on the covariance function C_Y and the 100 data locations $\{s_i\}$. Its patterned appearance is not surprising due to the regular spatial locations of the data, and the uncertainty of $Y^*(s_0)$ increases the further s_0 is from a data location (as expected).

4.1.2 Kriging (Optimal Spatial Prediction)

Kriging is a spatial prediction methodology due to Matheron (1963), with roots in the geodesy work of Gauss carried out early in the nineteenth century. A very early derivation of optimal prediction (in the temporal domain rather

than the spatial domain) that results in the simple kriging predictor, can be found in Davis (1952). Matheron coined the term *kriging* in honor of D. G. Krige, a South African mining engineer [see Cressie (1990) for an account of the origins of kriging]. It is well known to geologists, hydrologists, soil scientists, ecologists, and so forth, as a statistical methodology based on the theory of stochastic processes. Matheron, and many that followed (e.g., Journé and Huijbregts, 1978; Isaaks and Srivastava, 1989), assumed implicitly that any observation on the process is taken with zero measurement error, an assumption that in general is untenable. There is a way to view kriging in terms of a hierarchical model (HM).

The (hidden) process of interest $\{Y(\mathbf{s}) : \mathbf{s} \in D_s\}$ is a spatial random process satisfying $0 < \text{var}(Y(\mathbf{s})) < \infty$, for all $\mathbf{s} \in D_s$. The data $\mathbf{Z}$ are noisy, incomplete observations on $Y(\cdot)$ given by (4.3), with measurement error $\varepsilon(\cdot)$ that has mean zero and variance $\sigma_\varepsilon^2 \in [0, \infty)$. It is simplest to assume for the moment that the mean function,

$$\mu_Y(\mathbf{s}) \equiv E(Y(\mathbf{s})), \quad \mathbf{s} \in D_s, \quad (4.17)$$

is known and that the covariance function,

$$C_Y(\mathbf{u}, \mathbf{v}) \equiv \text{cov}(Y(\mathbf{u}), Y(\mathbf{v})), \quad \mathbf{u}, \mathbf{v} \in D_s, \quad (4.18)$$

and the measurement-error variance, σ_ε^2 , are known. Notice that here, $C_Y(\mathbf{u}, \mathbf{v})$ is not necessarily a function of $(\mathbf{u} - \mathbf{v})$; we are attempting to be as general as possible. The (nonstationary) variogram is derivable as $2\gamma_Y(\mathbf{u}, \mathbf{v}) \equiv \text{var}(Y(\mathbf{u}) - Y(\mathbf{v})) = C_Y(\mathbf{u}, \mathbf{u}) + C_Y(\mathbf{v}, \mathbf{v}) - 2C_Y(\mathbf{u}, \mathbf{v})$.

We require the posterior distribution, $[Y(\cdot)|\mathbf{Z}, \mu_Y, C_Y, \sigma_\varepsilon^2]$, where $\mathbf{Z} \equiv (Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_m))'$, to make inference on $Y(\cdot)$. This is obtained in an HM setting from (i) a data model, $[\mathbf{Z}|Y(\cdot), \sigma_\varepsilon^2]$, and (ii) a process model, $[Y(\cdot)|\mu_Y, C_Y]$. However, in its original formulation, geostatistics eschewed distributional assumptions. Diggle, Tawn, and Moyeed (1998) coined the term *model-based geostatistics* to describe a BHM formulation of kriging with the distributional assumptions needed to derive the posterior distribution. In the rest of this section, we present the original formulation of kriging *first* and *then* compare it to an EHM and a BHM formulation of the same.

Simple Kriging

The methodology known as *simple kriging* looks in the class of all heterogeneously linear predictors, $\{\ell'Z + k : \ell \in \mathbb{R}^m, k \in \mathbb{R}\}$, to obtain an optimal spatial predictor of $Y(\mathbf{s}_0)$; $\mathbf{s}_0 \in D_s$. Recall from Section 2.2 that $E(Y(\mathbf{s}_0)|\mathbf{Z})$ is optimal in terms of minimizing the mean squared prediction error. Likewise, the simple kriging predictor is the heterogeneously linear combination of the data that minimizes the mean squared prediction error,

$$\text{MSPE}(\ell, k) \equiv E(Y(\mathbf{s}_0) - \ell'Z - k)^2, \quad (4.19)$$

with respect to ℓ and k . It is generally *not* appropriate to minimize $E(Z(s_0) - \ell'Z - k)^2$, as one often sees in the geostatistics literature; from an HM point of view, this ignores the data model and implicitly assumes that $\sigma_\varepsilon^2 = 0$. One could look upon the derivations we give below as “kriging done right,” under the general assumption that $\sigma_\varepsilon^2 \geq 0$.

Write

$$\text{MSPE}(\ell, k) = \text{var}(Y(s_0) - \ell'Z - k) + \{E(Y(s_0) - \ell'Z - k)\}^2.$$

The second term is $\{\mu_Y(s_0) - \ell'\mu_Y - k\}^2$, where $\mu_Y \equiv (\mu_Y(s_1), \dots, \mu_Y(s_m))'$, which is minimized (in fact, is equal to 0) if k satisfies

$$k = \mu_Y(s_0) - \ell'\mu_Y. \quad (4.20)$$

Now, since k is not random, the first term is

$$C_Y(s_0, s_0) - 2\ell'\text{cov}(Y(s_0), Z)' + \ell'C_Z\ell, \quad (4.21)$$

where the $m \times m$ covariance matrix is given by $C_Z \equiv (C_Z(s_i, s_j))$;

$$C_Z(u, v) \equiv \begin{cases} C_Y(v, v) + \sigma_\varepsilon^2, & u = v, \\ C_Y(u, v), & u \neq v; \end{cases} \quad (4.22)$$

and

$$\begin{aligned} \text{cov}(Y(s_0), Z) &= (C_Y(s_0, s_1), \dots, C_Y(s_0, s_m)) \\ &\equiv c_Y(s_0)'. \end{aligned} \quad (4.23)$$

Unless $\sigma_\varepsilon^2 = 0$, there is a non-zero difference between $c_Y(s_0)$ in (4.23) and $c_Z(s_0)' \equiv \text{cov}(Z(s_0), Z) = (C_Z(s_0, s_1), \dots, C_Z(s_0, s_m))$, the latter quantity being the one used incorrectly in some classical geostatistics kriging equations.

Upon differentiating (4.21) with respect to ℓ and setting the result equal to $\mathbf{0}$, we obtain

$$-2c_Y(s_0) + 2C_Z\ell = \mathbf{0}; \quad (4.24)$$

that is, the optimal coefficients are given by $\ell^* \equiv C_Z^{-1}c_Y(s_0)$, and hence the optimal constant term is, $k^* \equiv \mu_Y(s_0) - c_Y(s_0)'C_Z^{-1}\mu_Y$. Finally then, the simple-kriging predictor is

$$Y^*(s_0) \equiv \mu_Y(s_0) + c_Y(s_0)'C_Z^{-1}(Z - \mu_Y), \quad (4.25)$$

where $c_Y(s_0)'$ is given by (4.23). Furthermore, upon substituting (4.20) and (4.24) into (4.19), the minimized mean squared prediction error, often called the (simple) kriging variance, is

$$\sigma_{Y,sk}^2(s_0) \equiv \text{MSPE}(\ell^*, k^*) = C_Y(s_0, s_0) - c_Y(s_0)'C_Z^{-1}c_Y(s_0), \quad (4.26)$$

where the subscript “sk” on the left-hand side denotes *simple kriging*. When $\sigma_\varepsilon^2 = 0$, these formulas are the same as those found in, for example, Journé and Huijbregts (1978, p. 562).

As an example, consider the linear model (4.1), where

$$\mu_Y(\mathbf{s}) = \mathbf{x}(\mathbf{s})'\boldsymbol{\beta}, \quad \mathbf{s} \in D_s, \quad (4.27)$$

for $\mathbf{x}(\mathbf{s}) \equiv (x_1(\mathbf{s}), \dots, x_p(\mathbf{s}))'$, a p -dimensional vector of known covariates, and for $\boldsymbol{\beta} \equiv (\beta_1, \dots, \beta_p)'$, a p -dimensional vector of (for the moment, *known*) regression parameters. Traditional choices in geostatistics are, for example in $\mathbb{R}^2$, the spatial covariates $x_1(\mathbf{s}) = 1$, $x_2(\mathbf{s}) = s_1$, $x_3(\mathbf{s}) = s_2$, $x_4(\mathbf{s}) = s_1^2, \dots$, where $\mathbf{s} \equiv (s_1, s_2)'$. Polynomials, splines, and wavelets are special cases of spatial covariates (sometimes called basis functions). Basis functions model trend terms, but it should be realized that they can be augmented by or replaced with other covariates that attempt to explain directly the underlying physical/chemical/biological/economic/etc. mechanisms of the process $Y(\cdot)$. The principal requirement of a covariate $x_j(\mathbf{s})$ is that it is known at *all* $\mathbf{s} \in D_s$. Substituting (4.27) into (4.25), the simple-kriging predictor is

$$Y^*(\mathbf{s}_0) = \mathbf{x}(\mathbf{s}_0)'\boldsymbol{\beta} + \mathbf{c}_Y(\mathbf{s}_0)'\mathbf{C}_Z^{-1}(\mathbf{Z} - \mathbf{X}\boldsymbol{\beta}), \quad (4.28)$$

where $\mathbf{c}_Y(\mathbf{s}_0)$ is given by (4.23), $\mathbf{C}_Z = \text{var}(\mathbf{Z})$, and recall from (4.2) that $\mathbf{X} \equiv (x_j(\mathbf{s}_i))$ is an $m \times p$ matrix of covariate values at the locations where the data $\mathbf{Z}$ are observed. The simple-kriging variance for the predictor (4.28) is still given by (4.26); it does not involve the covariates, since $\boldsymbol{\beta}$ is assumed (for the moment) known.

A small example is illustrative of the calculations that are involved in (4.25) and (4.26). First, let the known mean $\mu_Y(\mathbf{s}) \equiv 18$; $\mathbf{s} \in D_s$, which is everywhere constant, and let the variance, $\text{var}(Y(\mathbf{s})) = 1$; $\mathbf{s} \in D_s$. Suppose $m = 2$; then $\boldsymbol{\mu}_Y = (18, 18)'$. Regarding the covariances, $C_Y(\mathbf{s}_0, \mathbf{s}_1)$, $C_Y(\mathbf{s}_0, \mathbf{s}_2)$, and $C_Y(\mathbf{s}_1, \mathbf{s}_2)$, we suppose that prediction location $\mathbf{s}_0$ is different from $\mathbf{s}_1$ and $\mathbf{s}_2$, that $\mathbf{s}_1$ is closer to $\mathbf{s}_0$ than is $\mathbf{s}_2$, and that $\sigma_\varepsilon^2 = 0.5$. We assume an exponential covariance function that results in

$$\begin{aligned} \mathbf{c}_Y(\mathbf{s}_0) &= (C_Y(\mathbf{s}_0, \mathbf{s}_1), C_Y(\mathbf{s}_0, \mathbf{s}_2))' \equiv (0.61, 0.22)', \\ \mathbf{C}_Z &= \begin{bmatrix} C_Y(\mathbf{s}_1, \mathbf{s}_1) & C_Y(\mathbf{s}_2, \mathbf{s}_1) \\ C_Y(\mathbf{s}_1, \mathbf{s}_2) & C_Y(\mathbf{s}_2, \mathbf{s}_2) \end{bmatrix} + \sigma_\varepsilon^2 \mathbf{I} \\ &= \begin{bmatrix} 1 & 0.37 \\ 0.37 & 1 \end{bmatrix} + 0.5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1.5 & 0.37 \\ 0.37 & 1.5 \end{bmatrix}, \end{aligned}$$

and

$$\begin{aligned} Y^*(\mathbf{s}_0) &= 18 + (0.61, 0.22) \begin{bmatrix} 1.5 & 0.37 \\ 0.37 & 1.5 \end{bmatrix}^{-1} (\mathbf{Z} - (18, 18)') \\ &= 18 + 0.3945\{Z(\mathbf{s}_1) - 18\} + 0.0494\{Z(\mathbf{s}_2) - 18\}, \end{aligned}$$

$$\begin{aligned}\sigma_{Y,sk}^2(\mathbf{s}_0) &= 1 - (0.61, 0.22) \begin{bmatrix} 1.5 & 0.37 \\ 0.37 & 1.5 \end{bmatrix}^{-1} (0.61, 0.22)' \\ &= 0.7485.\end{aligned}$$

Notice that $Z(\mathbf{s}_1)$ has a larger simple-kriging coefficient ($\ell_1^* = 0.395$) than $Z(\mathbf{s}_2)$ ($\ell_2^* = 0.049$) because $\mathbf{s}_1$ is closer to $\mathbf{s}_0$ than is $\mathbf{s}_2$. Notice also that $\sigma_{Y,sk}^2(\mathbf{s}_0)$ does not depend on the data $\mathbf{Z}$, and it is less than $\text{var}(Y(\mathbf{s}_0)) = 1$.

The Hierarchical Model (HM) and Simple Kriging

We now return to the HM formulation given at the beginning of Section 4.1. Recall that both the data model and the process model assumed Gaussian processes; but, in deriving the kriging equations (4.25) and (4.26), we avoided model assumptions and instead made assumptions about linearity of the predictor. We now provide the connection between the two approaches. The earlier data model is repeated here.

Data model: Conditional on σ_ϵ^2 , and for $i = 1, \dots, m$,

$$Z(\mathbf{s}_i) | Y(\mathbf{s}_i), \sigma_\epsilon^2 \sim \text{ind. Gau}(Y(\mathbf{s}_i), \sigma_\epsilon^2).$$

Recall that $\mathbf{Z} \equiv (Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_m))'$. Now the earlier process model is generalized slightly to allow for nonstationary covariances.

Process model: Conditional on $\boldsymbol{\beta}$ and $C_Y(\cdot, \cdot)$, $Y(\cdot)$ is a Gaussian process with the following properties:

$$E(Y(\cdot)) = \mathbf{x}(\cdot)' \boldsymbol{\beta}, \text{ and we write } \text{cov}(Y(\mathbf{u}), Y(\mathbf{v})) \equiv C_Y(\mathbf{u}, \mathbf{v}).$$

Because all joint and conditional distributions are Gaussian, the optimal spatial predictor $E(Y(\mathbf{s}_0) | \mathbf{Z}, \boldsymbol{\beta}, C_Y(\cdot, \cdot), \sigma_\epsilon^2)$ is (heterogeneously) linear in the data. Consequently, the Gaussian assumptions imply that the simple-kriging predictor (4.28) is exactly the optimal spatial predictor in the class of *all measurable functions* of $\mathbf{Z}$.

We now demonstrate this equivalence algebraically, because it is illustrative of what can be done with more complicated spatial HMs. Based on the data model and the process model given just above, we seek the posterior distribution of $Y(\mathbf{s}_0)$ given $\mathbf{Z}$. It is implicit that any unknown parameters are given too; recall we are assuming that $\boldsymbol{\beta}$, $C_Y(\cdot, \cdot)$, and σ_ϵ^2 are known, and hence for the algebra that follows we can drop them from the notation. Then,

$$Y(\mathbf{s}_0) | \mathbf{Z} \sim \text{Gau}(E(Y(\mathbf{s}_0) | \mathbf{Z}), \text{var}(Y(\mathbf{s}_0) | \mathbf{Z})), \quad (4.29)$$

where the conditional mean and conditional variance are given, for example, by Rencher (2002, p. 88):

$$\begin{aligned}E(Y(\mathbf{s}_0) | \mathbf{Z}) &= \mathbf{x}(\mathbf{s}_0)' \boldsymbol{\beta} + \mathbf{c}_Y(\mathbf{s}_0)' \mathbf{C}_Z^{-1} (\mathbf{Z} - \mathbf{X} \boldsymbol{\beta}) \\ \text{var}(Y(\mathbf{s}_0) | \mathbf{Z}) &= C_Y(\mathbf{s}_0, \mathbf{s}_0) - \mathbf{c}_Y(\mathbf{s}_0)' \mathbf{C}_Z^{-1} \mathbf{c}_Y(\mathbf{s}_0).\end{aligned} \quad (4.30)$$

It is illustrative in this rather simple HM to derive the formulas (4.29) and (4.30) from first principles. Associated with the data vector $\mathbf{Z}$ is the vector $\mathbf{Y}$ obtained from the hidden process:

$$\mathbf{Y} \equiv (Y(s_1), \dots, Y(s_m))'. \quad (4.31)$$

Then the data model implies that $\mathbf{Z}|\mathbf{Y} \sim \text{Gau}(\mathbf{Y}, \sigma_\varepsilon^2 \mathbf{I})$. From the process model given just above, we obtain

$$\begin{bmatrix} Y(s_0) \\ \mathbf{Y} \end{bmatrix} \sim \text{Gau} \left(\begin{bmatrix} \mathbf{x}(s_0) \\ \mathbf{X} \end{bmatrix} \boldsymbol{\beta}, \begin{bmatrix} C_Y(s_0, s_0) & \mathbf{c}_Y(s_0)' \\ \mathbf{c}_Y(s_0) & \boldsymbol{\Sigma}_Y \end{bmatrix} \right), \quad (4.32)$$

where $\boldsymbol{\Sigma}_Y \equiv \text{var}(\mathbf{Y})$. Standard results for the multivariate Gaussian distribution yield

$$\begin{aligned} Y(s_0)|\mathbf{Y} &\sim \text{Gau}(\mathbf{x}(s_0)' \boldsymbol{\beta} + \mathbf{c}_Y(s_0)' \boldsymbol{\Sigma}_Y^{-1} (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta}), C_Y(s_0, s_0) \\ &\quad - \mathbf{c}_Y(s_0)' \boldsymbol{\Sigma}_Y^{-1} \mathbf{c}_Y(s_0)), \end{aligned} \quad (4.33)$$

and $\mathbf{Y} \sim \text{Gau}(\mathbf{X} \boldsymbol{\beta}, \boldsymbol{\Sigma}_Y)$.

Our interest is in the posterior distribution, $[\mathbf{Y}|\mathbf{Z}]$. Bayes' Theorem gives

$$[\mathbf{Y}|\mathbf{Z}] \propto [\mathbf{Z}|\mathbf{Y}][\mathbf{Y}],$$

and hence from the HM, and after some algebra, we derive

$$\mathbf{Y}|\mathbf{Z} \sim \text{Gau}((\sigma_\varepsilon^{-2} \mathbf{I} + \boldsymbol{\Sigma}_Y^{-1})^{-1} (\sigma_\varepsilon^{-2} \mathbf{Z} + \boldsymbol{\Sigma}_Y^{-1} \mathbf{X} \boldsymbol{\beta}), (\sigma_\varepsilon^{-2} \mathbf{I} + \boldsymbol{\Sigma}_Y^{-1})^{-1}). \quad (4.34)$$

Using well known Sherman-Morrison-Woodbury matrix identities,

$$\begin{aligned} (\mathbf{A}' \mathbf{B}^{-1} \mathbf{A} + \mathbf{D}^{-1})^{-1} &= \mathbf{D} - \mathbf{D} \mathbf{A}' (\mathbf{A} \mathbf{D} \mathbf{A}' + \mathbf{B})^{-1} \mathbf{A} \mathbf{D} \\ (\mathbf{A}' \mathbf{B}^{-1} \mathbf{A} + \mathbf{D}^{-1})^{-1} \mathbf{A}' \mathbf{B} &= \mathbf{D} \mathbf{A}' (\mathbf{B} + \mathbf{A} \mathbf{D} \mathbf{A}')^{-1}, \end{aligned}$$

we can write

$$E(\mathbf{Y}|\mathbf{Z}) = \mathbf{X} \boldsymbol{\beta} + \mathbf{K}(\mathbf{Z} - \mathbf{X} \boldsymbol{\beta}) \quad (4.35)$$

$$\text{var}(\mathbf{Y}|\mathbf{Z}) = (\mathbf{I} - \mathbf{K}) \boldsymbol{\Sigma}_Y, \quad (4.36)$$

where $\mathbf{K}$ is the "gain,"

$$\mathbf{K} \equiv \boldsymbol{\Sigma}_Y (\sigma_\varepsilon^2 \mathbf{I} + \boldsymbol{\Sigma}_Y)^{-1}. \quad (4.37)$$

Now we return to prediction of $Y(s_0)$. In general, the posterior distribution of $Y(s_0)$ can be obtained from

$$[Y(s_0)|\mathbf{Z}] = \int [Y(s_0)|\mathbf{Y}][\mathbf{Y}|\mathbf{Z}] d\mathbf{Y}. \quad (4.38)$$

In the Gaussian case, the components of the integrand are given by (4.33) and (4.34). More directly, $[Y(s_0)|Z]$ is Gaussian, and hence one only has to derive the first two (posterior) moments to characterize the distribution. The posterior mean can be derived as follows:

$$\begin{aligned} E(Y(s_0)|Z) &= E\{E(Y(s_0)|Y, Z)|Z\} \\ &= E\{\mathbf{x}(s_0)' \boldsymbol{\beta} + \mathbf{c}_Y(s_0)' \boldsymbol{\Sigma}_Y^{-1} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) | Z\}, \end{aligned}$$

since $[Y(s_0)|Y, Z] = [Y(s_0)|Y]$. Hence, from (4.35) and (4.37), we obtain

$$\begin{aligned} E(Y(s_0)|Z) &= \mathbf{x}(s_0)' \boldsymbol{\beta} + \mathbf{c}_Y(s_0)' \boldsymbol{\Sigma}_Y^{-1} \mathbf{K}(\mathbf{Z} - \mathbf{X}\boldsymbol{\beta}) \\ &= \mathbf{x}(s_0)' \boldsymbol{\beta} + \mathbf{c}_Y(s_0)' (\sigma_\varepsilon^2 \mathbf{I} + \boldsymbol{\Sigma}_Y)^{-1} (\mathbf{Z} - \mathbf{X}\boldsymbol{\beta}) \\ &= Y^*(s_0), \end{aligned} \tag{4.39}$$

since $\mathbf{C}_Z = \boldsymbol{\Sigma}_Y + \sigma_\varepsilon^2 \mathbf{I}$.

From (4.33), (4.36), and (4.37), the posterior variance can be derived as follows:

$$\begin{aligned} \text{var}(Y(s_0)|Z) &= E\{\text{var}(Y(s_0)|Y)|Z\} + \text{var}\{E(Y(s_0)|Y)|Z\} \\ &= C_Y(s_0, s_0) - \mathbf{c}_Y(s_0)' \boldsymbol{\Sigma}_Y^{-1} \mathbf{c}_Y(s_0) + \text{var}\{\mathbf{c}_Y(s_0)' \boldsymbol{\Sigma}_Y^{-1} \mathbf{Y} | Z\} \\ &= C_Y(s_0, s_0) - \mathbf{c}_Y(s_0)' \boldsymbol{\Sigma}_Y^{-1} \mathbf{K} \mathbf{c}_Y(s_0) \\ &= C_Y(s_0, s_0) - \mathbf{c}_Y(s_0)' (\sigma_\varepsilon^2 \mathbf{I} + \boldsymbol{\Sigma}_Y)^{-1} \mathbf{c}_Y(s_0) \\ &= \sigma_{Y,sk}^2(s_0), \end{aligned} \tag{4.40}$$

again since $\mathbf{C}_Z = \boldsymbol{\Sigma}_Y + \sigma_\varepsilon^2 \mathbf{I}$. The result (4.40) should not be surprising because, assuming the HM given just above, we obtain

$$\begin{aligned} \sigma_{Y,sk}^2(s_0) &= E(Y(s_0) - Y^*(s_0))^2 \\ &= E\{(Y(s_0) - E(Y(s_0)|Z))^2 | Z\} \\ &= E\{\text{var}(Y(s_0)|Z)\} \\ &= \text{var}(Y(s_0)|Z); \end{aligned}$$

the last equality is a consequence of the Gaussian assumptions in the HM, which imply that the posterior variance does *not* depend on $\mathbf{Z}$.

In summary, the simple-kriging predictor given by (4.28), and the simple-kriging variance given by (4.26), can equally be derived from the HM given just above.

In the case where two (or more) values, $Y(s_{0,1})$ and $Y(s_{0,2})$, are to be predicted, the optimal predictors are still $Y^*(s_{0,1})$ and $Y^*(s_{0,2})$, respectively.

As well as calculating the mean squared prediction errors, $\sigma_{Y,sk}^2(s_{0,1})$ and $\sigma_{Y,sk}^2(s_{0,2})$, the mean cross-prediction errors can be calculated and shown to satisfy

$$E\{(Y(s_{0,1}) - Y^*(s_{0,1}))(Y(s_{0,2}) - Y^*(s_{0,2}))\} = E\{\text{cov}(Y(s_{0,1}), Y(s_{0,2})|\mathbf{Z})\}.$$

In Section 4.1.1, we gave a motivating example of simple kriging based on decadal temperature differences over the Americas. In fact, the original data (see Figure 4.6a) were produced from NCAR's Climate System Model, to which we deliberately added measurement error after subsampling. Figure 4.6b shows the noisy, corrupted data $\mathbf{Z}$ to which we applied simple kriging. Figures 4.6c and 4.6d show the simple kriging predictor $Y^*(\cdot)$ given by (4.28) and the simple kriging prediction standard error $\sigma_{Y,sk}(\cdot)$ obtained from (4.26), respectively.

Non-Gaussian Geostatistical Models

The HM given just above (4.29) has both a Gaussian data model and a Gaussian process model. As a consequence, the joint distribution of $\mathbf{Z}$ and $Y(\cdot)$ is Gaussian, as is the posterior distribution, $Y(\cdot)|\mathbf{Z}$ (assuming all parameters are known). We saw above that, in this case, to obtain the posterior distribution, we only need to calculate posterior means, variances, and covariances.

When the data $\mathbf{Z}$ are binary or are counts, a Gaussian data model is inappropriate. Diggle, Tawn, and Moyeed (1998) proposed that the *Gaussian* data model be replaced with an *exponential-family* data model. Let a random variable Z have density from the exponential family given by

$$\exp\{(z\eta - b(\eta))/a(\theta_1) + c(z, \theta_1)\}, \quad (4.41)$$

where, for the moment, θ_1 is considered to be known, and η is often referred to as the canonical parameter of the *exponential-family model* (EFM). For example, if $Z \sim \text{Gau}(\eta, \sigma_\epsilon^2)$, then $\theta_1 = \sigma_\epsilon^2$, $a(\theta_1) = \theta_1$, $b(\eta) = \eta^2/2$, and $c(z, \theta_1) = (-1/2)\{z^2/\theta_1 + \log(2\pi\theta_1)\}$; and if $Z \sim \text{Poi}(\exp(\eta))$, then there is no nuisance parameter θ_1 , $a = 1$, $b(\eta) = \exp(\eta)$, and $c(z) = -\log(z!)$. Other exponential-family models include the binomial distribution, the gamma distribution, and the inverse-Gaussian distribution (e.g., McCullagh and Nelder, 1989, p. 30).

Now define $\mu \equiv E(Z)$, which in general depends on η . Assuming the EFM, then it can be shown that (e.g., McCullagh and Nelder, 1989)

$$\mu(\eta) = b^{(1)}(\eta), \quad (4.42)$$

and $\text{var}(Z) = b^{(2)}(\eta)a(\theta_1)$, where the superscripts denote first and second derivatives with respect to η . The canonical link function, $g(\mu)$, has the property that

$$g(\mu(\eta)) = \eta. \quad (4.43)$$

For example, if $Z \sim \text{Gau}(\eta, \sigma_\varepsilon^2)$, then it can be verified that $g(\mu) = \mu$, the identity link; and if $Z \sim \text{Poi}(\exp(\eta))$, then $g(\mu) = \log \mu$, the log link. It is important to know this link because the *generalized linear model* defines the dependence of μ on covariates $\mathbf{x} \equiv (x_1, \dots, x_p)'$ through

$$\eta = g(\mu) = \mathbf{x}'\boldsymbol{\beta}, \quad (4.44)$$

where $\boldsymbol{\beta} \equiv (\beta_1, \dots, \beta_p)'$ are regression parameters.

We are now ready to define a non-Gaussian geostatistical model, using an HM.

Data model: Conditional on θ_1 , and for $i = 1, \dots, m$,

$$Z(\mathbf{s}_i) | Y(\mathbf{s}_i), \theta_1 \sim \text{ind. EFM},$$

with canonical parameter $Y(\mathbf{s}_i)$ and functions $a(\theta_1)$, $b(Y(\mathbf{s}_i))$, and $c(Z(\mathbf{s}_i), \theta_1)$ in (4.41).

Process model: Conditional on $\boldsymbol{\beta}$ and $C_Y(\cdot, \cdot)$, $Y(\cdot)$ is a Gaussian process with the following properties:

$$E(Y(\cdot)) = \mathbf{x}(\cdot)'\boldsymbol{\beta}, \text{ and we write } \text{cov}(Y(\mathbf{u}), Y(\mathbf{v})) \equiv C_Y(\mathbf{u}, \mathbf{v}).$$

A special case of the process model is where $Y(\cdot)$ has *no random variability*, and hence $Y(\mathbf{s}) = \mathbf{x}(\mathbf{s})'\boldsymbol{\beta}$; then the HM reduces to a model where the elements of $\mathbf{Z} \equiv (Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_m))'$ are independently sampled from a generalized linear model. With random spatial variability in $Y(\cdot)$, the HM allows the introduction of spatial dependence in a natural way through the canonical parameter of the EFM.

We have already seen that, in the special case of a Gaussian data model and Gaussian process model, the posterior distribution is Gaussian and the posterior means and variances can be calculated in closed form (assuming all parameters are known). We referred to these as the simple-kriging predictor and the simple-kriging (prediction) variance, respectively. For the more general HM with an EFM data model, the posterior distribution is not available in closed form, and a Monte Carlo algorithm is needed to obtain the posterior distribution. Cressie (2000) gives the MCMC algorithms for the HM given just above. Diggle, Tawn, and Moyeed (1998) add a parameter model to the HM and develop MCMC algorithms for the resulting BHM, and Wikle (2010a) discusses computational issues associated with non-Gaussian BHMs.

Another way to define *non-Gaussian* geostatistical models is to modify the *process model* in the HM. Suppose we write the covariance function C_Y as

$$C_Y(\mathbf{u}, \mathbf{v}) = \sigma(\mathbf{u})\sigma(\mathbf{v})\rho_Y(\mathbf{u}, \mathbf{v}), \quad \mathbf{u}, \mathbf{v} \in \mathbb{R}^d, \quad (4.45)$$

where $\rho_Y(\mathbf{u}, \mathbf{u}) \equiv 1$, and hence $C_Y(\mathbf{u}, \mathbf{u}) \equiv \sigma^2(\mathbf{u})$, for $\mathbf{u} \in \mathbb{R}^d$. Palacios and Steel (2006) suggest that $\sigma(\cdot)$ might be further modeled as a spatial process.

Using appropriate conditional distributions, the following HM results in non-Gaussian (but continuous) behavior for $Z(\cdot)$.

Data model: Conditional on σ_ε^2 , and for $i = 1, \dots, m$,

$$Z(\mathbf{s}_i) | Y(\mathbf{s}_i), \sigma_\varepsilon^2 \sim \text{ind. Gau}(Y(\mathbf{s}_i), \sigma_\varepsilon^2).$$

Process model 1: Conditional on β , $\sigma^2(\cdot)$, and $\rho_Y(\cdot, \cdot)$, $Y(\cdot)$ is a Gaussian process with the following properties:

$$E(Y(\cdot)) = \mathbf{x}(\cdot)' \beta, \quad \text{and we write} \quad \text{cov}(Y(\mathbf{u}), Y(\mathbf{v})) \equiv \sigma(\mathbf{u})\sigma(\mathbf{v})\rho_Y(\mathbf{u}, \mathbf{v}).$$

Process model 2: Conditional on $C_\omega(\cdot, \cdot)$, $\sigma(\cdot)$ is a log Gaussian process with the following properties:

$$E(\sigma(\mathbf{s})) \equiv 1, \quad \text{and we write} \quad \text{cov}(\sigma(\mathbf{u}), \sigma(\mathbf{v})) \equiv \exp(C_\omega(\mathbf{u}, \mathbf{v}) - 1), \quad \text{where } C_\omega(\cdot, \cdot) \text{ is the covariance function for } \omega(\cdot) \equiv \log \sigma(\cdot).$$

Notice that $\omega(\cdot) \equiv \log \sigma(\cdot)$ is a Gaussian process, and if we put $E(\omega(\mathbf{s})) = (-1/2)C_\omega(\mathbf{s}, \mathbf{s})$, $\mathbf{s} \in D$, then on the exponential scale the process $\sigma(\cdot) \equiv \exp(\omega(\cdot))$ does have mean 1, as specified in process model 2. Hence, if the Gaussian process $\omega(\cdot)$ is chosen to have constant variance σ_ω^2 , then $E(\omega(\mathbf{s})) \equiv (-1/2)\sigma_\omega^2$, a constant.

We note that some modelers may wish to consider the model on $\sigma(\cdot)$, given in process model 2, in a parameter level of the HM. Indeed this illustrates how one person's processes are another's parameters. At this point, the choice is somewhat semantic, but we wish to emphasize that $Y(\cdot)$ is conditioned on $\sigma(\cdot)$, and we are using the spatial variability of $\sigma(\cdot)$ simply to generate a non-Gaussian process. Hence, we leave it in the process model in this example.

Implementation of HMs with a geostatistical process model typically involves discretizing D_s onto a fine grid of n cells, resulting in a process that can be written as the n -dimensional vector $\mathbf{Y}$ (e.g., Figure 4.5a). Consequently, the data model may involve an incidence matrix $\mathbf{H}$ that assigns each $Z(\mathbf{s}_0)$ to one of the n cells. Then a statistical analysis often proceeds through an MCMC algorithm (e.g., Cressie, 2000; Wikle, 2010a).

The Kriging Predictor

We have already seen that nonparametric assumptions of the sort made in classical geostatistics produce a simple-kriging predictor and a kriging variance that can be interpreted as the posterior mean and posterior variance, respectively, in a Gaussian–Gaussian HM. We now take a small detour away from HMs to derive more general versions of kriging under different nonparametric assumptions. We shall then return to consider a BHM to show that these various kriging predictors can be obtained from the posterior distribution.

Observe two things from (4.25) and (4.27). First, the mean of the process $Y(\cdot)$ cannot be computed if the regression parameter vector β is unknown.

(We assume in the rest of this presentation of kriging that the covariance function $C_Y(\cdot, \cdot)$ is known.) Second, the simple kriging predictor $Y^*(s_0)$ satisfies $E(Y^*(s_0)) = E(Y(s_0)) = \mathbf{x}(s_0)' \boldsymbol{\beta}$, for all possible values of $\boldsymbol{\beta} \in \mathbb{R}^p$. We refer to this property as (*uniform*) *unbiasedness* of the predictor $Y^*(s_0)$. The two observations are related. Typically, $\boldsymbol{\beta}$ is unknown, making the simple kriging predictor, $Y^*(s_0) \equiv \boldsymbol{\ell}' \mathbf{Z} + k^*$, given by (4.28), unusable. But notice that $\boldsymbol{\beta}$ only appears in the optimal constant term, k^* . So, if we change from the class of heterogeneously linear predictors, $\{\boldsymbol{\ell}' \mathbf{Z} + k : \boldsymbol{\ell} \in \mathbb{R}^m, k \in \mathbb{R}\}$, to the class of homogeneously linear predictors, $\{\boldsymbol{\lambda}' \mathbf{Z} : \boldsymbol{\lambda} \in \mathbb{R}^m\}$, and *impose* unbiasedness, then there may be a computable optimal spatial predictor that is usable. In fact there is, and it is called the *universal kriging predictor*. [The word “universal” was used by Matheron and his school of geostatisticians to denote the generalization from assuming a known mean to assuming (4.27), where the regression parameter $\boldsymbol{\beta} \in \mathbb{R}^p$ is fixed but unknown.]

Ordinary Kriging

In what is to follow, we give the derivation of kriging assuming the special case (4.11), namely $E(Y(\mathbf{s})) = \mu$, for all $\mathbf{s} \in D_s$, which is a special case of (4.27) with $p = 1$, $\mathbf{x}(\cdot) = 1$, and $\boldsymbol{\beta} = \mu$. This is known as *ordinary kriging* in the geostatistics literature. (Universal kriging is also discussed below, but without derivation.) Here, we derive ordinary kriging of $Y(s_0)$ in terms of covariance functions, and we just state the formulas in terms of semivariograms. The ordinary-kriging predictor is the homogeneously linear combination of the data $\mathbf{Z}$ that minimizes the mean squared prediction error,

$$\text{MSPE}(\boldsymbol{\lambda}) \equiv E(Y(s_0) - \boldsymbol{\lambda}' \mathbf{Z})^2, \quad (4.46)$$

with respect to coefficients $\boldsymbol{\lambda}$, *subject to* the unbiasedness condition, $E(\boldsymbol{\lambda}' \mathbf{Z}) = E(Y(s_0)) = \mu$, for all $\mu \in \mathbb{R}$.

Since $E(Z(\mathbf{s})) = E(Y(\mathbf{s})) = \mu$, then $E(\mathbf{Z}) = \mu \mathbf{1}$, where $\mathbf{1} \equiv (1, \dots, 1)'$ is the m -dimensional vector of 1s. Hence, the unbiasedness condition is equivalent to

$$\boldsymbol{\lambda}' \mathbf{1} = 1; \quad (4.47)$$

that is, the sum of the ordinary-kriging coefficients is 1. Sometimes $\lambda_1, \dots, \lambda_m$ are referred to as kriging weights, but we find this misleading since they can be positive or negative; we refer to them as *kriging coefficients*.

Using the method of Lagrange multipliers to incorporate the constraint (4.47), we minimize the objective function,

$$\text{MSPE}(\boldsymbol{\lambda}) - 2\kappa(\boldsymbol{\lambda}' \mathbf{1} - 1), \quad (4.48)$$

with respect to $\boldsymbol{\lambda}$ and the Lagrange multiplier κ . Evaluating (4.48), we obtain

$$C_Y(s_0, s_0) - 2\boldsymbol{\lambda}' \mathbf{c}_Y(s_0) + \boldsymbol{\lambda}' \mathbf{C}_Z \boldsymbol{\lambda} - 2\kappa(\boldsymbol{\lambda}' \mathbf{1} - 1), \quad (4.49)$$

whose partial derivative with respect to λ and κ , set equal to zero, results in

$$\begin{aligned} -\mathbf{c}_Y(\mathbf{s}_0) + \mathbf{C}_Z \lambda &= \kappa \mathbf{1} \\ \lambda' \mathbf{1} &= 1. \end{aligned} \quad (4.50)$$

Recall that $\mathbf{C}_Z \equiv \text{var}(\mathbf{Z})$ and $\mathbf{c}_Y(\mathbf{s}_0)$ is given by (4.23).

Let the solution to (4.50) be λ^* and κ^* . Then, from the first equation, we obtain

$$\lambda^* = \mathbf{C}_Z^{-1}(\mathbf{c}_Y(\mathbf{s}_0) + \kappa^* \mathbf{1}),$$

and, from the second equation, $\kappa^* = (1 - \mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{c}_Y(\mathbf{s}_0)) / (\mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{1})$. Finally, the *ordinary-kriging predictor* is

$$\hat{Y}(\mathbf{s}_0) \equiv \{\mathbf{c}_Y(\mathbf{s}_0) + \mathbf{1}(1 - \mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{c}_Y(\mathbf{s}_0)) / (\mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{1})\}' \mathbf{C}_Z^{-1} \mathbf{Z} \equiv (\lambda^*)' \mathbf{Z}. \quad (4.51)$$

The minimized mean squared prediction error, often called the (*ordinary*) *kriging variance*, is obtained by substituting λ^* into (4.46):

$$\begin{aligned} \sigma_{Y,ok}^2(\mathbf{s}_0) &\equiv \text{MSPE}(\lambda^*) \\ &= C_Y(\mathbf{s}_0, \mathbf{s}_0) - \mathbf{c}_Y(\mathbf{s}_0)' \mathbf{C}_Z^{-1} \mathbf{c}_Y(\mathbf{s}_0) \\ &\quad + (1 - \mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{c}_Y(\mathbf{s}_0))^2 / (\mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{1}), \end{aligned} \quad (4.52)$$

where the subscript “*ok*” on the left-hand side of (4.52) denotes ordinary kriging. Comparing (4.52) to (4.26), we see that the unbiasedness restriction in ordinary kriging results in $\sigma_{Y,ok}^2(\mathbf{s}_0) \geq \sigma_{Y,sk}^2(\mathbf{s}_0)$. This is intuitively reasonable, since for ordinary kriging μ is unknown, which leads to greater prediction uncertainty.

Ordinary-kriging prediction intervals can also be computed from (4.51) and (4.52), although their accuracy is in doubt when the data are non-Gaussian. Schelin and Sjöstedt-de Luna (2010) give kriging prediction intervals based on a semiparametric bootstrap.

While it is not obvious, it is seen below that (4.51) can be written equivalently as

$$\hat{Y}(\mathbf{s}_0) = \hat{\mu}_{gls} + \mathbf{c}_Y(\mathbf{s}_0)' \mathbf{C}_Z^{-1} (\mathbf{Z} - \hat{\mu}_{gls} \mathbf{1}), \quad (4.53)$$

where $\hat{\mu}_{gls}$ is the generalized-least-squares estimator of μ , namely

$$\hat{\mu}_{gls} \equiv (\mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{Z}) / (\mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{1}). \quad (4.54)$$

By comparing (4.53) with the simple kriging predictor (4.25), where $\mu_Y(\cdot) \equiv \mu$, it is clear that ordinary kriging is just simple kriging with the optimal (generalized-least-squares) estimator $\hat{\mu}_{gls}$ substituted for μ .

The expression (4.53) is no accident; the result is due to Goldberger (1962) and (in the spatial context) Gotway and Cressie (1993), who show that for any linear estimator, $\bar{\mu} \equiv \mathbf{a}'\mathbf{Z}$, the linear predictor, $\bar{Y}(\mathbf{s}_0) \equiv \bar{\mu} + \mathbf{c}_Y(\mathbf{s}_0)'\mathbf{C}_Z^{-1}(\mathbf{Z} - \bar{\mu}\mathbf{1})$, satisfies

$$E(Y(\mathbf{s}_0) - \bar{Y}(\mathbf{s}_0))^2 = \text{MSPE}(\ell^*, k^*) + (1 - \mathbf{1}'\mathbf{C}_Z^{-1}\mathbf{c}_Y(\mathbf{s}_0))^2 E(\mu - \bar{\mu})^2.$$

Gotway and Cressie also show that there always exists a $\bar{\mu}$ ($= \mathbf{a}'\mathbf{Z}$, for some $\mathbf{a}$) such that *any* homogeneously linear predictor can be written as $\bar{Y}(\mathbf{s}_0)$. Hence, $\bar{Y}(\mathbf{s}_0)$ is an unbiased predictor of $Y(\mathbf{s}_0)$, if and only if $\bar{\mu}$ is an unbiased estimator of μ . Furthermore, the optimal homogeneously linear (ordinary-kriging) predictor results from choosing the $\bar{\mu}$ that minimizes $E(\mu - \bar{\mu})^2$. Provided $\bar{\mu}$ is unbiased, $E(\mu - \bar{\mu})^2 = \text{var}(\bar{\mu}) = \mathbf{a}'\mathbf{C}_Z\mathbf{a}$. It is a well known result from estimation theory that this *best* linear unbiased estimator (BLUE) of μ is the generalized-least-squares estimator, $\hat{\mu}_{gls}$, given by (4.54). That is, once the simple-kriging coefficients $\mathbf{c}_Y(\mathbf{s}_0)'\mathbf{C}_Z^{-1}$ are chosen, the *optimality* of the (homogeneously) linear spatial *predictor* is in one-to-one correspondence with the *optimality* of the linear *estimator* of the mean. By giving up both linearity and unbiasedness of the predictor, it is possible to find a predictor whose mean squared prediction error is *smaller* than the ordinary-kriging variance (Gotway and Cressie, 1993).

So far, we have presented simple kriging and ordinary kriging in terms of the covariance function $C_Y(\cdot, \cdot)$. There is an equivalent formulation of ordinary kriging (but not of simple kriging) in terms of the semivariogram,

$$\gamma_Y(\mathbf{u}, \mathbf{v}) = (1/2)\{C_Y(\mathbf{u}, \mathbf{u}) + C_Y(\mathbf{v}, \mathbf{v}) - 2C_Y(\mathbf{u}, \mathbf{v})\}. \quad (4.55)$$

In terms of semivariograms, we have

$$\text{MSPE}(\boldsymbol{\lambda}) = 2\boldsymbol{\lambda}'\tilde{\gamma}_Y(\mathbf{s}_0) - \boldsymbol{\lambda}'\boldsymbol{\Gamma}_Z\boldsymbol{\lambda}, \quad (4.56)$$

where the $m \times m$ matrix $\boldsymbol{\Gamma}_Z \equiv (\gamma_Z(\mathbf{s}_i, \mathbf{s}_j))$;

$$\gamma_Z(\mathbf{u}, \mathbf{v}) = \gamma_Y(\mathbf{u}, \mathbf{v}) + \sigma_\epsilon^2 I(\mathbf{u} \neq \mathbf{v});$$

and $\tilde{\gamma}_Y(\mathbf{s}_0) \equiv (\gamma_Y(\mathbf{s}_0, \mathbf{s}_1) + \sigma_\epsilon^2/2, \dots, \gamma_Y(\mathbf{s}_0, \mathbf{s}_m) + \sigma_\epsilon^2/2)'$; see Cressie (1993, Section 3.2.1).

Again, using the method of Lagrange multipliers to impose the restriction $\boldsymbol{\lambda}'\mathbf{1} = 1$, we wish to minimize (4.48) with respect to $\boldsymbol{\lambda}$ and κ . Using (4.56) in (4.48) and setting the partial derivatives with respect to $\boldsymbol{\lambda}$ and κ equal to zero, we obtain equivalent ordinary-kriging equations in terms of the semivariogram:

$$\hat{Y}(\mathbf{s}_0) = \{\tilde{\gamma}_Y(\mathbf{s}_0) + \mathbf{1}(1 - \mathbf{1}'\boldsymbol{\Gamma}_Z^{-1}\tilde{\gamma}_Y(\mathbf{s}_0))/(\mathbf{1}'\boldsymbol{\Gamma}_Z^{-1}\mathbf{1})\}'\boldsymbol{\Gamma}_Z^{-1}\mathbf{Z}, \quad (4.57)$$

$$\sigma_{Y,ok}^2(\mathbf{s}_0) = \tilde{\gamma}_Y(\mathbf{s}_0)'\boldsymbol{\Gamma}_Z^{-1}\tilde{\gamma}_Y(\mathbf{s}_0) - (1 - \mathbf{1}'\boldsymbol{\Gamma}_Z^{-1}\tilde{\gamma}_Y(\mathbf{s}_0))^2/(\mathbf{1}'\boldsymbol{\Gamma}_Z^{-1}\mathbf{1}). \quad (4.58)$$

Notice that simple kriging and ordinary kriging require either the covariance function or the semivariogram to be known. In practice, they have to be estimated, which means that, strictly speaking, they no longer yield optimal predictors.

Universal Kriging

When the constant-mean assumption (4.11) is generalized to the linear-model assumption (4.27), namely $E(Y(\cdot)) = \mathbf{x}(\cdot)' \boldsymbol{\beta}$, we have already noted that optimal (homogeneously) linear spatial prediction is known as *universal kriging* in the geostatistics literature. Recall from (4.28) the definitions of $\mathbf{X}$, $\mathbf{C}_Z$, and $\mathbf{c}_Y(\mathbf{s}_0)$. Then, it is straightforward to show that the universal-kriging predictor of $Y(\mathbf{s}_0)$ is

$$\hat{Y}(\mathbf{s}_0) = \mathbf{x}(\mathbf{s}_0)' \hat{\boldsymbol{\beta}}_{gls} + \mathbf{c}_Y(\mathbf{s}_0)' \mathbf{C}_Z^{-1} (\mathbf{Z} - \mathbf{X} \hat{\boldsymbol{\beta}}_{gls}), \quad (4.59)$$

where $\hat{\boldsymbol{\beta}}_{gls} \equiv (\mathbf{X}' \mathbf{C}_Z^{-1} \mathbf{X})^{-1} \mathbf{X}' \mathbf{C}_Z^{-1} \mathbf{Z}$, is the generalized-least-squares estimator of $\boldsymbol{\beta}$. The result is analogous to (4.53) and a direct consequence of a decomposition of the mean squared prediction error for the linear-model case (see, e.g., Cressie, 1993, p. 174). The decomposition also implies that the (universal) kriging variance is

$$\begin{aligned} \sigma_{Y,uk}^2(\mathbf{s}_0) &= \mathbf{C}_Y(\mathbf{s}_0, \mathbf{s}_0) - \mathbf{c}_Y(\mathbf{s}_0)' \mathbf{C}_Z^{-1} \mathbf{c}_Y(\mathbf{s}_0) + (\mathbf{x}(\mathbf{s}_0) \\ &\quad - \mathbf{X}' \mathbf{C}_Z^{-1} \mathbf{c}_Y(\mathbf{s}_0))' (\mathbf{X}' \mathbf{C}_Z^{-1} \mathbf{X})^{-1} (\mathbf{x}(\mathbf{s}_0) - \mathbf{X}' \mathbf{C}_Z^{-1} \mathbf{c}_Y(\mathbf{s}_0)). \end{aligned} \quad (4.60)$$

The equivalent formulation of universal kriging in terms of the semivariogram only holds if $\mathbf{x}(\cdot)' \boldsymbol{\beta}$ has an intercept term (i.e., $x_1(\mathbf{s}) \equiv 1$). In that case, simply replace $\mathbf{c}_Y(\mathbf{s}_0)$ with $-\tilde{\gamma}_Y(\mathbf{s}_0)$ and $\mathbf{C}_Z$ with $-\Gamma_Z$ in (4.59) and (4.60), to obtain formulas for $\hat{Y}(\mathbf{s}_0)$ and $\sigma_{Y,uk}^2(\mathbf{s}_0)$, respectively.

BHM and Kriging

There is a BHM that yields the universal kriging predictor, $\hat{Y}(\mathbf{s}_0)$. Let a BHM be defined by the Gaussian data model and the Gaussian process model given at the beginning of this section (Section 4.1). The HM becomes Bayesian by specifying a parameter model. Here we specify an improper distribution for $\boldsymbol{\beta}$; we continue to assume that variances and covariances are known.

Parameter model: $\boldsymbol{\beta} \sim U((-\infty, \infty)^p)$, which is an improper uniform distribution on $\mathbb{R}^p$. (Parameters $\mathbf{C}_Y$ and σ_ϵ^2 are assumed known.)

Then, $E(Y(\mathbf{s}_0)|\mathbf{Z}) = \hat{Y}(\mathbf{s}_0)$, and $\text{var}(Y(\mathbf{s}_0)|\mathbf{Z}) = \sigma_{Y,uk}^2(\mathbf{s}_0)$, are the relevant universal-kriging quantities (Omre and Halvorsen, 1989).

As a special case, ordinary kriging is obtained from the following parameter model.

Parameter model: $\mu \sim U(-\infty, \infty)$, which is an improper uniform distribution on $\mathbb{R}$. (Parameters C_Y and σ_ε^2 are assumed known.)

Other, more general parameter models that put (prior) distributions on β and on covariance parameters θ can be specified. Early examples can be found in Kitanidis (1986) and in Diggle, Tawn, and Moyeed (1998); Banerjee, Carlin, and Gelfand (2004) and Wikle (2010a) can be consulted for more recent developments.

The simple-, ordinary-, and universal-kriging formulas are all given above in terms of nonstationary covariance functions (and semivariograms); hence they are appropriate for a stationary covariance function, $C_Y(\mathbf{h})$, or a stationary semivariogram, $\gamma_Y(\mathbf{h})$. Stationarity is not a requirement for a BHM, but it can be important for an EHM, which is the approach typically used in geostatistical analyses. Stationarity provides a type of pseudo-replication of dependencies at given spatial lags $\mathbf{h}$, from which C_Y (and γ_Y) can be estimated (e.g., Cressie, 1993, Chapter 2).

Splines and Kriging

There is a close relationship between the universal-kriging predictor and thin-plate smoothing splines. This is made most transparent through the dual kriging equations; see the discussion given by Cressie (1993, pp. 180–183) and Kent and Mardia (1994). Both presentations adopt an EHM approach to inference. From the point of view of kriging, thin-plate smoothing splines *pre-select* a particular semivariogram (or more generally a generalized covariance function) without letting the data decide which semivariogram gives the best fit (in contrast to kriging; e.g., see Figure 4.5b). When using an approach based on thin-plate smoothing splines, the data are used to determine the size of a measurement-error variance, which is classically assumed to be the only source of randomness. The use of splines as smoothers, rather than as predictors, means that there is no analogous notion to the kriging variance, although Wahba (1983) did give a Bayesian interpretation of smoothing splines (from which posterior variances could be calculated). Of course, different spline basis functions could be used as spatial basis functions in the Spatial Random Effects model (4.79) below.

4.1.3 Change-of-Support

The support of

$$Y(B) \equiv (1/|B|) \int_B Y(\mathbf{u}) \, d\mathbf{u},$$

is simply B , the region over which $Y(\cdot)$ is averaged, where $|B| > 0$ is the d -dimensional volume of B . The quantity on the right-hand side is a stochastic integral, and it may be interpreted in terms of an integral of each realization coming from the probability space that defines the stochastic process $Y(\cdot)$. Alternatively, a finite collection $\{Y(\mathbf{s}_{0,j})\}$ defined on $\{\mathbf{s}_{0,j}\}$ has a multivariate distribution; an appropriately weighted sum, converging in mean square to a limiting random variable, defines the right-hand side (Yaglom, 1962, p. 23). The notion of *support* is an extremely important one in modeling a physical process; one can think of $|B|$ as the spatial resolution at which the phenomenon is being studied, although data often come as $\mathbf{Z} = (Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_m))'$, at point support (or more generally at *different* supports). In Chapter 1, we noted that change-of-support is analogous to Simpson's Paradox, and sometimes it helps to think of it this way (e.g., Ma, 2009).

In this section, as previously, we give some classical geostatistical approaches to this problem, followed by a formulation in terms of a BHM. Kriging adapts very easily to accommodate the change from predicting $Y(\mathbf{s}_0)$ with point support $\mathbf{s}_0$, to predicting $Y(B)$ with block support B . For example, in the formulas (4.28) and (4.51) for kriging predictors of $Y(\mathbf{s}_0)$, $\mathbf{x}(\mathbf{s}_0)$, and $\mathbf{c}_Y(\mathbf{s}_0)$ are modified very simply to

$$\mathbf{x}(B) \equiv (1/|B|) \left(\int_B x_1(\mathbf{u}) d\mathbf{u}, \dots, \int_B x_p(\mathbf{u}) d\mathbf{u} \right)'$$

and

$$\mathbf{c}_Y(B) \equiv (1/|B|) \left(\int_B C_Y(\mathbf{u}, \mathbf{s}_1) d\mathbf{u}, \dots, \int_B C_Y(\mathbf{u}, \mathbf{s}_m) d\mathbf{u} \right)',$$

respectively, resulting in the simple-kriging predictor $Y^*(B)$ and the ordinary-kriging predictor $\hat{Y}(B)$. The same modification is made in (4.26) and (4.52), and $C_Y(\mathbf{s}_0, \mathbf{s}_0)$ is replaced with $C(B, B) \equiv (1/|B|)^2 \int_B \int_B C_Y(\mathbf{u}, \mathbf{v}) d\mathbf{u} d\mathbf{v}$, to yield $\sigma_{Y,sk}^2(B)$ and $\sigma_{Y,ok}^2(B)$, respectively. For more details, see Cressie (1993, pp. 172–173).

Now consider the prediction of $g(Y(B))$, where g is a nonlinear function. Then the essential quantity we need is the conditional distribution, $\Pr(Y(B) \leq y | \mathbf{Z})$, which is the posterior distribution of $Y(B)$; any optimal predictor of $g(Y(B))$ is obtained from it. Disjunctive kriging and indicator kriging attempt to estimate a similar quantity, $\Pr(Z(B) \leq z | \mathbf{Z})$, based on bivariate distributional properties of the (possibly transformed) process; see Cressie (1993, Section 5.2). However, these methods have serious drawbacks, associated with failing to distinguish between $Z(B)$ and $Y(B)$, and with difficulty in estimating block-to-block and point-to-block spatial dependencies of indicator variables from the point-to-point spatial dependencies indicated by the data.

Aldworth and Cressie (2003) present a methodology they call *covariance-matching constrained kriging* (CMCK). Write the predictand as $\mathbf{Y}(\mathbf{B}) \equiv (Y(B_1), \dots, Y(B_m))'$, where $\mathbf{B} \equiv \{B_1, \dots, B_m\}$. Then CMCK looks for an

optimal homogeneously linear predictor, $\mathbf{Y}^@(\mathbf{B}) \equiv (Y^@(B_1), \dots, Y^@(B_m))' \equiv \mathbf{AZ}$, of $\mathbf{Y}(\mathbf{B})$ with constraints:

$$E(\mathbf{Y}^@(\mathbf{B})) = E(\mathbf{Y}(\mathbf{B})) \quad (4.61)$$

$$\text{var}(\mathbf{Y}^@(\mathbf{B})) = \text{var}(\mathbf{Y}(\mathbf{B})); \quad (4.62)$$

there are conditions to check that guarantee CMCK is a solution to the constrained optimization problem. This has strong similarities to the constrained empirical Bayes estimators introduced by Louis (1984) and extended by Ghosh (1992). The CMCK predictor of $\mathbf{g}(\mathbf{Y}(\mathbf{B}))$ is defined to be $\mathbf{g}(\mathbf{Y}^@(\mathbf{B}))$, for any vector function $\mathbf{g}(\cdot)$, and it is *approximately* unbiased.

The traditional change-of-support problem that has been formulated in the geostatistics literature (e.g., Journel and Huijbregts, 1978, Chapter VI) is to make inference on $\mathbf{Z}(\mathbf{B})$ from observations $\mathbf{Z}$, which, from our discussion in Section 4.1.2, does not recognize the general goal of filtering out measurement error. Generally, inference should be on $\mathbf{Y}(\mathbf{B})$ rather than on $\mathbf{Z}(\mathbf{B})$. Traditional approaches also assume that trend parameters and covariance parameters are fixed but unknown, and they effectively adopt an EHM. Gelfand, Zhu, and Carlin (2001) give one way to modify the traditional EHM approach by showing how a BHM can be used to make inference on $\mathbf{Z}(\mathbf{B})$ from observations $\mathbf{Z}$; they do *not* discuss inference on $\mathbf{Y}(\mathbf{B})$.

Wikle and Berliner (2005) have formulated the change-of-support problem using a BHM that includes measurement error. Inference is on $\mathbf{Y}(\mathbf{B})$, based on data at levels of aggregation different from $\mathbf{B} = \{B_j\}$ (finer or coarser), say $\mathbf{Z}(\mathbf{A})$ and $\mathbf{Z}(\mathbf{C})$. They assume that for any $s \in D_s$, there is some $B_j \subset D_s$ such that $Y(s) = Y(B_j) + \zeta(s)$, where $\zeta(\cdot)$ is a spatial process with covariance function, $C_\zeta(\mathbf{u}, \mathbf{v})$, and the distribution of $\zeta(s)$ does not depend on $\{Y(B_j)\}$. For the finer-scale process, they derive

$$Y(A_i) = \mathbf{G}(A_i)' \mathbf{Y}(\mathbf{B}) + (1/|A_i|) \int_{A_i} \zeta(\mathbf{u}) d\mathbf{u}, \quad (4.63)$$

where the j th element of $\mathbf{G}(A_i)$ is $|B_j \cap A_i|/|A_i|$; $j = 1, \dots, n$. Notice that

$$\text{cov}(Y(A_i), Y(A_{i'}) | \mathbf{Y}(\mathbf{B})) = (|A_i| |A_{i'}|)^{-1} \int_{A_i} \int_{A_{i'}} C_\zeta(\mathbf{u}, \mathbf{v}) d\mathbf{u} d\mathbf{v}, \quad (4.64)$$

which results in covariance matrix $\mathbf{C}_\zeta^A$. Observations $\mathbf{Z}(\mathbf{A})$ at the finer scale are made with measurement error; Wikle and Berliner assume that, conditional on $\mathbf{Y}(\mathbf{A})$, $\mathbf{Z}(\mathbf{A})$ has mean $\mathbf{Y}(\mathbf{A})$ and measurement-error covariance matrix $\mathbf{C}_\varepsilon^A$. This, along with (4.63) and (4.64), leads to the following BHM based on data $\mathbf{Z}(\mathbf{A})$:

Data model: Conditional on $\mathbf{Y}(\mathbf{B})$ and variance–covariance parameters the data vector $\mathbf{Z}(\mathbf{A})$ has mean

$$\mathbf{G}(\mathbf{A})'\mathbf{Y}(\mathbf{B}),$$

where the i th row of $\mathbf{G}(\mathbf{A})$ is $\mathbf{G}(\mathbf{A}_i)$; and $\mathbf{Z}(\mathbf{A})$ has covariance matrix,

$$\mathbf{C}_{\mathbf{Z}}^{\mathbf{A}} = \mathbf{C}_{\zeta}^{\mathbf{A}} + \mathbf{C}_{\mathbf{F}}^{\mathbf{A}}.$$

The conditional distribution is specified (e.g., a joint Gaussian distribution).

Process model: Conditional on process-model parameters, the joint distribution of $\mathbf{Y}(\mathbf{B})$ is specified (e.g., a joint Gaussian distribution).

Parameter model: The joint distribution of all unknown parameters is specified.

The distributions in the data model and the process model are often assumed to be Gaussian, but they do not have to be; this hierarchical formulation of change-of-support allows more flexibility in modeling, because (fully specified) non-Gaussian distributions are just as welcome, provided they adapt well to obtaining the posterior distribution (e.g., through simulating full conditional distributions in an MCMC algorithm). Once the conditional distributions in the BHM above are specified, an MCMC algorithm can be derived and implemented. After “burn in,” the MCMC yields realizations $\{\mathbf{Y}(\mathbf{B})^{(\ell)} : \ell = 1, \dots, L\}$ from the posterior distribution,

$$[\mathbf{Y}(\mathbf{B})|\mathbf{Z}(\mathbf{A})],$$

and inference on nonlinear $\mathbf{g}(\mathbf{Y}(\mathbf{B}))$ is straightforward. For example, the conditional distribution,

$$[\exp\{Y(\mathbf{B}_1)\}|\mathbf{Z}(\mathbf{A})],$$

is obtained simply by computing histograms from $\{\exp\{Y(\mathbf{B}_1)^{(\ell)}\} : \ell = 1, \dots, L\}$. Wikle and Berliner (2005) give the BHM for the general case where both finer-scale and coarser-scale data are available, and they use MCMC to simulate from the posterior distribution, $[\mathbf{Y}(\mathbf{B})|\mathbf{Z}(\mathbf{A}), \mathbf{Z}(\mathbf{C})]$.

4.1.4 Spatial Moving Average (SMA) Models

This section on geostatistical processes centers around modeling the covariance function or the variogram of a spatial process. Covariance functions sometimes

come from physical processes, such as the exponential covariance function that is based on a proportional growth process, or the spherical covariance function that is based on the volume of intersecting spheres of equal radius, or the linear variogram that is based on Brownian motion (of gas molecules). When building HMs, it is very useful to have flexible classes of models at hand that exhibit spatial dependence.

In either a classical geostatistical setting or an HM setting, spatial statistical dependence is ubiquitously captured by the covariance function. Let $\{Y(\mathbf{s}) : \mathbf{s} \in D_s \subset \mathbb{R}^d\}$ be a spatial process, where $\text{var}(Y(\mathbf{s})) < \infty; \mathbf{s} \in D_s$. Recall that $C_Y(\mathbf{s}, \mathbf{x}) = \text{cov}(Y(\mathbf{s}), Y(\mathbf{x}))$ or, in the second-order-stationary case, $C_Y(\mathbf{h}) = \text{cov}(Y(\mathbf{s}), Y(\mathbf{s} + \mathbf{h}))$. The semivariogram is $\gamma_Y(\mathbf{s}, \mathbf{x}) = (1/2)\text{var}(Y(\mathbf{s}) - Y(\mathbf{x}))$ or, in the intrinsically stationary case, $\gamma_Y(\mathbf{h}) = (1/2)\text{var}(Y(\mathbf{s} + \mathbf{h}) - Y(\mathbf{s}))$.

Recall that both $C_Y(\cdot)$ and $\gamma_Y(\cdot)$ have a spectral representation. Thus, by generating classes of models for the spectral distribution, we can generate classes of models for $C_Y(\cdot)$ or $\gamma_Y(\cdot)$ by inverting a Fourier transform. For example, the Matérn covariance function given by (4.6) with $\sigma_0^2 = 0$ and $\sigma_1^2 = 1$, has spectral density given by

$$f_Y(\boldsymbol{\omega}) = (\theta_1/\pi^{1/2})^d (\Gamma(\theta_2 + d/2)/\Gamma(\theta_2))(1 + \theta_1^2 \|\boldsymbol{\omega}\|^2)^{-(\theta_2 + d/2)}, \quad (4.65)$$

where $\int f_Y(\boldsymbol{\omega}) d\boldsymbol{\omega} = 1$. Now the spectral representation of a covariance function is a decomposition of the covariation into an integral of the spectral density f_Y , weighted by cosines at various frequencies; see (4.15). Furthermore, the infinitesimal quantity $f_Y(\boldsymbol{\omega}) d\boldsymbol{\omega}$ can be attributed to an infinitesimal orthogonal random component. That is, in two dimensions, consider the spatial process defined by

$$Y(\mathbf{s}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{i\boldsymbol{\omega}'\mathbf{s}} V_Y(d\boldsymbol{\omega}), \quad \mathbf{s} \in \mathbb{R}^2, \quad (4.66)$$

where $V_Y(\cdot)$ is a complex-valued random process with independent increments, $E(|V_Y(d\boldsymbol{\omega})|^2) = F_Y(d\boldsymbol{\omega})$, and $F_Y(d\boldsymbol{\omega}) = f_Y(\boldsymbol{\omega}) d\boldsymbol{\omega}$. Then $Y(\cdot)$ has stationary covariance function $C_Y(\cdot)$ given by (4.15).

The Spatial Moving Average (SMA) models use (4.66) to *motivate* the following construction of spatial dependence in a spatial process in $\mathbb{R}^d$. Consider starting directly with standard zero-mean, d -dimensional Brownian motion, $W(\cdot)$, and define the process,

$$Y(\mathbf{s}) \equiv \int k(\mathbf{s}, \mathbf{u}) W(d\mathbf{u}), \quad \mathbf{s} \in D_s, \quad (4.67)$$

where the kernel $k(\cdot, \cdot)$ satisfies $\int k(\mathbf{s}, \mathbf{u})^2 d\mathbf{u} < \infty$ for all $\mathbf{s} \in D_s$. We say that $Y(\cdot)$ defined by (4.67) is a *spatial moving average* with kernel $k(\cdot, \cdot)$.

Notice that $W(\cdot)$ is a (Gaussian) process with independent increments and $E(\{W(d\mathbf{u})\}^2) = d\mathbf{u}$, which, as is seen below, makes computation of the covariance function $C_Y(\cdot, \cdot)$ straightforward.

To explain “SMA,” recall from Section 3.4.4 the Moving Average (MA) time-series model,

$$Y_t = \sum_{i=1}^{\infty} a_i \varepsilon_{t-i+1},$$

which is a linear combination of *iid* (Gaussian) random variables $\{\varepsilon_j\}$. Now a discretized version of (4.67) is

$$Y(\mathbf{s}) = \sum_{i=1}^{\infty} k(\mathbf{s}, \mathbf{u}_i) \varepsilon(\mathbf{u}_i),$$

where $\{\mathbf{u}_i\}$ is a dense grid of spatial locations on D_s and $\varepsilon(\cdot)$ is a (Gaussian) white-noise process. Both the MA time series and the SMA spatial process are linear combinations of *iid* random variables. Notice that, in general, they *cannot* be written in terms of a Markov random field with sparse neighborhood structure (Section 4.2.2). In Section 4.2.8, we make the comment that SMA models align more naturally with geostatistical models than CAR or SAR models.

The representation (4.67) has also been called a *process convolution (PC)*; see Higdon (1998). Recall that the convolution $a \circ b$ of two functions a and b is defined as

$$(a \circ b)(\mathbf{s}) \equiv \int a(\mathbf{u})b(\mathbf{s} - \mathbf{u}) d\mathbf{u}, \quad \mathbf{s} \in \mathbb{R}^d.$$

We see that (4.67) is a slightly more general *random* version of this. The advantage of having $k(\mathbf{s}, \mathbf{u})$ not be a function of $(\mathbf{s} - \mathbf{u})$ is obvious, since (4.67) can generate larger, nonstationary classes of (Gaussian) processes.

In this section, we have paid particular attention to covariance properties of various $Y(\cdot)$. From (4.67), spatial covariances for SMA models can be easily calculated:

$$\begin{aligned} C_Y(\mathbf{s}, \mathbf{x}) &\equiv \text{cov}(Y(\mathbf{s}), Y(\mathbf{x})) = \int \int k(\mathbf{s}, \mathbf{u})k(\mathbf{x}, \mathbf{v})E(W(d\mathbf{u})W(d\mathbf{v})) \\ &= \int k(\mathbf{s}, \mathbf{u})k(\mathbf{x}, \mathbf{u}) d\mathbf{u}, \end{aligned} \quad (4.68)$$

because of orthogonality of the increments of $W(\cdot)$. In general, this is *not* a function of $(\mathbf{s} - \mathbf{x})$. Furthermore, the semivariogram is

$$\gamma(\mathbf{s}, \mathbf{x}) = (1/2)C_Y(\mathbf{s}, \mathbf{s}) + (1/2)C_Y(\mathbf{x}, \mathbf{x}) - C_Y(\mathbf{s}, \mathbf{x}), \quad (4.69)$$

which, in general, is not a function of $(\mathbf{s} - \mathbf{x})$ either. That is, in general, SMA models are *not* stationary. Higdon, Swall, and Kern (1999) give examples of kernels $k(\cdot, \cdot)$ that lead to nonstationary spatial models. The kernel $k(\cdot, \cdot)$ can be parameterized, which in turn leads to a parametric form for the covariance

function (4.68) and its associated semivariogram (4.69). Alternatively, specifying the kernel in terms of support points leads to a “predictive process” approach (Banerjee et al., 2008, Finley et al. 2009).

Recall that classical geostatistics is often involved with building models of stationary covariance functions or stationary semivariograms. Are there choices of kernels $k(\cdot, \cdot)$ that yield these classical geostatistical models? For example, can (4.68) be used to obtain the spherical covariance function referred to in Section 4.1.1? In general, it cannot, although Cressie and Pavlicova (2002) show how a $k(\cdot, \cdot)$ can be chosen so that (4.68) is stationary and has the same range, sill, and derivative at the origin as the spherical covariance function. When simulating geostatistical models, the advantage of using the SMA model (4.67) is immediately obvious: Classical geostatistical simulations are much more cumbersome [e.g., the turning-bands method due to Matheron (1973); or Gaussian sequential simulation due to Deutsch and Journel (1992)] than SMA-based simulations.

If we assume that the kernel $k(\cdot, \cdot)$ in (4.67) is continuous and square integrable on $\mathbb{R}^d \times \mathbb{R}^d$, then, from Wahba (1990, p. 3), it can be written as

$$k(\mathbf{s}, \mathbf{u}) = \sum_{j=1}^{\infty} v_j \phi_j(\mathbf{s}) \phi_j(\mathbf{u}), \quad (4.70)$$

where $v_1 \geq v_2 \geq \dots$ are eigenvalues and $\{\phi_j(\cdot) : j = 1, 2, \dots\}$ are orthonormal eigenvectors (i.e., $\int \phi_j(\mathbf{u}) \phi_k(\mathbf{u}) d\mathbf{u} = I(j = k)$) that solve the Fredholm integral equation:

$$\int k(\mathbf{s}, \mathbf{u}) \phi_j(\mathbf{u}) d\mathbf{u} = v_j \phi_j(\mathbf{s}), \quad j = 1, 2, \dots \quad (4.71)$$

From (4.67),

$$\begin{aligned} Y(\mathbf{s}) &= \sum_{j=1}^{\infty} v_j \phi_j(\mathbf{s}) \int \phi_j(\mathbf{u}) W(d\mathbf{u}) \\ &\equiv \sum_{j=1}^{\infty} v_j V_j \phi_j(\mathbf{s}), \end{aligned} \quad (4.72)$$

where $\{V_j\}$ are zero-mean orthonormal random variables, as we now demonstrate:

$$\begin{aligned} E(V_j \cdot V_k) &= \int \int \phi_j(\mathbf{u}) \phi_k(\mathbf{v}) E(dW(\mathbf{u}) \cdot dW(\mathbf{v})) \\ &= \int \phi_j(\mathbf{u}) \phi_k(\mathbf{u}) d\mathbf{u} \\ &= I(j = k). \end{aligned} \quad (4.73)$$

Furthermore, because $W(\cdot)$ is Brownian motion, $\{V_j\}$ in (4.72) is a sequence of *iid* standard Gaussian random variables. If we write (4.72) as

$$Y(\mathbf{s}) = \sum_{j=1}^{\infty} \alpha_j \phi_j(\mathbf{s}),$$

where $\alpha_j \equiv v_j V_j$, then $\{\alpha_j\}$ are independent random variables such that $\text{var}(\alpha_j) = v_j^2$. Notice that the orthonormal eigenvectors $\{\phi_j(\cdot)\}$ capture the spatial structure directly.

Due to the orthonormality given by (4.73), the covariance function of $Y(\cdot)$ defined by (4.67) or (4.72) follows easily as

$$C_Y(\mathbf{s}, \mathbf{x}) = \sum_{j=1}^{\infty} v_j^2 \phi_j(\mathbf{s}) \phi_j(\mathbf{x}), \quad (4.74)$$

which is often referred to as a Karhunen–Loève expansion. Now, any positive-definite function that is continuous and square integrable on $\mathbb{R}^d \times \mathbb{R}^d$ has a Karhunen–Loève expansion with nonnegative eigenvalues (e.g., Papoulis, 1965, pp. 457–461). Through (4.67), we have shown that such covariance functions can be obtained from a process $Y(\cdot)$ that is a spatial moving average of Brownian motion.

The result (4.74) is a consequence of Mercer's Theorem: For any positive-definite function $C(\mathbf{s}, \mathbf{x})$, there are *nonnegative* eigenvalues $\{\kappa_j : j = 1, 2, \dots\}$ and eigenfunctions $\{\phi_j(\cdot) : j = 1, 2, \dots\}$ corresponding to the eigensystem, $\int C(\mathbf{s}, \mathbf{x}) \phi(\mathbf{s}) d\mathbf{s} = \kappa \phi(\mathbf{x})$, such that $C(\mathbf{s}, \mathbf{x}) = \sum_{j=1}^{\infty} \kappa_j \phi_j(\mathbf{s}) \phi_j(\mathbf{x})$, where the domain is $a \leq x, s \leq b$, for $a < b$.

The representation (4.72) is interesting in its own right. Notice that $Y(\cdot)$ is a linear combination of deterministic basis functions $\{\phi_j(\cdot) : j = 1, 2, \dots\}$; the randomness in the process $Y(\cdot)$ comes from the randomness in the *coefficients* $\{\alpha_j = v_j V_j : j = 1, \dots, n\}$, which do not depend on spatial location.

Let $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ be a set of n sites, at which $Y(\cdot)$ is evaluated. The resulting vector $\mathbf{Y} \equiv (Y(\mathbf{s}_1), \dots, Y(\mathbf{s}_n))'$ has covariance matrix, $\Sigma_Y \equiv \text{var}(\mathbf{Y})$. Now, a principal components analysis yields

$$\Sigma_Y = \Psi \text{diag}(\lambda_1, \dots, \lambda_n) \Psi', \quad \Psi' \Psi = \mathbf{I} = \Psi \Psi', \quad (4.75)$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$. Write $\Psi = (\psi_1, \dots, \psi_n)$ and the k th column of Ψ as

$$\psi_k = (\psi_k(\mathbf{s}_1), \dots, \psi_k(\mathbf{s}_n))', \quad k = 1, \dots, n. \quad (4.76)$$

The vectors $\{\psi_1, \dots, \psi_n\}$ are sometimes referred to as the *empirical orthogonal functions* (EOFs), although strictly speaking the orthogonal functions should not be called “empirical” until an estimate $\hat{\Sigma}_Y$ is used in place of Σ_Y

above (Section 5.3). Define $a_k \equiv \psi'_k \mathbf{Y}$, $k = 1, \dots, n$. Then the orthogonality allows us to write

$$Y(s_i) = \sum_{k=1}^n a_k \psi_k(s_i), \quad i = 1, \dots, n, \quad (4.77)$$

where $a_1, \dots, a_n$ are uncorrelated with variances given by $\lambda_1, \dots, \lambda_n$, respectively. Again, we see that $\mathbf{Y}$ is a random linear combination of basis vectors $\{\psi_j; j = 1, \dots, n\}$. That is, the decomposition in (4.77) of $\{Y(s_i)\}$ in terms of EOFs is a finite-spatial-location version of the expressions (4.67) or (4.72).

Consider the general spectral expansion,

$$Y(s_i) = \sum_{k=1}^n \alpha_k \phi_k(s_i), \quad i = 1, \dots, n, \quad (4.78)$$

where $\{\phi_k(\cdot)\}$ are complete and orthonormal on $\{s_1, \dots, s_n\}$. As discussed in Section 3.5.2, one distinct advantage of such a spectral decomposition is the decorrelation it induces. For example, recall that if $\{\phi_k(s_i)\}$ are Fourier basis functions, then $\{\alpha_k\}$ are approximately uncorrelated. This property can be used to perform efficient spatial prediction on a discrete lattice for very large n . Consider $\mathbf{Y} \sim \text{Gau}(\mathbf{0}, \Sigma_Y)$, where $\mathbf{Y}$ is an n -dimensional vector. Given that $\mathbf{Y} = \Phi \boldsymbol{\alpha}$ and $\boldsymbol{\alpha} = \Phi' \mathbf{Y}$, we note that $\boldsymbol{\alpha} \sim \text{Gau}(\mathbf{0}, \Sigma_\alpha)$, where $\Sigma_\alpha \equiv \Phi' \Sigma_Y \Phi$. With the choice of basis functions such that Σ_α is diagonal (or reasonably approximated by a diagonal) matrix, say $\mathbf{D}$, one can show that the matrix inverses used in the kriging formulations only require the inverse of a diagonal matrix, *greatly* simplifying computations. In addition, for the case where Φ is made up of discrete Fourier basis functions or certain classes of wavelet basis functions, then discrete Fourier transforms or multiresolution transforms (or their associated inverse transforms) can be used in place of matrix multiplication involving Φ (or Φ'), so that Φ need not actually be stored. This allows the spectral approach to spatial prediction to be used on *very large* spatial domains.

The spectral approach to spatial prediction can be used when the form of the covariance function is known and stationary (e.g., Wikle, 2002b; Royle and Wikle, 2005; Paciorek, 2007; Lindgren et al., 2011) or when it is unknown and/or nonstationary (e.g., Davis, 1952; Nychka, Wikle, and Royle, 2002; Pavlicova, Santner, and Cressie, 2008; Yue and Speckman, 2010). For example, in the case of the former, assume that a Matérn covariance structure is specified, in which case Σ_Y will depend on parameters $\boldsymbol{\theta}$ as shown in (4.6). In this case, if Φ is made up of Fourier basis functions, then the diagonal matrix $\mathbf{D}$ is also parameterized in terms of the same parameters by using the spectral representation of the Matérn covariance function (e.g., see Royle and Wikle, 2005). Perhaps more importantly, in the BHM framework, such spectral parameterizations can be used in the process model when one has non-Gaussian data models (Wikle, 2002b; Hooten, Larsen, and Wikle, 2003; Royle and Wikle, 2005; Paciorek, 2007). This is a theme we pick up again in the dynamical spatio-temporal setting (Section 7.1.3).

Return to the case where spatial index $\mathbf{s} \in D_s$; one possible parameterization of $k(\mathbf{s}, \mathbf{u})$ in (4.67) might be to restrict $v_{p+1} = v_{p+2} = \dots = 0$, resulting in

$$k(\mathbf{s}, \mathbf{u}) = \sum_{j=1}^p v_j \phi_j(\mathbf{s}) \phi_j(\mathbf{u}) \quad \text{and} \quad Y(\mathbf{s}) = \sum_{j=1}^p v_j V_j \phi_j(\mathbf{s}). \quad (4.79)$$

This generates a class of *models* whose parameter space is defined by the integer p and positive real numbers $\{v_j: j = 1, \dots, p\}$. The randomness in $Y(\cdot)$ comes from (4.79). The model (4.79) should *not* be confused with (4.75) and (4.77), which give the EOF *representation* obtained from the covariance matrix Σ_Y .

Finally, we emphasize that SMA models, or their discrete versions, are usually defined as moving averages of *uncorrelated* random variables. By reducing the number of random variables but allowing them to be correlated, a low-dimensional but highly flexible class of spatial processes can be generated.

Let $\alpha \equiv (\alpha_1, \dots, \alpha_{p_\alpha})'$ be a p_α -dimensional random vector with $\text{var}(\alpha) = \Sigma_\alpha$. Then the smooth *Spatial Random Effects* model is

$$Y(\mathbf{s}) = \sum_{i=1}^{p_\alpha} \phi_i(\mathbf{s}) \alpha_i = \phi(\mathbf{s})' \alpha, \quad \mathbf{s} \in D_s, \quad (4.80)$$

for $\phi(\cdot) \equiv (\phi_i(\cdot): i = 1, \dots, p_\alpha)'$ a p_α -dimensional vector of specified basis functions (which may or may not be orthogonal). In this book, we continue to call (4.80) a spectral decomposition, even though the basis functions may no longer be orthogonal and $\{\alpha_i\}$ may no longer be independent.

The covariance function of $Y(\cdot)$ is nonstationary and given by

$$C_Y(\mathbf{s}, \mathbf{x}) = \phi(\mathbf{s})' \Sigma_\alpha \phi(\mathbf{x}). \quad (4.81)$$

This spatial covariance model (with bisquare spatial basis functions) was used by Cressie and Johannesson (2006, 2008) for kriging of very large datasets, resulting in a methodology they called *Fixed Rank Kriging*. Optimal spatial prediction (Section 4.1.2) based on the covariance function (4.81) is computationally efficient, representing an improvement in order of computation over covariance tapering (Furrer, Genton, and Nychka, 2006). For discussion on various choices of basis functions, see Wikle (2010b).

4.1.5 Multivariate Geostatistical Processes

So far, this chapter has concentrated on a univariate spatial process $Y(\cdot)$. Phenomena typically do not act alone, although they might be studied in isolation. There have been two paths taken to model multivariate spatial data: The *joint-distribution path* has predominated, but we believe that the *conditional-distribution path* comes closer to modeling the etiology of how phenomena interact.

Let $\mathbf{Y}(\cdot) \equiv (Y_1(\cdot), \dots, Y_K(\cdot))'$ denote a vector of K spatial processes observed over the same spatial domain D_s . The joint-distribution path

involves defining and estimating measures of cross-dependence, such as the cross-variogram, the pseudo-cross-variogram, the cross-covariance function, and the generalized cross-covariance function. Künsch, Papritz, and Bassi (1997) are the clearest on the relationship between these measures, showing that the most satisfactory summary is the generalized cross-covariance function.

The conditional-distribution path is natural when one has scientific knowledge about how $Y_2(\cdot)$ behaves given knowledge about $Y_1(\cdot)$, where for illustration we have considered the case $K = 2$. Then, because of the probability decomposition, $[Y_1(\cdot), Y_2(\cdot)] = [Y_2(\cdot)|Y_1(\cdot)][Y_1(\cdot)]$, it is more natural to model $Y_2(\cdot)$ *conditionally* on $Y_1(\cdot)$ rather than jointly with $Y_1(\cdot)$. Royle and Berliner (1999) propose this approach and it is discussed further in the paper by Gelfand et al. (2004). We would like to return to this discussion and offer a fresh perspective on it.

A basic goal of spatial statistical modeling in a multivariate setting is the specification of a *valid* model for $\mathbf{Y}(\cdot)$ based on first and second moments. That is, for *any* positive integer n_k and locations $\{\mathbf{s}_{k1}, \dots, \mathbf{s}_{kn_k}\}$, $k = 1, \dots, K$, the $(\sum_{k=1}^K n_k) \times (\sum_{k=1}^K n_k)$ covariance matrix,

$$\text{var}(Y_1(\mathbf{s}_{11}), \dots, Y_1(\mathbf{s}_{1n_1}), \dots, Y_K(\mathbf{s}_{K1}), \dots, Y_K(\mathbf{s}_{Kn_K})),$$

is *always* positive-definite. Any multivariate spatial model must achieve this. Only then can the model be compared to others, based on such criteria as interpretability, simplicity, and ease of use in inference (including ease of computing). A major problem with the joint-distribution path has been the difficulty in specifying parametric models of between-variable spatial dependence that result in a valid multivariate spatial model for $\mathbf{Y}(\cdot)$. Ver Hoef and Barry (1998) and Ver Hoef, Cressie, and Barry (2004) achieve this through spatial moving average (SMA) models (Section 4.1.4). Banerjee, Carlin, and Gelfand (2004, Chapter 7) and Gelfand et al. (2004) achieve this through the Linear Model of Coregionalization (LMC), given here for $K = 2$. Define

$$\mathbf{Y}(\mathbf{s}) \equiv \mathbf{A}\mathbf{W}(\mathbf{s}) \equiv (\mathbf{a}_1 \ \mathbf{a}_2) \begin{pmatrix} W_1(\mathbf{s}) \\ W_2(\mathbf{s}) \end{pmatrix}, \quad \mathbf{s} \in D_s, \quad (4.82)$$

where $\mathbf{Y}(\cdot) \equiv (Y_1(\cdot), Y_2(\cdot))'$, $\mathbf{A} \equiv (\mathbf{a}_1, \mathbf{a}_2)$ is a 2×2 matrix, and $W_1(\cdot)$ and $W_2(\cdot)$ are two independent stationary random processes with mean 0 and variance 1. Consequently,

$$\text{cov}(\mathbf{Y}(\mathbf{s}), \mathbf{Y}(\mathbf{x})) = \mathbf{a}_1 \mathbf{a}_1' \rho_1(\mathbf{s} - \mathbf{x}) + \mathbf{a}_2 \mathbf{a}_2' \rho_2(\mathbf{s} - \mathbf{x}),$$

where $\rho_k(\cdot)$ is the correlation (also covariance) function associated with $W_k(\cdot)$; $k = 1, 2$.

From the expression above, it is easy to see that the LMC has a very strong symmetry assumption, namely,

$$\text{cov}(Y_1(\mathbf{s}), Y_2(\mathbf{x})) = \text{cov}(Y_1(\mathbf{x}), Y_2(\mathbf{s})).$$

For example, when applied to (the average maximum and minimum) temperature and precipitation on a given day, this assumption says that the covariance of temperature in St. Louis, MO, with precipitation in Columbus, OH, is *equal to* the covariance of precipitation in St. Louis, MO, with temperature in Columbus, OH. This is a strong constraint to put on the joint distribution, and it limits the modeler's ability to capture properly the between-variable spatial dependence. Gelfand et al. (2004) emphasize the LMC's computational advantages in a BHM.

The conditional-distribution path has a distinct advantage over the joint-distribution path, since it allows a valid spatial multivariate model to be built with relative ease, as we set out below. We choose $K = 2$, and hence $\mathbf{Y}(\mathbf{s}) \equiv (Y_1(\mathbf{s}), Y_2(\mathbf{s}))'$; $\mathbf{s} \in D_s$, since the exposition is clearest in this case. Then, from scientific considerations, there is often a natural ordering for conditioning. Here we model the process $Y_1(\cdot)$ given $Y_2(\cdot)$ for illustration, although if there is no natural ordering, one could simply be chosen. Now consider the processes defined on the discretized finite grid, $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$, and write $\mathbf{Y}_k \equiv (Y_k(\mathbf{s}_1), \dots, Y_k(\mathbf{s}_n))'$; $k = 1, 2$. The conditional-probability path is based on the decomposition,

$$[\mathbf{Y}_1, \mathbf{Y}_2] = [\mathbf{Y}_2 | \mathbf{Y}_1] [\mathbf{Y}_1],$$

where for the purposes of illustration we assume the left-hand side has a joint Gaussian distribution with mean zero. Hence,

$$\begin{aligned} \mathbf{Y}_2 | \mathbf{Y}_1 &\sim \text{Gau}(\mathbf{B}\mathbf{Y}_1, \mathbf{\Sigma}_{2|1}) \\ \mathbf{Y}_1 &\sim \text{Gau}(\mathbf{0}, \mathbf{\Sigma}_{11}). \end{aligned} \tag{4.83}$$

The *mechanistic* model of $\mathbf{Y}_2 | \mathbf{Y}_1$ determines the form of $\mathbf{B}$ (which is any $n \times n$ matrix) and $\mathbf{\Sigma}_{2|1}$ (which is any $n \times n$ symmetric positive-definite matrix). Finally, $\mathbf{\Sigma}_{11}$ is specified (as any $n \times n$ positive-definite matrix) to complete the model.

We wish to emphasize that (4.83) requires specification of only two positive-definite matrices, something that spatial statistical modelers are very experienced with. There is *no* requirement of specifying any cross-dependence, since it is already captured in the regression relationship, $E(\mathbf{Y}_2 | \mathbf{Y}_1) = \mathbf{B}\mathbf{Y}_1$, and $\mathbf{B}$ is completely general. This is key, since validity of the joint model is guaranteed no matter how $\mathbf{B}$ is chosen. In fact, from (4.83), the joint model's moments follow straightforwardly as

$$\begin{aligned} \text{var}(\mathbf{Y}_2) &= \mathbf{\Sigma}_{2|1} + \mathbf{B}\mathbf{\Sigma}_{11}\mathbf{B}', \\ \text{var}(\mathbf{Y}_1) &= \mathbf{\Sigma}_{11}, \end{aligned}$$

and

$$\text{cov}(\mathbf{Y}_1, \mathbf{Y}_2) = \mathbf{B}\mathbf{\Sigma}_{11}.$$

It is this last quantity, $\text{cov}(\mathbf{Y}_1, \mathbf{Y}_2)$ that causes modelers taking the joint-probability path so much trouble. They, too, have to specify two positive-definite matrices ($\text{var}(\mathbf{Y}_1)$ and $\text{var}(\mathbf{Y}_2)$), but whatever model they choose for $\text{cov}(Y_1(\cdot), Y_2(\cdot))$, the $2n \times 2n$ matrix,

$$\begin{bmatrix} \text{var}(\mathbf{Y}_1) & \text{cov}(\mathbf{Y}_1, \mathbf{Y}_2) \\ \text{cov}(\mathbf{Y}_2, \mathbf{Y}_1) & \text{var}(\mathbf{Y}_2) \end{bmatrix}$$

has to be symmetric and positive-definite for the model to be valid. Notice that after choosing two positive-definite matrices, the conditional-distribution path (4.83) results in a bivariate model that is automatically valid, no matter what $\mathbf{B}$ is chosen. See the Bibliographic Notes in Section 4.5 for further discussion.

4.1.6 Other Topics

Change-of-Resolution Spatial Models

In this topic, we present an example of a spatial HM that has strong computational and change-of-support features. Our goal is to illustrate the breadth of the hierarchical-modeling approach. Geostatistical processes can be aggregated from the point level to regions; Section 4.1.3 shows how the mean and covariance functions of the regions are determined by the point-level geostatistical model. In practice, computational problems can arise when data are massive and there are change-of-support issues associated with data being of differing resolutions than that of the process to be predicted. Multiresolution models give a consistent way to build the process model over partitions of the spatial region $D_s \subset \mathbb{R}^d$, that nest successively within each other. Multiresolutional models are a powerful way to look at data with different spatial scales of variation (e.g., Ferreira and Lee, 2007; Gramacy and Lee, 2008; Pasanen and Holmström, 2008; Jansen, Nason, and Silverman, 2009).

The following exposition is adapted from Huang, Cressie, and Gabrosek (2002). Let $\{D(i, r) : i = 1, \dots, n_r\}$, for *each* resolution-index $r = 0, \dots, R$, be a family of *disjoint partitions* of D_s such that

$$D(i, r) = \bigcup_{j=1}^{m_r} D(\text{ch}_j(i, r)), \quad (i, r) \in \mathcal{N}_{R-1},$$

where $\{\text{ch}_j(i, r) : j = 1, \dots, m_r\}$ indexes the m_r *children cells* of $D(i, r)$; $m_0, \dots, m_{R-1}$ are the *aperture (i.e., resolution) changes*; and $\mathcal{N}_k \equiv \{(i, r) : i = 1, \dots, n_r, r = 0, \dots, k\}$. Hence, there are multiresolution partitions of D_s with a tree-structured parent–children relationship between cells (pixels) at adjacent resolutions.

Let $\{Y(s) : s \in D_s\}$ be a real-valued univariate spatial process (i.e., a geostatistical process) that is defined on D_s . Based on the partitioning $\{D(i, r)\}$, define the multiresolution, aggregated Y -process as

$$Y(i, r) \equiv \frac{1}{v(i, r)} \int_{D(i, r)} Y(s) ds, \quad (i, r) \in \mathcal{N}_R,$$

where $v(i, r) \equiv |D(i, r)|$ is the volume of $D(i, r)$. Hence, $\{Y(\cdot, r)\}$ represents the resolution- r values associated with the process $Y(\cdot)$; $r = 0, \dots, R$. Furthermore, denote by

$$\mathbf{Y}(ch(i, r)) \equiv (Y(ch_1(i, r)), \dots, Y(ch_{m_r}(i, r)))', \quad (i, r) \in \mathcal{N}_{R-1},$$

the vector of the aggregated process on the children cells of (i, r) , which are values defined at resolution $(r + 1)$. The vector of all resolution- r values is denoted by

$$\mathbf{Y}(r) \equiv (Y(1, r), \dots, Y(n_r, r))', \quad r = 0, \dots, R.$$

In what follows, we define the *data model* via a linear statistical relationship between data and the hidden process Y , and we define the *process model* as a coarse-to-fine, change-of-resolution autoregression in Y .

At each resolution $(r + 1)$, we have (potentially) observed data $\{\mathbf{Z}(ch(i, r))\}$ that are assumed to be related to the process Y at resolution $(r + 1)$, according to an additive-measurement-error model.

Data model: Conditional on σ^2 , $\mathbf{V}(ch(i, r))$, and $\mathbf{Y}(ch(i, r))$,

$$\mathbf{Z}(ch(i, r)) = \mathbf{H}(ch(i, r))\mathbf{Y}(ch(i, r)) + \mathbf{v}(ch(i, r)),$$

$(i, r) \in \mathcal{V} \subseteq \mathcal{N}_{R-1}$, where $\mathbf{H}(ch(i, r))$ is a known incidence matrix and $\mathbf{v}(ch(i, r)) \sim \text{ind. Gau}(\mathbf{0}, \sigma^2 \mathbf{V}(ch(i, r)))$, independently of $\mathbf{Y}(\cdot)$.

The data vector $\mathbf{Z}(ch(i, r))$ is sometimes the same length as $\mathbf{Y}(ch(i, r))$, with each element being a direct measurement of the corresponding element of $\mathbf{Y}(ch(i, r))$; in that case, $\mathbf{H}(ch(i, r)) = \mathbf{I}$. However, $\mathbf{Z}(ch(i, r))$ may have some missing data (i.e., some of the elements of $\mathbf{Y}(ch(i, r))$ may be unobserved); in that case, $\mathbf{H}(ch(i, r))$ is an incidence matrix of zeros and ones that matches each of the observed elements of $\mathbf{Z}(ch(i, r))$ to its corresponding element of $\mathbf{Y}(ch(i, r))$. The measurement-error covariance matrices $\{\sigma^2 \mathbf{V}(ch(i, r))\}$ are typically diagonal, corresponding to independent measurement errors.

Instead of specifying the spatial covariance structure of $Y(\cdot)$ directly, spatial dependence is introduced in the process model indirectly through the following coarse-to-fine-resolution model:

Process model: Conditional on $\mathbf{W}(ch(i, r))$,

$$\mathbf{Y}(ch(i, r)) = Y(i, r)\mathbf{1} + \boldsymbol{\omega}(ch(i, r)); \quad (i, r) \in \mathcal{N}_{R-1},$$

where $\boldsymbol{\omega}(ch(i, r)) \sim \text{ind. Gau}(\mathbf{0}, \sigma^2 \mathbf{W}(ch(i, r)))$, independently of $Y(i, r)$.

The spatial dependence in $\mathbf{Y}(\cdot)$ depends on the variation in the process-model errors $\{\boldsymbol{\omega}(ch(i, r))\}$, where smaller variation results in stronger spatial

dependence. As such, the spatial dependence is indirectly specified through the covariance matrices $\{\sigma^2 \mathbf{W}(ch(i, r))\}$; while σ^2 is known through the data model, the elements of $\mathbf{W}$ contain parameters that require estimation (EHM) or a prior distribution (BHM). The process model is then fully specified, once a distribution is specified at the coarsest resolution.

Process model, ctd: Conditional on $\mathbf{a}(0)$, σ^2 , and $\mathbf{R}(0)$,

$$\mathbf{Y}(0) \sim \text{Gau}(\mathbf{a}(0), \sigma^2 \mathbf{R}(0)).$$

This resolution-0 process model, along with a recursive use of the coarse-to-fine-resolution process model, yields the distribution of Y (at all aggregated resolutions) to be multivariate Gaussian:

$$\mathbf{Y}(ch(i, r)) \sim \text{Gau}(\mathbf{a}(ch(i, r)), \sigma^2 \mathbf{R}(ch(i, r))), \quad (i, r) \in \mathcal{N}_{R-1},$$

where $\mathbf{a}(ch(i, r))$ and $\mathbf{R}(ch(i, r))$ are derived via successive application of the coarse-to-fine-resolution process model just above. Given the data $\{\mathbf{Z}(ch(i, r))\}$, we are able to derive the *posterior distribution* of Y at all resolutions using a change-of-resolution Kalman filter (Chou, Willsky, and Nikoukhah, 1994). The derivation consists of two steps, the leaves-to-root filtering step and the root-to-leaves smoothing step (Huang, Cressie, and Gabrosek, 2002). In a sense, the change-of-resolution in the process model sets up an ordering in space to which a type of Kalman filter can be applied. A spatio-temporal version of the multiresolution model presented below can be found in Section 7.2.7.

Example: Total Column Ozone

Total column ozone (TCO) data from the Total Ozone Mapping Spectrometer (TOMS) instrument from October 5, 1998 are shown in Figure 4.7. These data are part of a spatio-temporal dataset analyzed by Johannesson, Cressie, and Huang (2007); the figure shows a considerable amount of missing data associated with different orbits of the satellite on that day.

At the very finest resolution (resolution R), the pixels are 1° latitude $\times$ 1.25° longitude; thus, there are $n_R = 288 \times 180 = 51,840$ pixels that cover the globe. The (level 3) data released by the U.S. National Aeronautics and Space Administration (NASA) are at this resolution; there is none at any coarser resolutions $R - 1, \dots, 0$. Thus, the data model above is quite simple: It is only nontrivial at resolution R , where the matrix $\mathbf{H}$ serves to annihilate pixels whose data are missing.

The posterior mean of Y at resolution R is shown in the top panel of Figure 4.8, and the corresponding posterior standard deviation is shown in the bottom panel of Figure 4.8. (Parameter estimates are “plugged into” the posterior mean and standard deviation.)

Parameters specified in the process model are estimated using composite-likelihood methods, and hence this multiresolution modeling takes an EHM

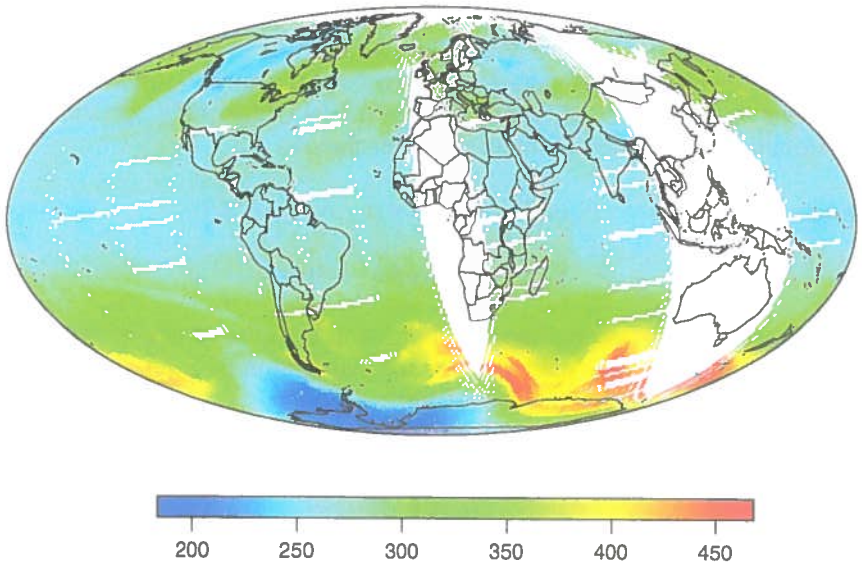


Figure 4.7 Total column ozone (TCO) data retrieved from the TOMS instrument on October 5, 1988. Regions of the globe in white depict places where no data were retrieved. Low (high) TCO values are shown in blue (red).

approach. More details of the model and the computational algorithms associated with it can be found in Huang, Cressie, and Gabrosek (2002), Tzeng, Huang, and Cressie (2005), and Johannesson, Cressie, and Huang (2007).

What makes this EHM particularly attractive is its computational advantage; the computational cost of deriving the posterior distribution is linear in the number of parent cells, namely $O(\sum_{r=0}^{R-1} m_r^3 n_r)$. Fixed Rank Kriging referred to in Section 4.1.4 is also linear in the data size, namely $O(n_R)$. These are in contrast to many traditional spatial modeling approaches, for example, classical kriging presented in Section 4.1.2, where the computational cost is cubic in the size of the data, namely $O(n_R^3)$.

Spatial Sampling Design

Fedorov (1972) is an early reference to research carried out in spatial sampling design; it has become an area of considerable activity in the last ten years. The monograph by Müller (2000) gives a snapshot of the literature at the beginning of the period, and Le and Zidek (2006, Chapter 11) is devoted to environmental network design. Two comprehensive articles should be consulted for different approaches, one taking a design-based approach (Stevens and Olsen, 2004) and the other taking a model-based approach (Diggle, Menezes, and Su, 2010) to spatial sampling. There have been many articles in the last 20 years, including Cressie, Gotway, and Grondona (1990), Overton and Stehman (1996), Brus and de Gruijter (1997), Cressie (1998b), Aldworth and Cressie (1999), Berliner, Lu, and Snyder (1999), Wikle and Royle (1999), Martin (2001), Thompson (2002),

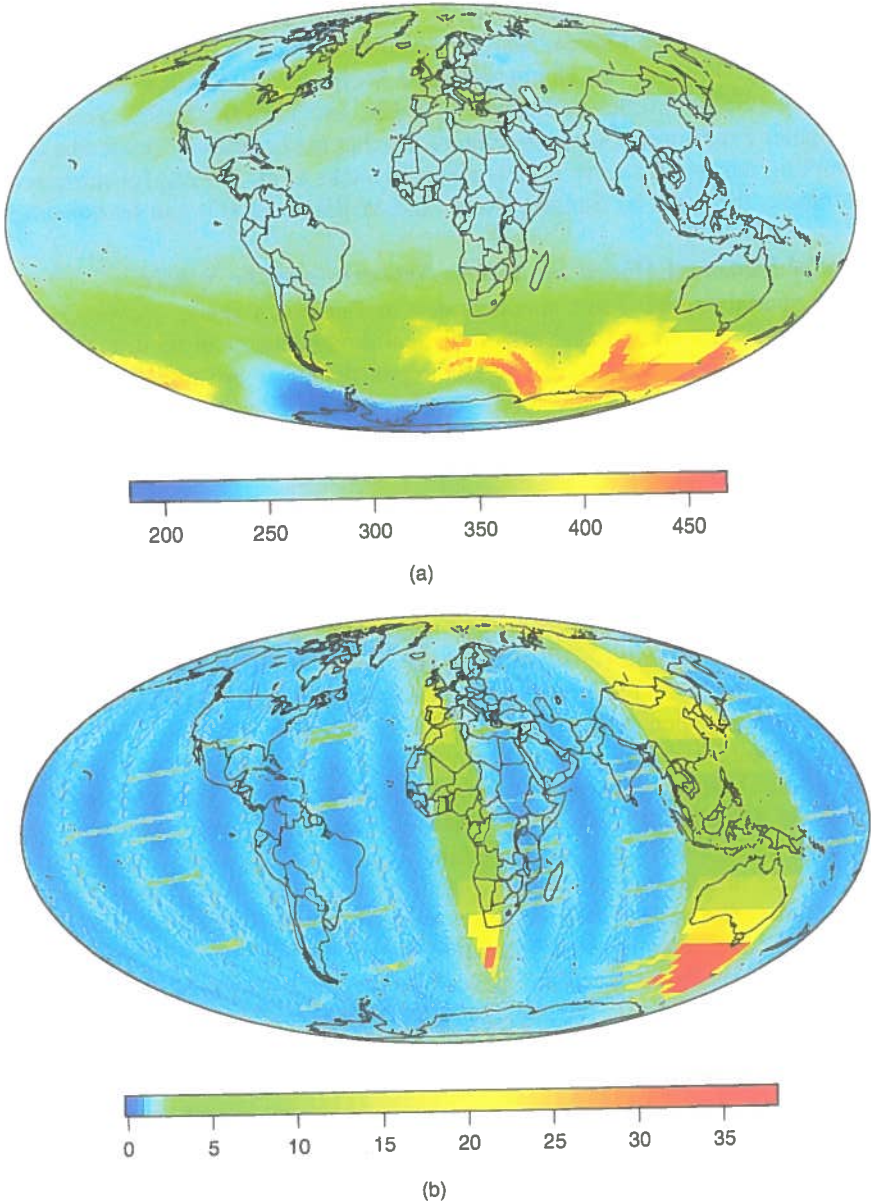


Figure 4.8 The figure shows the result of the change-of-resolution Kalman filter applied to the TOMS data retrieved on October 5, 1988, shown in Figure 4.7. **Top panel:** Posterior mean of the latent process Y at its finest possible resolution. **Bottom panel:** Posterior standard deviation of the latent process Y ; notice large standard deviations in regions of the globe where data are missing.

Ver Hoef (2002), Müller (2005), Wikle and Royle (2005), Diggle and Lophave (2006), Zimmerman (2006), Marchant and Lark (2007), Irvine, Gitelman, and Hoeting (2007), Dobbie, Henderson, and Stevens (2008), Munoz, Lesser, and Ramsay (2008), Müller and Stehlík (2010), and Nowak (2010). The area of computer experiments has a large literature that is very closely related to spatial sampling design. An early review paper is Sacks et al. (1989); for more recent literature, the book by Santner, Notz, and Williams (2003) can be consulted.

Spatial Statistical Models on Stream Networks and the Sphere

Most spatial models have spatial index in Euclidean space. However, when the “space” is a stream network, statistical dependence can be directional and spatial models are needed where Euclidean distances are replaced by stream distances. Monestiez et al. (2005) developed a probability model that reflects the unidirectional flow of the stream network from source to mouth. However, they have computational difficulties applying it to kriging at all stream locations based on data at monitoring sites interspersed along the network. Cressie et al. (2006) and Ver Hoef, Peterson, and Theobald (2006) use spatial moving average (SMA) models on a generic stream network to define valid spatial models that incorporate stream flow and allow kriging in a straightforward manner. Various applications of the SMA approach are summarized in Ver Hoef and Peterson (2010) and the accompanying discussion and rejoinder. That paper also features models of spatial dependence involving different types of directional dependence; fish can swim upstream.

Banerjee (2005) and Jun and Stein (2008) consider covariance functions on the sphere. Distances that are no longer Euclidean require special care because a simple substitution into a (Euclidean-distance) isotropic covariance function can yield invalid models; for example, the validity of the spatio-temporal covariance function used by Cressie and Majure (1997) was assumed but may not be true. Curriero (2006) discusses the use of non-Euclidean distance measures in geostatistics; even in Euclidean space, anisotropy of a spatial process might lead one to use non-Euclidean distances (e.g., Boisvert, Manchuk, and Beutsch, 2009).

Conditional Simulation

Conditional simulation is a methodology originally developed within geostatistics (e.g., Journé, 1974) that allows one to see the variability of a spatial process at locations between the observations. In a sense, it is now obsolete, since the HM approach makes clear that the target simulation is from the posterior distribution $[Y(s_0)|Z, \theta]$ or $[Z(s_0)|Z, \theta]$.

In its simplest form, conditional simulation relies on an *orthogonal* decomposition of the spatial process into the kriging predictor plus the residual process (e.g., Cressie, 1993, Section 3.6.2). There is an implicit assumption that the process being simulated is Gaussian, since if it were not, then the conditionally simulated quantity is *not* the posterior distribution. In spite of these limitations, conditional simulation has been developed (in largely *ad hoc* ways)

to handle ever more complex problems; the software GSLIB (Deutsch and Journel, 1992, and later versions and editions) and SGeMS (Remy, Boucher, and Wu, 2009) depend on stochastic simulation algorithms that will almost certainly not yield the posterior distributions of the unobserved parts of the spatial process, and they do not account for the measurement error in the observed parts. Recent research in conditional simulation tends to ignore the desirability of the posterior distribution; for example, Emery (2005) develops a conditional-simulation algorithm for a bivariate gamma isofactorial model, and Strebelle (2002) introduces the notion of multiple-point statistics to improve conditional simulation for non-Gaussian processes by modeling the process' moments beyond the covariance function (in his terminology, the covariance is a two-point statistic).

4.2 LATTICE PROCESSES

Recall that lattice processes $\{Y(\mathbf{s}) : \mathbf{s} \in D_s\}$ are defined on a finite or countable subset D_s of $\mathbb{R}^d$. They are used to model lattice data, $\mathbf{Z} \equiv (Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_m))'$, where $\{\mathbf{s}_1, \dots, \mathbf{s}_m\}$ are reference locations for data defined on discrete spatial features (e.g., grid nodes, pixels, small areas). Define the spatial-neighborhood matrix $\mathbf{W} \equiv (w_{ij})$ as an $m \times m$ matrix with zero down the diagonals and $w_{ij} = 1$ if location $\mathbf{s}_j$ is considered "close to" location $\mathbf{s}_i$. Here, the definition of "close to" is made specific by the modeler; for example, it might be whenever $\|\mathbf{s}_i - \mathbf{s}_j\| \leq r$, for some specified "radius of influence" $r > 0$.

Moran (1950) considered the simple case of $\mathbb{R}^1$, $\mathbf{s}_i = i$; $i = 1, \dots, m$ and $w_{ij} = w_{ji} = 1$ if $0 < |s_i - s_j| \leq 1$. Then for detrended data $\mathbf{Z}$ satisfying $E(\mathbf{Z}) = \mathbf{0}$, he defined the statistic,

$$I \equiv \mathbf{Z}'\mathbf{W}\mathbf{Z}/\mathbf{Z}'\mathbf{Z},$$

which he used to test the null hypothesis that $Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_m)$ are iid. Cli and Ord (1981) proposed the use of more general $\mathbf{W}$, where $\mathbf{W}$ does not have to be symmetric and off-diagonal elements do not have to be 1, and named the statistic Moran's I . This has become a ubiquitous statistic for testing the null hypothesis of the absence of spatial autocorrelation in $\mathbf{Z}$, although more powerful alternative has recently been suggested (Li, Calder, and Cress 2007).

The statistic I is directly analogous to the statistic F defined in terms of the empirical semivariogram in Section 4.1.1. Under the null hypothesis, both statistics have a permutation distribution or an asymptotic distribution that allows an initial check to see whether the spatial data warrant a spatial model. Figure 4.9 is a plot of cantons and their reference locations (centroids) $\{\mathbf{s}_i : i = 1, \dots, 268\}$ in the Midi-Pyrénées region of south-west France. Define $w_{ij} = 1$ if and only if canton i shares a boundary with canton j . In this case, the data are prescription amounts per doctor consultation for each canton. Without detrending, a test based on Moran's I rejects the null hypothesis of independence.

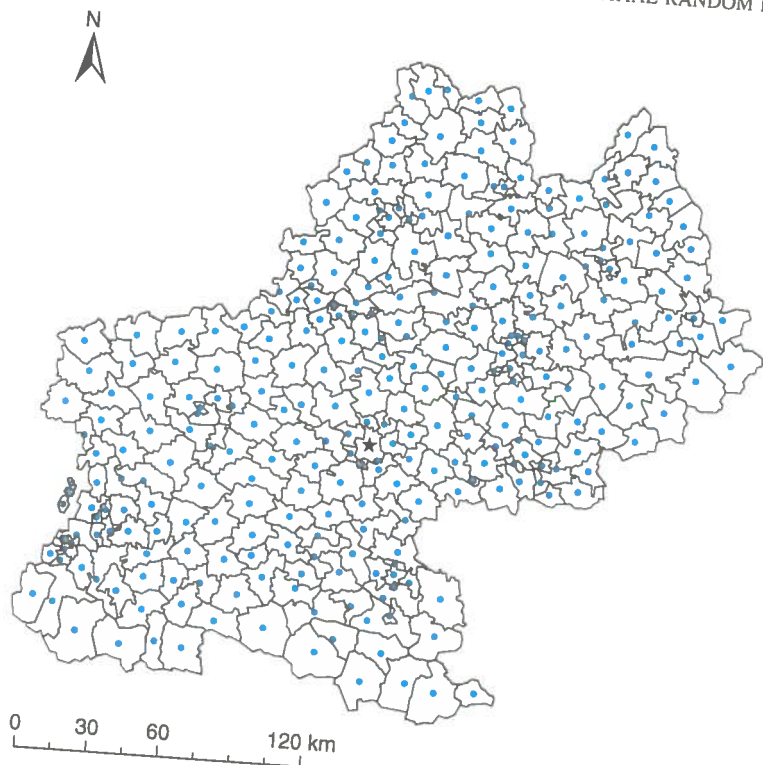


Figure 4.9 Boundaries and spatial locations of cantons in the Midi-Pyrénées region of southwest France. The “star” denotes the canton containing Toulouse.

A spatial lattice in $\mathbb{R}^d$ is made up of a finite or countable number of locations $D_s \equiv \{s_1, s_2, \dots\}$. Figure 4.9 is an example of an irregular spatial lattice. Regular spatial lattices are often achieved by discretization or aggregation of a continuous spatial domain; for example, see Figure 4.5a, or consider pixelization of an image. In the rest of this section, we assume $|D_s| < \infty$, and we build lattice models for the second level (i.e., the process-model level) of the hierarchy.

A Markov Model in One Dimension

Spatial processes in two or higher dimensions are fundamentally different from those in one dimension. The difference arises because it is easy to completely order elements in $\mathbb{R}^1$, but in $\mathbb{R}^d$, for $d \geq 2$, complete orderings are not as natural. Even partial orderings are difficult to motivate when modeling purely spatial data; we shall return to partial orderings in Section 4.2.8.

We now introduce spatial dependence in $\mathbb{R}^1$ and discuss the extent to which it can be generalized to lattice processes in $\mathbb{R}^d$. Any two elements $a, b \in \mathbb{R}^1$ are always related through the (complete) ordering, “ $\leq$ ” (“less than or equal to”).

If $a \leq b$ but $a \neq b$, then we write $a < b$ to mean that a is (strictly) less than b . We can use this complete ordering when defining a random process on $\mathbb{R}^1$. For mathematical simplicity, we shall consider the finite subset $D_s = \{1, 2, \dots, n\}$ consisting of the first n integers of $\mathbb{R}^1$ and define the process $\{Y(s) : s \in D_s\}$.

Suppose that we specify the distribution of $Y(s)$ in terms of values of Y at locations *less than* s . For example, consider the first-order *unilateral* spatial process,

$$Y(s) = \phi Y(s-1) + \varepsilon(s), \quad s = 2, \dots, n, \quad (4.84)$$

where $|\phi| < 1$; $\varepsilon(s)$ is independent of $Y(s-1)$, for $s = 2, \dots, n$; and $\varepsilon(s) \sim iid\ Gau(0, \sigma^2)$. Finally, assume

$$Y(1) \sim Gau(0, \sigma^2/(1 - \phi^2)), \quad (4.85)$$

to obtain a stationary Gaussian process on $\{1, 2, \dots, n\}$. This simple example was already considered in the introduction to Chapter 2, and it has a well known covariance function, $\text{cov}(Y(s), Y(s+h)) \propto \phi^h$. (The analogous temporal process is called a first-order autoregressive process; see Section 3.4.3.) It is not difficult to show that (4.84) and (4.85) imply that $\mathbf{Y} \equiv (Y(1), \dots, Y(n))'$ is jointly Gaussian with $E(\mathbf{Y}) = \mathbf{0}$,

$$\text{var}(\mathbf{Y}) = (\sigma^2/(1 - \phi^2)) \cdot \begin{bmatrix} 1 & \phi & \dots & \phi^{n-1} \\ \phi & 1 & \dots & \phi^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ \phi^{n-1} & \phi^{n-2} & \dots & 1 \end{bmatrix}, \quad (4.86)$$

and $\text{corr}(Y(s), Y(s+1)) = \phi$.

There is an equivalent way to specify the model (4.84), although *a priori* it is not clear that they do result in the same probability distribution; the equivalence is demonstrated using the result (4.96) below. For $s = 2, \dots, n-1$, assume

$$\begin{aligned} &Y(s)|Y(1), \dots, Y(s-1), Y(s+1), \dots, Y(n) \\ &\sim Gau((\phi/(1 + \phi^2))(Y(s-1) + Y(s+1)), \sigma^2/(1 + \phi^2)), \end{aligned} \quad (4.87)$$

and

$$Y(1)|Y(2), \dots, Y(n) \sim Gau(\phi Y(2), \sigma^2), \quad (4.88)$$

$$Y(n)|Y(1), \dots, Y(n-1) \sim Gau(\phi Y(n-1), \sigma^2), \quad (4.89)$$

where $|\phi| < 1$. That is, apart from the edge locations $s = 1$ and $s = n$, the conditional distributions are Gaussian with a mean determined by the sum of the two values of Y at the neighboring locations. The equivalence between

(4.87)–(4.89) and (a temporal version of) (4.84)–(4.85) has already been discussed in Chapter 2. We have a different point to make in comparing the two *spatial* lattice processes.

Based on the theory given below for Markov random fields, in particular for the conditional autoregressive (CAR) models in Section 4.2.5, (4.87)–(4.89) imply that $\mathbf{Y}$ is jointly Gaussian with $E(\mathbf{Y}) = \mathbf{0}$ and

$$(\text{var}(\mathbf{Y}))^{-1} = (1/\sigma^2) \begin{bmatrix} 1 & -\phi & 0 & 0 & \dots & 0 & 0 & 0 \\ -\phi & 1 + \phi^2 & -\phi & 0 & \dots & 0 & 0 & 0 \\ 0 & -\phi & 1 + \phi^2 & -\phi & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & -\phi & 1 + \phi^2 & -\phi \\ 0 & 0 & 0 & 0 & \dots & 0 & -\phi & 1 \end{bmatrix}.$$

This shows that the *inverse* of the covariance matrix, $\text{var}(\mathbf{Y})$, is sparse, which is typical for a CAR model. For this particular model defined on $\{1, \dots, n\} \subset \mathbb{R}^1$, it is possible to find an exact expression for the inverse of $(\text{var}(\mathbf{Y}))^{-1}$. Then,

$$\{(\text{var}(\mathbf{Y}))^{-1}\}^{-1} = \text{right-hand side of (4.86)},$$

which can be verified by multiplying both matrices to obtain the $n \times n$ identity matrix. Hence, although specified completely differently, both the first-order unilateral spatial process (4.84) and (4.85) and the CAR model (4.87)–(4.89) are the *same* statistical model. (They are both Gaussian with mean zero and the same covariance matrix.) Thus, seemingly different conditional probability specifications can lead to equivalent lattice models; more general results of this nature are given in Section 4.2.8.

Still in one-dimensional space, it is straightforward to show that the Markov chain defined by

$$[Y(s)|Y(1), \dots, Y(s-1)] = [Y(s)|Y(s-1)], \quad s = 2, 3, \dots, \quad (4.90)$$

and $[Y(1)]$, can be equivalently characterized by

$$[Y(2), \dots, Y(s)|Y(1)] = \prod_{i=2}^s P_i(Y(i), Y(i-1)), \quad s = 2, 3, \dots, \quad (4.91)$$

and $[Y(1)]$; in (4.91), P_i is a function that depends only on $Y(i)$ and $Y(i-1)$. Furthermore, a special case of Result 2 in Cressie and Davidson (1998) shows that the model defined by (4.90) on $\{1, 2, \dots, n\}$ can be written as a random process on $\{1, 2, \dots, n\}$, characterized by

$$\begin{aligned} & [Y(s)|\{Y(j): j = 1, \dots, n, j \neq s\}] \\ &= [Y(s)|Y(s-1), Y(s+1)], \quad s = 2, \dots, n-1, \end{aligned} \quad (4.92)$$

$[Y(1)|Y(2)]$, and $[Y(n)|Y(n-1)]$. That is, in one-dimensional space, the Markov assumption (4.90) implies (4.91) and (4.92), and either (4.91) or (4.92) implies (4.90).

In two or more dimensions, the locations are no longer completely ordered, and it is not *a priori* clear how to generalize the “lack of memory” assumption (4.90), nor whether there are Markov notions analogous to (4.91) and (4.92). This is taken up in Sections 4.2.1–4.2.5 and 4.2.8.

4.2.1 Markov-Type Models in Space

In looking for Markov-type models in space, in particular on the two-dimensional integer lattice, various authors have attempted to generalize (4.90) or (4.91). Undoubtedly, the most successful generalization has been of (4.90), where the notion of (Markov) neighborhoods is critical (Besag, 1974); for continuous, non-skewed data, the CAR models (Section 4.2.5) are a popular choice. In Section 4.2.2 below, we give a probabilistic definition of the lattice models in $\mathbb{R}^d$ that are called *Markov random fields* (MRFs).

Alternatively, Whittle (1963) attempted a generalization of (4.91); however, the class of *partially ordered Markov models* (POMMs) developed by Cressie and Davidson (1998) turns out to be the appropriate generalization of (4.91). Its probabilistic definition requires a *partial* order on the spatial locations D_s . A brief discussion of POMMs is given in Section 4.2.8.

In another generalization, this time of (4.84), Whittle (1954) gave a simultaneous specification of a spatial process that under Gaussian assumptions has been called a *simultaneous autoregressive (SAR) model* (e.g., Anselin, 1988). We give a brief discussion of such models, including unilateral versions, in Section 4.2.7.

As an illustration of an MRF, we define here (briefly) the *CAR model* on $D_s = \{s_1, \dots, s_n\} \subset \mathbb{R}^d$: Let the sequence of conditional distributions $[Y(s_i)|\mathbf{Y}_{-i}]$; $i = 1, \dots, n$ each be Gaussian with mean and variance,

$$E(Y(s_i)|\mathbf{Y}_{-i}) = \sum_{s_j \in N(s_i)} c_{ij} Y(s_j), \quad (4.93)$$

$$\text{var}(Y(s_i)|\mathbf{Y}_{-i}) = \tau_i^2, \quad (4.94)$$

where $\mathbf{Y}_{-i} \equiv (Y(s_1), \dots, Y(s_{i-1}), Y(s_{i+1}), \dots, Y(s_n))'$, and $N(s_i)$ is a set of prespecified locations that define the *neighborhood* of s_i , $i = 1, \dots, n$. Inspection of (4.93) shows from where the CAR model gets its name; a detailed treatment is given in Section 4.2.5.

Notice from (4.93) and (4.94) that the conditional distribution, $[Y(s_i)|\mathbf{Y}_{-i}]$, depends functionally on $\{Y(s_j) : s_j \in N(s_i)\}$, which is a Markov-type property in space defined in terms of the neighborhood $N(s_i)$. This Markov property is discussed in considerable detail in Section 4.2.2. Notice that specifying $[Y(s_i)|\mathbf{Y}_{-i}]$, $i = 1, \dots, n$, as Gaussian with mean (4.93) and variance (4.94),

does *not guarantee existence* of the joint distribution $[Y]$. For example, consider the homogeneous case of $\tau_1^2 = \dots = \tau_n^2$ in (4.94); then it only takes specifying $c_{12} \neq c_{21}$ in (4.93) for there to be *no* joint distribution $[Y]$. Regularity conditions on $\{c_{ij}\}$ and $\{\tau_i^2\}$ that do guarantee existence are given in Section 4.2.5.

Because $N(s_i)$ does not include the location s_i , it makes sense in (4.93) to define $c_{ii} = 0$. Furthermore, if we define $c_{ij} = 0$, for $s_j \notin N(s_i)$; $i = 1, \dots, n$, then we can write

$$E(Y(s_i)|Y_{-i}) = \sum_{j=1}^n c_{ij}Y(s_j), \quad i = 1, \dots, n. \quad (4.95)$$

Define the $n \times n$ matrix $C \equiv (c_{ij})$, whose (i,j) th element is c_{ij} ; and define the $n \times n$ diagonal matrix $M \equiv \text{diag}(\tau_1^2, \dots, \tau_n^2)$. Besag (1974) shows that if $M^{-1}(I - C)$ is symmetric and positive-definite, then the joint distribution of the CAR model is

$$Y \sim \text{Gau}(0, (I - C)^{-1}M). \quad (4.96)$$

Notice that $\text{var}(Y)^{-1} = M^{-1}(I - C)$, which is sparse (i.e., few nonzero elements) if C is sparse. Modelers typically choose c_{ij} to be nonzero for s_j “close to” s_i ; otherwise it is zero. (This is in contrast to the geostatistical models presented in Section 4.1, where $\text{var}(Y)$ is often sparse.)

To see how a Gaussian MRF (or CAR) model could be used as a *process model* in an HM, consider the following example (often used in small-area estimation; see Section 4.2.6).

Data model: For $i = 1, \dots, n$,

$$Z(s_i)|Y(s_i) \sim \text{ind. Poi}(\exp\{Y(s_i)\}),$$

where $\text{Poi}(v)$ denotes a Poisson distribution with mean v .

Process model: Conditional on β , τ^2 , ϕ , and for $Y \equiv (Y(s_1), \dots, Y(s_n))'$,

$$Y|\beta, \tau^2, \phi \sim \text{Gau}(X\beta, \tau^2(I - \phi H)^{-1}),$$

where $X \equiv (x_j(s_i))$ is an $n \times p$ matrix derived from the p -dimensional covariate $x(s_i)$ at location s_i , $i = 1, \dots, n$; H is a known matrix with zeros down the diagonal that is often the spatial neighborhood matrix (i.e., $h_{ij} = 1$ iff $s_j \in N(s_i)$, equivalently iff $s_i \in N(s_j)$); and the parameter space of ϕ ensures that the matrix inverse exists.

Clearly, the process-model variance is a special case of that in (4.96).

Another way to write the process model is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}, \quad (4.97)$$

where $\boldsymbol{\beta}$ is fixed but unknown, and $\boldsymbol{\delta}|\tau^2, \phi \sim \text{Gau}(\mathbf{0}, \tau^2(\mathbf{I} - \phi\mathbf{H})^{-1})$. Notice that this is a particular case of a general linear model with $\text{var}(\boldsymbol{\delta}) = \tau^2(\mathbf{I} - \phi\mathbf{H})^{-1}$. However, we prefer the specification through $[\mathbf{Y}|\boldsymbol{\beta}, \tau^2, \phi]$ rather than (4.97), since the former generalizes easily to more complex, nonadditive decompositions of variation. More general HMs (not necessarily Poisson distributions in the data model and not necessarily Gaussian distributions in the process model) can be found in Sections 4.2.2 and 4.2.6.

Inference for the HM given above is mostly concerned with prediction of $\mathbf{Y}$ from data $\mathbf{Z}$, and inference on parameters $\boldsymbol{\beta}$, τ^2 , and ϕ . Consider the EHM, where point estimators of $\boldsymbol{\beta}$, τ^2 , and ϕ are sought. The marginal likelihood is

$$\mathcal{L}(\boldsymbol{\beta}, \tau^2, \phi) \equiv [\mathbf{Z}|\boldsymbol{\beta}, \tau^2, \phi] = \int [\mathbf{Z}|\mathbf{Y}][\mathbf{Y}|\boldsymbol{\beta}, \tau^2, \phi] d\mathbf{Y};$$

however, the integration is most often analytically intractable. Maximum likelihood estimates $\hat{\boldsymbol{\theta}}$ of $\boldsymbol{\theta} \equiv (\boldsymbol{\beta}, \tau^2, \phi)'$ have been developed by simulating from the HM and simultaneously optimizing the likelihood (e.g., Geyer and Thompson, 1992; Lee, Kaiser, and Cressie, 2001). These estimates can then be substituted into the EHM's posterior distribution. For example, the (squared-error-loss) optimal predictor of the true rate $\exp\{Y(s_1)\}$ at location s_1 is $E(\exp\{Y(s_1)\}|\mathbf{Z}, \boldsymbol{\beta}, \tau^2, \phi)$, which is estimated by $E(\exp\{Y(s_1)\}|\mathbf{Z}, \hat{\boldsymbol{\beta}}, \hat{\tau}^2, \hat{\phi})$. Unfortunately, the posterior expectation is not known in closed form, and this is typical for all HMs that do not have a Gaussian data model and a Gaussian process model; Clayton and Kaldor (1987) and Breslow and Clayton (1993) give useful approximations.

Suppose that the parameter estimate $\hat{\boldsymbol{\theta}}$ is obtained from the data $\mathbf{Z}$ and substituted in place of $\boldsymbol{\theta}$ in the posterior distribution, resulting in $[\mathbf{Y}|\mathbf{Z}, \hat{\boldsymbol{\theta}}]$; see Section 2.1.2. Cressie (2000) discusses MCMC algorithms that produce realizations $\mathbf{Y}^{(1)}, \dots, \mathbf{Y}^{(L)}$ from $[\mathbf{Y}|\mathbf{Z}, \hat{\boldsymbol{\theta}}]$, where the process model in the HM is a MRF. Then, to finish the example introduced in the previous paragraph, the (estimated) optimal spatial predictor of the true rate at s_1 is approximated by

$$(1/L) \sum_{\ell=1}^L \exp\{Y(s_1)^{(\ell)}\},$$

and this approximation improves as L increases (Section 2.3).

In contrast, the BHM requires specification of another level, namely the parameter model. For example, consider the parameter model used by Stern and Cressie (2000) in a disease-mapping problem.

Parameter model: The joint distribution of θ is

$$[\theta] = [\beta][\tau^2][\phi],$$

where $\beta \sim U((-\infty, \infty)^p)$; $[\tau^2] \propto \exp(-b/\tau^2)$, for l a small positive number (e.g., $b = 0.01$); and $\phi \sim U(0, \phi_{\max})$, for $\phi_{\max}$ the largest parameter value (> 0) such that $(\mathbf{I} - \phi\mathbf{H})$ is positive-definite for $0 < \phi < \phi_{\max}$.

This particular parameter model is based on choosing noninformative, maximum-entropy prior distributions (e.g., Cressie, Richardson, and Jaussent, 2004). In problems where there is scientific theory to guide the behavior of parameters (e.g., the regression coefficients β), one should choose more informative priors.

Recall the example of predicting the true rate at $\mathbf{s}_1$. The BHM's (squared-error loss) optimal predictor is $E(\exp\{Y(\mathbf{s}_1)|\mathbf{Z}\})$, where the parameters are integrated out (rather than estimated, as they were for the EHM). That is,

$$E(\exp\{Y(\mathbf{s}_1)|\mathbf{Z}\}) = \int [\exp\{Y(\mathbf{s}_1)|\mathbf{Z}, \beta, \tau^2, \phi\} [\beta, \tau^2, \phi | \mathbf{Z}]] d\beta d\tau^2 d\phi,$$

although the integration is generally analytically intractable. Once again, MCMC is overwhelmingly the method of choice to compute the optimal predictor, this time including uncertainty on θ through the parameter model. Then $(\mathbf{Y}^{(1)}, \theta^{(1)}), \dots, (\mathbf{Y}^{(L)}, \theta^{(L)})$ are samples from the posterior distribution $[\mathbf{Y}, \theta | \mathbf{Z}]$, and the optimal spatial predictor of the true rate at $\mathbf{s}_1$ is approximated, as for the EHM, by

$$\frac{1}{L} \sum_{l=1}^L \exp\{Y(\mathbf{s}_1)^{(l)}\}.$$

Details of MCMC implementation for a BHM can be found in Banerjee, Carlin, and Gelfand (2004, Section A.7).

4.2.2 The Markov Random Field (MRF)

In this section, we define a class of spatial models known as Markov random fields, with the intention of using them as process models in an HM. For temporal data or data in $D_s \subset \mathbb{R}^1$, the Markov models discussed at the beginning of Section 4.2 have been a popular way of introducing statistical dependence. Such models assume a complete ordering of D_s ; then the “present,” conditioned on the “past,” is assumed to depend only on the “immediate past.” Financial time series, such as quarterly trade balances, are often modeled with a component that is Markov (superimposed on a seasonal component and other large-scale effects); for that component, only its immediate-past value influences the present value.

For spatial data, an analogous notion can be formulated that does not rely on a complete ordering of the spatial locations, but it is based on which locations are “neighbors” of any particular location (Besag, 1974; Cressie, 1993, Section 6.4). For most of this section, we assume that the process $Y(\cdot)$ is defined on the lattice $D_s \equiv \{s_1, \dots, s_n\} \subset \mathbb{R}^d$. In general, $Y(\cdot)$ is characterized by its joint distribution,

$$[Y(s_1), \dots, Y(s_n)], \quad (4.98)$$

from which the conditional probabilities, $[Y(s_i) \mid \mathbf{Y}_{-i}]$; $i = 1, \dots, n$, can be calculated, where we recall that $\mathbf{Y}_{-i} \equiv (Y(s_1), \dots, Y(s_{i-1}), Y(s_{i+1}), \dots, Y(s_n))'$.

The *neighborhood* $N(s_i)$, of the i th location s_i , is formally defined to be the collection of all other locations s_j ($j \neq i$) such that

$$[Y(s_i) \mid \mathbf{Y}_{-i}] = [Y(s_i) \mid \mathbf{Y}(N(s_i))], \quad i = 1, \dots, n, \quad (4.99)$$

where $\mathbf{Y}(N(s_i)) \equiv (Y(\mathbf{u}) : \mathbf{u} \in N(s_i))'$. For example, suppose that the right-hand side of (4.99) is a Poisson distribution,

$$Y(s_i) \mid \mathbf{Y}(N(s_i)) \sim \text{Poi} \left(\exp \left\{ \alpha_i + \sum_{j \in N(s_i)} \theta_{ij} Y(s_j) \right\} \right), \quad i = 1, \dots, n,$$

where $\theta_{ij} = \theta_{ji}$ (the reason for this symmetry in the parameters $\{\theta_{ij}\}$ will become clearer in Section 4.2.3). This is called the *auto Poisson model* (Besag, 1974).

In (4.99), if $s_j \in N(s_i)$ then s_j is said to be a *neighbor* of s_i , and it is straightforward to show that if s_j is a neighbor of s_i , then s_i is a neighbor of s_j . In words, (4.99) says that the conditional distribution of the Y at the i th location, conditioned on Y at all other locations, in fact depends only on its neighboring values $\mathbf{Y}(N(s_i))$. In this sense, (4.99) is a spatial analogue of the temporal Markov property (4.90).

There is a way to represent the conditional dependencies in (4.99) using graph theory. An *undirected graph* is made up of vertices and (undirected) edges; we define an edge between vertices s_i and s_j to mean that $s_j \in N(s_i)$. Figure 4.10 shows the nodes and edges of an undirected graph for a simple example, where $n = 7$. Although a graphical representation shows the conditional *independencies* in a statistical model, it does not give the *type and strength of the dependencies*. Moreover, there are other ways to represent spatial dependence, through the covariance function or variogram (Section 4.1.1), or through a spatial moving average (Section 4.1.4). Further discussion of directed and undirected graphs can be found in Section 2.4.

Use of the spatial Markov property can be found in statistical image analysis (e.g., Geman and Geman, 1984) and small area estimation, particularly disease

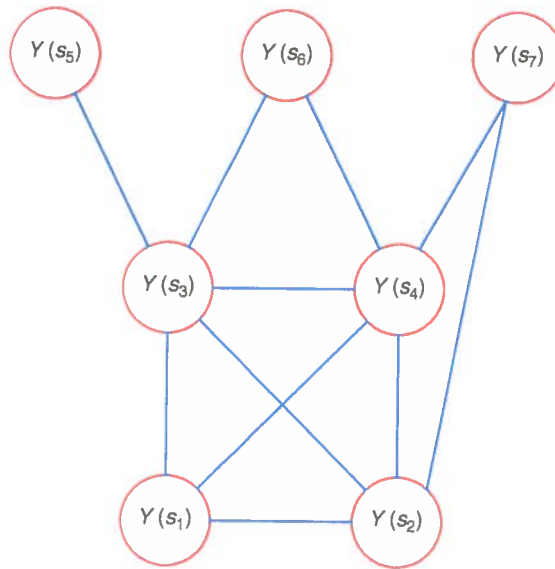


Figure 4.10 Undirected graph with seven vertices, showing the (undirected) edges between the vertices (this is the moral graph of Figure 4.14).

mapping (e.g., Besag, York, and Mollié, 1991). In image analysis, the spatial lattice D_s is typically a regular rectangular lattice of pixels; Geman and Geman (1984) proposed models based on (4.99) for the process model of (the hidden image) $\mathbf{Y}$ when the noisy image $\mathbf{Z}$ is observed. In small area estimation, the lattice is irregular and made up of location identifiers (e.g., centroids) of the small areas; for example, Besag, York, and Mollié (1991) used models based on (4.99) for the process model of (hidden) disease risks $\mathbf{Y}$ when (noisy) disease rates $\mathbf{Z}$ are observed in the Départements of France. More details on small area estimation are given in Section 4.2.6.

When the conditional probabilities on the right-hand side of (4.99) define the joint probability (4.98), the process $\{Y(s) : s \in D_s\}$ is called a *Markov random field (MRF)*. This definition can be appropriately generalized for countable D_s ; for example, see discussion of the Ising model in Section 4.2.4. A very attractive feature of working with an MRF is that modeling can be carried out at the *local* level by specifying the neighborhoods,

$$\{N(s_i) : i = 1, \dots, n\}, \quad (4.100)$$

and the conditional probabilities,

$$\{[Y(s_i) \mid \mathbf{Y}(N(s_i))]: i = 1, \dots, n\}. \quad (4.101)$$

However, such specifications have to be made consistently, so that at a *global* level the joint distribution (4.98) is well defined.

Much of the material presented in this section follows closely the notation and development of Besag (1974, Section 3) and Cressie (1993, Chapter 6). Define the support of the probability distribution of $\mathbf{Y}$ to be $\Omega \equiv \{\mathbf{y} : \Pr(\mathbf{y}) > 0\}$, where $\Omega \subset D_s$ and $\mathbf{y} \equiv (y(s_1), \dots, y(s_n))' \in \mathbb{R}^n$. Furthermore, define $\Omega_i \equiv \{y(s_i) : \Pr(y(s_i)) > 0\}$; $i = 1, \dots, n$. Note that here we use $\Pr(\cdot)$ to denote either a probability density or a probability mass function.

The *positivity condition* is said to be satisfied if $\Omega = \Omega_1 \times \dots \times \Omega_n$, which says that if each $y(s_i)$ individually has positive probability, then $\mathbf{y}$ has positive probability. This allows results given below to be proved simply, although in Section 4.2.3 we shall assume a much weaker condition (the MRF support condition of Kaiser and Cressie, 2000) and obtain similar results.

A result in Besag (1974), which we call Besag's Lemma, relates joint probabilities to conditional probabilities:

$$\frac{\Pr(\mathbf{y})}{\Pr(\mathbf{w})} = \prod_{i=1}^n \frac{\Pr(y(s_i) | y(s_1), \dots, y(s_{i-1}), w(s_{i+1}), \dots, w(s_n))}{\Pr(w(s_i) | y(s_1), \dots, y(s_{i-1}), w(s_{i+1}), \dots, w(s_n))},$$

$\mathbf{y}, \mathbf{w} \in \Omega. \quad (4.102)$

For example, if $n = 3$, the right-hand side of (4.102) is

$$\frac{\Pr(y(s_1) | w(s_2), w(s_3))}{\Pr(w(s_1) | w(s_2), w(s_3))} \cdot \frac{\Pr(y(s_2) | y(s_1), w(s_3))}{\Pr(w(s_2) | y(s_1), w(s_3))} \cdot \frac{\Pr(y(s_3) | y(s_1), y(s_2))}{\Pr(w(s_3) | y(s_1), y(s_2))}.$$

It is a simple matter to verify that this is equal to the left-hand side of (4.102).

The relation (4.102) has sometimes been called Brook's Lemma, where the article by Brook (1964) is cited. However, after close inspection of that article and that of Besag (1974), it would appear that the result should be properly credited to Besag. Notice that the right-hand side of (4.102) depends on the ordering of the locations, but the left-hand side does not. Thus, specification of the conditional probabilities in (4.99) such that a joint distribution exists, is delicate. If it can be done coherently, then the joint probability is available from (4.102), up to a multiplicative normalizing constant.

In the rest of this section, we address how a Markov random field can be specified and discuss its properties. Recall from (4.99) the definition of neighbor and neighborhood. A *clique* is defined to be a set of locations in D_s that consists either of a single site or of a set of sites that are *all* neighbors of each other. For example, if $n = 3$, $s_1 \in N(s_2)$, $s_1 \in N(s_3)$, but $s_2 \notin N(s_3)$, then the cliques are $\{s_1\}$, $\{s_2\}$, $\{s_3\}$, $\{s_1, s_2\}$, and $\{s_1, s_3\}$. Cliques are central to a particular representation of $[\mathbf{Y}]$ given by the Hammersley–Clifford Theorem, which is stated below.

In what follows, we assume that $\mathbf{0} \in \Omega$. This makes some of the algebra below easier, but Kaiser and Cressie (2000) show how the results based on a possibly *nonzero* reference value, $y_0 \in \Omega$, follow in a similar manner.

Define

$$Q(\mathbf{y}) \equiv \log\{\Pr(\mathbf{y})/\Pr(\mathbf{0})\}, \quad \mathbf{y} \in \Omega, \quad (4.102)$$

and call $Q(\cdot)$ the *negpotential function*. Note that (4.103) involves *ratio* of probabilities. For example, if $\mathbf{Y} \sim \text{Gau}(\mathbf{0}, \Sigma)$, then $Q(\mathbf{y}) = -(1/2)\mathbf{y}'\Sigma^{-1}\mathbf{y}$; $\mathbf{y} \in \mathbb{R}^n$. For discrete data, this yields

$$\Pr(\mathbf{y}) = \exp\{Q(\mathbf{y})\} / \sum_{\mathbf{x} \in \Omega} \exp\{Q(\mathbf{x})\}, \quad \mathbf{y} \in \Omega, \quad (4.104)$$

where the sum exists and is finite because $\{\Pr(\mathbf{y}) : \mathbf{y} \in \Omega\}$ is a probability mass function that sums to 1. For continuous data, the summation in the denominator of (4.104) is replaced with integration. Then, for the example above, it is a simple matter to use $Q(\mathbf{y}) = -(1/2)\mathbf{y}'\Sigma^{-1}\mathbf{y}$, to recover the Gaussian density $\Pr(\mathbf{y})$, with mean $\mathbf{0}$ and covariance matrix Σ .

Let $\mathbf{j}_m \equiv (j(1), \dots, j(m))'$ denote $1 \leq m \leq n$ distinct index values from $\{1, 2, \dots, n\}$. Define the sets

$$T_m(p) \equiv \{\text{all distinct } p\text{-tuples formed from } \mathbf{j}_m\}, \quad 1 \leq p \leq m.$$

Then for all $m = 1, \dots, n$ and all $\mathbf{j}_m$, define the so-called *G-functions* through

$$\begin{aligned} & y(\mathbf{s}_{j(1)}) \dots y(\mathbf{s}_{j(m)}) G_{j(1), \dots, j(m)}(y(\mathbf{s}_{j(1)}), \dots, y(\mathbf{s}_{j(m)})) \\ & \equiv \sum_{t=0}^{m-1} \sum_{\mathbf{j}_{m-t} \in T_m(m-t)} (-1)^t Q(\mathbf{y} : y(\mathbf{s}_h) = 0; h \notin \mathbf{j}_{m-t}), \end{aligned} \quad (4.105)$$

where all arguments at which $Q(\cdot)$ is evaluated are assumed to be in the support Ω . For example, $y(\mathbf{s}_i)G_i(y(\mathbf{s}_i)) \equiv Q(0, \dots, 0, y(\mathbf{s}_i), 0, \dots, 0)$ and

$$\begin{aligned} y(\mathbf{s}_i)y(\mathbf{s}_j)G_{ij}(y(\mathbf{s}_i), y(\mathbf{s}_j)) & \equiv Q(0, \dots, 0, y(\mathbf{s}_i), 0, \dots, 0, y(\mathbf{s}_j), 0, \dots, 0) \\ & - Q(0, \dots, 0, y(\mathbf{s}_i), 0, \dots, 0, 0, \dots, 0) \\ & - Q(0, \dots, 0, 0, 0, \dots, 0, y(\mathbf{s}_j), 0, \dots, 0). \end{aligned} \quad (4.106)$$

Consequently, $Q(\cdot)$ can be expanded in terms of a subset of the *G-functions* (e.g., Cressie, 1993, Section 6.4.1). For any $\mathbf{y} \in \Omega$,

$$\begin{aligned} Q(\mathbf{y}) &= \sum_{i=1}^n y(\mathbf{s}_i)G_i(y(\mathbf{s}_i)) + \sum_{1 \leq i < j \leq n} y(\mathbf{s}_i)y(\mathbf{s}_j)G_{ij}(y(\mathbf{s}_i), y(\mathbf{s}_j)) \\ &+ \dots + y(\mathbf{s}_1)y(\mathbf{s}_2) \dots y(\mathbf{s}_n)G_{12 \dots n}(y(\mathbf{s}_1), y(\mathbf{s}_2), \dots, y(\mathbf{s}_n)). \end{aligned} \quad (4.107)$$

Relations (4.105) and (4.107) are analogous to the Mobius inversion formulas (e.g., Lauritzen, 1996, p. 239). Observe that (4.105) defines G -functions $G_{j(1), \dots, j(m)}(\cdot)$, for any $1 \leq m \leq n$ and any $j(1), \dots, j(m) \in \{1, \dots, n\}$, although only some of these functions appear in the expansion (4.107). For example, while (4.105) results in a valid definition for $G_{1,2,3}(\cdot)$, $G_{2,1,3}(\cdot)$, and so forth, only $G_{1,2,3}(\cdot)$ appears in (4.107). The joint probability $\{\Pr(\mathbf{y}): \mathbf{y} \in \Omega\}$, and hence $\{Q(\mathbf{y}): \mathbf{y} \in \Omega\}$, is *invariant to permutation of the indices*; consequently, the functions defined in (4.105) are invariant to permutation of the indices contained in each $\mathbf{j}_m$. It will be seen in Section 4.2.3 that this necessary condition, namely permutation invariance, is an important sufficient condition for construction of the joint probability from the conditional probabilities, via the G -functions.

The expression (4.107) is very general and, as yet, does not involve any neighborhood information. That appears in the Hammersley–Clifford Theorem, which says that, depending on the cliques (which depend on the neighborhoods $\{N(\mathbf{s}_i): i = 1, \dots, n\}$) of the MRF, many of the G -functions are in fact *zero*.

Hammersley–Clifford Theorem (Hammersley and Clifford, 1971). Suppose that the random process on Ω is a Markov random field with neighborhood structure $\{N(\mathbf{s}_i): i = 1, \dots, n\}$. Then the negpotential function Q given by (4.107) satisfies the following property:

If sites $i, j(2), \dots, j(m)$ do not form a clique, then $G_{i,j(2), \dots, j(m)}(\cdot) \equiv 0$.

Another result is that the G -functions in the expansion (4.107) can be expressed in terms of conditional probabilities. Cressie (1993, Section 6.4.1), shows that for $i = 1, \dots, n$,

$$y(\mathbf{s}_i)G_i(y(\mathbf{s}_i)) = \log \left[\frac{\Pr(y(\mathbf{s}_i) \mid \{0(\mathbf{s}_j): j \neq i\})}{\Pr(0(\mathbf{s}_i) \mid \{0(\mathbf{s}_j): j \neq i\})} \right], \quad (4.108)$$

where $0(\mathbf{s}_i)$ is shorthand to denote evaluation at $y(\mathbf{s}_i) = 0$; and

$$\begin{aligned} y(\mathbf{s}_i)y(\mathbf{s}_j)G_{ij}(y(\mathbf{s}_i), y(\mathbf{s}_j)) &= \log \left[\frac{\Pr(y(\mathbf{s}_i) \mid y(\mathbf{s}_j), \{0(\mathbf{s}_k): k \neq i, j\})}{\Pr(0(\mathbf{s}_i) \mid y(\mathbf{s}_j), \{0(\mathbf{s}_k): k \neq i, j\})} \right] \\ &\quad - \log \left[\frac{\Pr(y(\mathbf{s}_i) \mid \{0(\mathbf{s}_k): k \neq i\})}{\Pr(0(\mathbf{s}_i) \mid \{0(\mathbf{s}_k): k \neq i\})} \right]. \end{aligned} \quad (4.109)$$

The general expression for higher-order terms is given in Section 4.2.3.

There is a large class of models defined by truncating (4.107), after including the singleton and pairwise G -functions. This means only (4.108) and (4.109) are needed. Consider the *auto spatial models* of Besag (1974): For $i = 1, \dots, n$,

$$[Y(\mathbf{s}_i) | \mathbf{Y}_{-i}] \equiv \exp\{A_i(\mathbf{Y}_{-i})B_i(Y(\mathbf{s}_i)) + C_i(Y(\mathbf{s}_i)) + D_i(\mathbf{Y}_{-i})\}, \quad (4.110)$$

which are of exponential-family form. If

$$A_i(\mathbf{Y}_{-i}) = \alpha_i + \sum_{j=1}^n \theta_{ij}\{B_j(Y(\mathbf{s}_j)) - B_j(0)\}, \quad i = 1, \dots, n, \quad (4.111)$$

where $\theta_{ii} = 0$ and $\theta_{ij} = \theta_{ji}$, we see from (4.107) that, up to an additive constant,

$$\begin{aligned} Q(\mathbf{y}) = & \sum_{i=1}^n [\alpha_i \{B_i(y(\mathbf{s}_i)) - B_i(0)\} + C_i(y(\mathbf{s}_i))] \\ & + \sum_{1 \leq i < j \leq n} \theta_{ij} \{B_i(y(\mathbf{s}_i)) - B_i(0)\} \{B_j(y(\mathbf{s}_j)) - B_j(0)\}. \end{aligned} \quad (4.112)$$

That is, the auto spatial models given by (4.110) imply that $Q(\cdot)$ is truncated. Furthermore, we can specify $N(\mathbf{s}_i)$ in (4.110) and (4.111) conveniently by

$$N(\mathbf{s}_i) = \{\mathbf{s}_j : \theta_{ij} \neq 0\}; \quad (4.113)$$

hence $\{\theta_{ij}\}$ have a direct interpretation as spatial-dependence parameters.

For example, recall the *auto Poisson* model: For $i = 1, \dots, n$, we have

$$\begin{aligned} [Y(\mathbf{s}_i) | \mathbf{Y}(N(\mathbf{s}_i))] \\ = \exp\{(\alpha_i + \sum_{j=1}^n \theta_{ij} Y(\mathbf{s}_j)) Y(\mathbf{s}_i) - \log Y(\mathbf{s}_i)! - (\alpha_i + \sum_{j=1}^n \theta_{ij} Y(\mathbf{s}_j))\}, \end{aligned} \quad (4.114)$$

where $\theta_{ii} = 0$, $\theta_{ij} = \theta_{ji}$, and $\theta_{ik} = 0$ for $\mathbf{s}_k \notin N(\mathbf{s}_i)$. Then (4.112) becomes, up to an additive constant,

$$Q(\mathbf{y}) = \sum_{i=1}^n \{\alpha_i y(\mathbf{s}_i) - \log y(\mathbf{s}_i)!\} + \sum_{1 \leq i < j \leq n} \theta_{ij} y(\mathbf{s}_i) y(\mathbf{s}_j). \quad (4.115)$$

Recall that a necessary condition for (4.107) to define a joint probability distribution, is invariance of (4.105) to permutation of the

indices. When $Q(\cdot)$ is truncated, this amounts to $y(\mathbf{s}_i)y(\mathbf{s}_j)G_{ij}(y(\mathbf{s}_i), y(\mathbf{s}_j)) = y(\mathbf{s}_j)y(\mathbf{s}_i)G_{ji}(y(\mathbf{s}_j), y(\mathbf{s}_i))$; for the auto spatial model (4.110), this amounts to

$$\begin{aligned} & \theta_{ij}\{B_i(y(\mathbf{s}_i)) - B_i(0)\}\{B_j(y(\mathbf{s}_j)) - B_j(0)\} \\ & = \theta_{ji}\{B_j(y(\mathbf{s}_j)) - B_j(0)\}\{B_i(y(\mathbf{s}_i)) - B_i(0)\}, \end{aligned}$$

which is satisfied because we assumed $\theta_{ij} = \theta_{ji}$.

Another necessary condition for (4.107) to define a joint probability distribution is that $\exp(Q(\cdot))$ integrates (or sums) to 1 over Ω ; see (4.104). This may not always hold, as in the case of the auto Poisson given by (4.114): Clearly, if $\theta_{12} > 0$ (say) in (4.115), then $\exp(Q(\cdot))$ is not summable; that is, the auto Poisson model is defined only if all $\{\theta_{ij}\}$ are negative or zero (Besag, 1974).

The two necessary conditions of *permutation invariance* and *summability* turn out to be crucial for specification of a Markov random field from conditional probabilities. The full statement of this result is deferred to Section 4.2.3, where it is presented in its most general form.

In summary, the joint probability of $\mathbf{Y}$ yields $Q(\cdot)$, which yields the G -functions, which can be written in terms of the conditional probabilities. Since MRFs are defined locally (i.e., in terms of the conditional probabilities), our modeling strategy is based on having *checkable conditions* that allow us to go backwards through the sequence, from conditional probabilities, to G -functions, to $Q(\cdot)$, to the joint probability $\Pr(\cdot)$.

4.2.3 From the Conditional Probabilities to the Joint Probability

Notice that up to now we have *assumed* that a joint probability distribution (i.e., a MRF) for $\mathbf{Y}$ *exists* (e.g., in the Hammersley–Clifford Theorem), and we have derived properties of the MRF based on the conditional probabilities. We now consider the more difficult problem of finding the joint probability distribution (if it exists) from the conditional probability distributions used to define the MRF. Our approach is constructive and follows the presentation by Kaiser and Cressie (2000).

The derivation of the expression (4.109) involves an arbitrary choice of the i th site or the j th site to be featured in the (conditional) probability statement. Because the indices $1, 2, \dots, n$ are only used as convenient labels, it is clear that the conditional-probability consistency conditions must include a permutation-invariance condition. We shall see that other conditions involve summability (integrability) and supports of probability distributions.

The positivity condition, $\Omega = \Omega_1 \times \dots \times \Omega_n$, is sufficient to ensure that all of the arguments at which $Q(\cdot)$ is evaluated in (4.105) are contained in Ω . However, it is not necessary, as evidenced by examples given in Mousouris (1974). Let $\Omega_{j(1), \dots, j(m)}$ be the support of the marginal probability, $\Pr(y(\mathbf{s}_{j(1)}), \dots, y(\mathbf{s}_{j(m)}))$, and define $\Phi_i \equiv \Omega_{\{1, \dots, n\} \setminus \{i\}}$ to be the support of the marginal probability, $\Pr(\mathbf{y}_{-i})$. Then the *MRF support condition* (Kaiser

and Cressie, 2000) is defined as

$$\{0(\mathbf{s}_i)\} \times \Phi_i \subset \Omega, \quad i = 1, \dots, n, \quad (4.116)$$

where recall that $0(\mathbf{s}_i)$ denotes a value of zero at location $\mathbf{s}_i$. This condition is weaker than the positivity condition. A weaker condition yet is given by Besag (1994, Lemma 0.1). However, the MRF support condition is precisely what is needed to *construct* the joint probability from G -functions (which are defined in terms of the conditional probabilities).

Recall (4.108) and (4.109), expressions for the singleton and pairwise G -functions, respectively, in terms of the conditional probabilities. There is a general expression, which is established by Kaiser and Cressie (2000). Recall that $\mathbf{j}_m = (j(1), j(2), \dots, j(m))'$ is an index of distinct integers, where $1 \leq m \leq n$. Without loss of generality, let $j(1) = i$ for some $i \in \{1, \dots, n\}$. Also, define $\mathbf{j}_m^{(-i)} \equiv (j(2), \dots, j(m))'$, and let $T_m^{(-i)}(p)$ be the set of all distinct p -tuples formed from $\mathbf{j}_m^{(-i)}$, $1 \leq p \leq m-1$. For $1 \leq m \leq n$, define

$$\begin{aligned} & y(\mathbf{s}_i) \dots y(\mathbf{s}_{j(m)}) G_{i, \dots, j(m)}(y(\mathbf{s}_i), \dots, y(\mathbf{s}_{j(m)})) \\ & \equiv \sum_{t=1}^{m-1} \sum_{\mathbf{j}_{m-t}^{(-i)} \in T_m^{(-i)}(m-t)} \left[(-1)^{t-1} \log \left[\frac{\Pr(y(\mathbf{s}_i) | \{y(\mathbf{s}_h) : h \in \mathbf{j}_{m-t}^{(-i)}\}, \{0(\mathbf{s}_k) : k \notin \mathbf{j}_{m-t}\})}{\Pr(0(\mathbf{s}_i) | \{y(\mathbf{s}_h) : h \in \mathbf{j}_{m-t}^{(-i)}\}, \{0(\mathbf{s}_k) : k \notin \mathbf{j}_{m-t}\})} \right] \right. \\ & \quad \left. + (-1)^{m-1} \log \left[\frac{\Pr(y(\mathbf{s}_i) | \{0(\mathbf{s}_k) : k \neq i\})}{\Pr(0(\mathbf{s}_i) | \{0(\mathbf{s}_k) : k \neq i\})} \right] \right], \end{aligned} \quad (4.117)$$

if $m \geq 2$; and if $m = 1$, only the last term in (4.117) appears. This definition is valid for all $\mathbf{j}_m$, but we use it in $Q(\cdot)$ given by (4.107) for ordered indices, $i < j(2) < \dots < j(m)$.

Kaiser and Cressie's construction gives conditions under which the relations (4.117), followed by (4.107), followed by (4.104), yield a joint probability whose conditional probabilities are the ones that are used to initiate the construction.

Theorem (Kaiser and Cressie, 2000). Assume that a set of conditional distributions, $\{[Y(\mathbf{s}_i) | \mathbf{Y}(N(\mathbf{s}_i))]: i = 1, \dots, n\}$, have been specified and that the MRF support condition (4.116) holds. Then a joint probability, $[\mathbf{Y}]$, having those conditional distributions exists, and it may be specified up to a normalizing constant through the application of (4.117), (4.107), and (4.104), if and only if

- (i) The G -functions constructed from the specified conditional probabilities using (4.117) are each invariant under permutation of their associated indices.

- (ii) The negpotential function $Q(\cdot)$ constructed from the G -functions using (4.107) and (4.117), and defined on support $\Omega \equiv \{\mathbf{y}: \exp\{Q(\mathbf{y})\} > 0\}$, satisfies the summability (or more generally a Lebesgue–Stieltjes integrability) condition,

$$\sum_{t \in \Omega} \exp\{Q(t)\} < \infty. \quad (4.118)$$

Several observations about this theorem will help in making model choices for specifying an MRF. The permutation-invariance condition of (i) is straightforward to check for any particular model specification. For example, the auto spatial models (Sections 4.2.2 and 4.2.4) defined through the one-parameter exponential family of conditional distributions, $[Y(\mathbf{s}_i)|\mathbf{Y}(N(\mathbf{s}_i))]$, given by (4.110) and (4.111), satisfy the permutation-invariance condition if $\theta_{ij} = \theta_{ji}$; $i, j = 1, \dots, n$. More generally, Lee, Kaiser, and Cressie (2001) give a generalization of MRF auto spatial models that allows for nonzero G -functions for cliques of any size; those models satisfy (i). We have already seen that the auto Poisson requires all $\{\theta_{ij}\}$ to be negative or zero in order for (ii) to be satisfied.

4.2.4 Some Examples of Auto Spatial Models

We have already considered two examples of auto spatial models. The *auto Gaussian* model, more commonly called a conditional autoregressive (CAR) model, is given by (4.93) and (4.94); discussion of CAR models will be deferred to Section 4.2.5. The *auto Poisson* model for count data is given by (4.114).

When data are *binary* (taking values 0 and 1) and $G_A \equiv 0$ for $|A| \geq 3$, where A is an index set of ordered integers in $\{1, \dots, n\}$, then Besag (1974) shows that for $i = 1, \dots, n$,

$$[Y(\mathbf{s}_i)|\mathbf{Y}(N(\mathbf{s}_i))] = \frac{\exp\{\alpha_i y(\mathbf{s}_i) + \sum_{s_j \in N(\mathbf{s}_i)} \theta_{ij} y(\mathbf{s}_i) y(\mathbf{s}_j)\}}{1 + \exp\{\alpha_i + \sum_{s_j \in N(\mathbf{s}_i)} \theta_{ij} y(\mathbf{s}_j)\}}, \quad y(\mathbf{s}_i) \in \{0, 1\}, \quad (4.119)$$

where $\theta_{ij} = \theta_{ji}$. This spatial model is often called an *auto logistic model*. Statistical independence in the model is achieved by putting $\theta_{ij} \equiv 0$. Furthermore, any large-scale inhomogeneities can be modeled via $\alpha \equiv (\alpha_1, \dots, \alpha_n)'$. For example, put $\alpha = \mathbf{X}\beta$, where $\mathbf{X}$ is an $n \times p$ matrix of known explanatory variables and β is a p -dimensional vector of unknown regression coefficients. When $\alpha = (\alpha, \dots, \alpha)'$, the model is homogeneous in large-scale effects, leaving the spatial-dependence parameters $\{\theta_{ij}\}$ to model any spatial inhomogeneities.

The *Ising model* is defined by (4.119) on a regular lattice with locations given by the countable set $D_s = \{(u, v): u, v = \dots, -1, 0, 1, \dots\} \subset \mathbb{R}^2$;

assume homogeneity of $\{\alpha(u, v)\}$ and assume a “nearest-neighbor” neighborhood structure, $N(u, v) = \{(u - 1, v), (u + 1, v), (u, v - 1), (u, v + 1)\}$. Thus, the numerator of (4.119) becomes

$$\begin{aligned} \exp[y(u, v)\{\alpha + \gamma_1(y(u - 1, v) + y(u + 1, v)) \\ + \gamma_2(y(u, v - 1) + y(u, v + 1))\}], \end{aligned}$$

Where γ_1 and γ_2 represent a reparameterization. The model was originally presented by Ising (1925) under the added assumption that $\gamma_1 = \gamma_2$. Because the lattice D_s is countably infinite, a delicate mathematical question remains as to whether (4.119) yields a well defined probability measure for the countably infinite set, D_s . Indeed, there may be regions of the parameter space where two different probability measures give rise to the *same* conditional probabilities, (4.119) (physicists call this phenomenon *phase transition*). Furthermore, suppose that $\gamma_1 = \gamma_2 = \gamma$; then critical values of γ exist, below which the limiting nearest-neighbor correlation is zero, but above which the process exhibits long-range dependence. These issues are discussed by, for example, Pickard (1987).

Other MRFs based on (4.110) and (4.111) are the *auto gamma model* and the *auto binomial model*. The *Winsorized auto Poisson model* (Kaiser and Cressie, 1997) is a modification of the auto Poisson that allows positive spatial dependence; see Møller and Rubak (2010) for a different lattice process that achieves the same result. The *auto beta model* (Kaiser and Cressie, 2000) comes from a bivariate exponential family of conditional distributions applied to the joint-distribution construction given in Section 4.2.3. Arguably, the most useful of all MRFs is the auto Gaussian model, or *CAR model*. [Cressie (1993), called it a conditionally specified Gaussian model, to contrast it with a simultaneously specified Gaussian model, but the terminology of CAR and SAR models, respectively, remains popular.]

4.2.5 The CAR Model

When data are continuous, a natural model to choose for the conditional distributions $\{[Y(\mathbf{s}_i) | \mathbf{Y}(N(\mathbf{s}_i))]: i = 1, \dots, n\}$ given by (4.110) and (4.111) is the Gaussian (or normal) distribution. For $i = 1, \dots, n$, we have

$$\Pr(y(\mathbf{s}_i) | \mathbf{y}(N(\mathbf{s}_i))) = (2\pi\tau_i^2)^{-1/2} \exp[-\{y(\mathbf{s}_i) - \theta_i\}^2 / 2\tau_i^2], \quad (4.120)$$

where it is understood that θ_i is a function of $\mathbf{y}(N(\mathbf{s}_i))$ and that the left-hand side is the conditional density function of $Y(\mathbf{s}_i)$, given $\mathbf{Y}(N(\mathbf{s}_i)) = \mathbf{y}(N(\mathbf{s}_i))$. In (4.120),

$$\begin{aligned} \theta_i(\mathbf{y}(N(\mathbf{s}_i))) &= E(Y(\mathbf{s}_i) | \mathbf{y}(N(\mathbf{s}_i))) \\ &= \mu(\mathbf{s}_i) + \sum_{\mathbf{s}_j \in N(\mathbf{s}_i)} c_{ij}(y(\mathbf{s}_j) - \mu(\mathbf{s}_j)), \end{aligned} \quad (4.121)$$

and $\mu(s_i) \equiv E(Y(s_i))$. The general condition $\theta_{ij} = \theta_{ji}$ for auto spatial models (4.111), applied to (4.120) and (4.121), implies that $c_{ij}\tau_j^2 = c_{ji}\tau_i^2$. Furthermore, $c_{ii} = 0$, and $c_{ik} = 0$ for $s_k \notin N(s_i)$; $i, j = 1, \dots, n$.

Let $\mathbf{C} = (c_{ij})$ and $\mathbf{M} = \text{diag}(\tau_1^2, \dots, \tau_n^2)$. Then Besag (1974) shows that, provided $(\mathbf{I} - \mathbf{C})^{-1}\mathbf{M}$ is symmetric and positive-definite, the Gaussian conditional distributions (4.120) define a joint Gaussian distribution given by

$$\mathbf{Y} \sim \text{Gau}(\boldsymbol{\mu}, (\mathbf{I} - \mathbf{C})^{-1}\mathbf{M}). \quad (4.122)$$

The MRF given by (4.120) is an auto Gaussian model, but it is known more commonly as a *conditional autoregressive (CAR) model*.

The CAR model in particular, and MRFs in general, are an attractive way to handle spatial statistical dependencies. For each location s_i in the lattice D_s , a univariate (*conditional*) distribution is posited for $Y(s_i)$; $i = 1, \dots, n$. Thus, potentially complicated joint statistical dependencies are modeled simply through a collection of conditional dependencies. Because the CAR model is a Gaussian MRF, it is particularly appropriate to discuss the relationship between the CAR model and the geostatistical models described in Section 4.1.

On the lattice $D_s = \{s_1, \dots, s_n\}$, a geostatistical model specifies the mean and covariance matrix of $\mathbf{Y} \equiv (Y(s_1), \dots, Y(s_n))'$:

$$E(\mathbf{Y}) = \boldsymbol{\mu}_Y \quad \text{and} \quad \text{var}(\mathbf{Y}) = \boldsymbol{\Sigma}_Y. \quad (4.123)$$

Further specification of the geostatistical model often involves $E(Y(s_i)) \equiv \mu_Y(s_i) = \mathbf{x}(s_i)'\boldsymbol{\beta}$ (linear model) and $\text{cov}(Y(s_i), Y(s_j)) = (\boldsymbol{\Sigma}_Y)_{ij} = C_Y(s_j - s_i)$ (covariance stationarity). The intuitive notion of spatial dependence, that nearby values tend to be more alike than values far apart, is captured in the geostatistical model through

$$C_Y(s_j - s_i) \simeq 0 \quad (\text{or} = 0), \quad \text{for } \|s_j - s_i\| \text{ "large."} \quad (4.124)$$

The same intuitive notion is captured in the CAR model,

$$E(\mathbf{Y}) = \boldsymbol{\mu}_Y \quad \text{and} \quad \text{var}(\mathbf{Y}) = (\mathbf{I} - \mathbf{C})^{-1}\mathbf{M}, \quad (4.125)$$

through

$$c_{ij} \simeq 0 \quad (\text{or} = 0), \quad \text{for } \|s_j - s_i\| \text{ "large."} \quad (4.126)$$

That is, for geostatistical models, $\text{var}(\mathbf{Y})$ is "sparse," and for CAR models, $(\text{var}(\mathbf{Y}))^{-1}$ is "sparse."

Algebraically, there is a one-to-one relationship between a geostatistical model on D_s and a CAR model on D_s . Given a CAR model, $\text{Gau}(\boldsymbol{\mu}_Y, (\mathbf{I} - \mathbf{C})^{-1}\mathbf{M})$, the geostatistical model is $\text{Gau}(\boldsymbol{\mu}_Y, \boldsymbol{\Sigma}_Y)$, where $\boldsymbol{\Sigma}_Y \equiv (\mathbf{I} - \mathbf{C})^{-1}\mathbf{M}$. And, given a geostatistical model $\text{Gau}(\boldsymbol{\mu}_Y, \boldsymbol{\Sigma}_Y)$, define $(\boldsymbol{\Sigma}_Y)^{-1} \equiv (\sigma^{(ij)})$, $\mathbf{M} \equiv$

$\{\text{diag}(\sigma^{(11)}, \dots, \sigma^{(nn)})\}^{-1}$, and $\mathbf{C} \equiv \mathbf{I} - \mathbf{M}\Sigma_Y^{-1}$. Then $\mathbf{C}$ is an $n \times n$ matrix with zeros down its diagonal, $\mathbf{M}$ is a diagonal matrix with positive entries, and the CAR model is $\text{Gau}(\mu_Y, (\mathbf{I} - \mathbf{C})^{-1}\mathbf{M})$. The two models really are different because if the geostatistical model is sparse, the CAR model is typically not and *vice versa*.

Because the CAR model possesses computational advantages (taking the inverse of the covariance matrix of $\mathbf{Y}$ is trivial), it is often of interest to approximate a geostatistical model with a sparse CAR model. This problem can be formulated as minimizing a “distance” between Σ_Y (given) and $(\mathbf{I} - \mathbf{C})^{-1}\mathbf{M}$, with respect to $\mathbf{M}$ and a sparse $\mathbf{C}$. Rue and Held (2005, Chapter 5) discuss this problem in considerable detail.

When scientific interest centers on the large-scale effects, the idea is to use a few extra small-scale (spatial-dependence) parameters so that the large-scale parameters are estimated more efficiently. Suppose that $\mathbf{Y}$ is modeled according to a CAR model and that the large-scale variation is modeled as $E(\mathbf{Y}) = \mu_Y = \mathbf{X}\beta$, where recall that $\mathbf{X}$ is an $n \times p$ matrix of covariates and β is a p -dimensional vector of fixed (but unknown) regression coefficients. The classical Gaussian linear model is $\mathbf{Y} \sim \text{Gau}(\mathbf{X}\beta, \tau^2\mathbf{I})$. Compare this to (4.122) with $\mu = \mathbf{X}\beta$, which can be written equivalently as

$$\mathbf{Y} = \mathbf{X}\beta + \delta, \quad (4.127)$$

where the error δ is an n -dimensional random vector with distribution,

$$\delta \sim \text{Gau}(\mathbf{0}, (\mathbf{I} - \mathbf{C})^{-1}\mathbf{M}).$$

Notice that when $\mathbf{C} = \mathbf{0}$, an $n \times n$ matrix with all zero entries, and $\mathbf{M} = \tau^2\mathbf{I}$, the CAR (regression) model (4.127) reduces to the classical linear model with *iid* Gaussian errors.

When $\mathbf{M}$ and $\mathbf{C}$ are unknown, as is usually the case, they are often expressed in terms of a few (small-scale) parameters, at least one of which captures the spatial dependence; for example, we use $\mathbf{C} = \phi\mathbf{H}$ in Section 4.2.6. The parameter space of $\mathbf{C}$ is $\{\mathbf{C}: (\mathbf{I} - \mathbf{C})^{-1}\mathbf{M} \text{ is symmetric and positive-definite}\}$. When $\mathbf{C} = \phi\mathbf{H} \equiv \phi \cdot (h_{ij})$ and $\tau_i^2 h_{ji} = \tau_j^2 h_{ij}$, the parameter space becomes

$$\{\phi: \phi_{\min} < \phi < \phi_{\max}\}, \quad (4.128)$$

where $\phi_{\min} \equiv (\lambda_{(1)})^{-1} < 0$, $\phi_{\max} \equiv (\lambda_{(n)})^{-1} > 0$, and $\lambda_{(1)} \leq \dots \leq \lambda_{(n)}$ are the ordered eigenvalues of $\mathbf{M}^{-1/2}\mathbf{H}\mathbf{M}^{1/2}$ (Cressie, 1993, p. 559). Other parameterizations of $\mathbf{C}$ and $\mathbf{M}$ are given below and in the Bibliographic Notes in Section 4.5.

The inhomogeneity in the conditional variances, $\{\tau_i^2\}$, can often be modeled through $\mathbf{M} = \tau^2\mathbf{\Delta}$, where the $n \times n$ diagonal matrix $\mathbf{\Delta}$ is known and has positive diagonal entries. In that case, a simple transformation yields

a homogeneous CAR. Define $\tilde{\mathbf{Y}} \equiv \Delta^{-1/2}\mathbf{Y}$, $\tilde{\mathbf{X}} \equiv \Delta^{-1/2}\mathbf{X}$, $\tilde{\delta} \equiv \Delta^{-1/2}\delta$, and $\tilde{\mathbf{H}} \equiv \Delta^{-1/2}\mathbf{H}\Delta^{1/2}$. Then (4.127) is equivalent to

$$\tilde{\mathbf{Y}} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \tilde{\delta}, \quad (4.129)$$

where $\tilde{\delta} \sim \text{Gau}(\mathbf{0}, \tau^2(\mathbf{I} - \phi\tilde{\mathbf{H}}))$, and the full parameter space is $\{(\boldsymbol{\beta}', \tau^2, \phi) : \boldsymbol{\beta} \in \mathbb{R}^p, \tau^2 > 0, \text{ and } \phi \in (\phi_{\min}, \phi_{\max})\}$. The likelihood, $\mathcal{L}(\boldsymbol{\beta}, \tau^2, \phi)$, is most conveniently written in terms of the joint distribution of $\mathbf{Y}$, and it can be maximized via a combination of analytical and numerical procedures. As we demonstrate in Section 4.2.6, there are alternative ways to do inference on a CAR model that is part of an HM.

On the *regular countable* spatial lattice $\{(u, v) : u, v = \dots, -1, 0, 1, \dots\}$ in $\mathbb{R}^2$, spatial dependence in a constant-mean homogeneous CAR model is expressed through

$$\begin{aligned} E(Y(u, v) \mid \{y(k, l) : (k, l) \neq (u, v)\}) \\ = \mu + \sum_i \sum_j \gamma(i, j)(y(u - i, v - j) - \mu), \end{aligned}$$

where $\gamma(i, j) = \gamma(-i, -j)$ and $\gamma(0, 0) = 0$. The first-order dependence model is the special case, $\gamma(1, 0) = \gamma(-1, 0) \equiv \gamma_1$, $\gamma(0, 1) = \gamma(0, -1) \equiv \gamma_2$, and all other $\gamma(i, j)$'s are zero, which is similar to the Ising model (for binary data) given in Section 4.2.4. For this first-order dependence model, it can be seen that the correlation between two observations decays rather slowly with increasing distance between their lattice sites (Besag and Kooperberg, 1995).

For an *irregular finite* spatial lattice, the $\{c_{ij}\}$ in (4.121) could be modeled as some function of distance, $d_{ij} \equiv \|\mathbf{s}_i - \mathbf{s}_j\|$, between the locations $\mathbf{s}_i$ and $\mathbf{s}_j$. Note that, although $\mathbf{C} \equiv (c_{ij})$ depends only on $\{d_{ij} : 1 \leq i < j \leq n\}$, the covariance, $(\boldsymbol{\Sigma}_Y)_{ij} \equiv \text{cov}(Y(\mathbf{s}_i), Y(\mathbf{s}_j))$, does *not* in general depend on $\|\mathbf{s}_i - \mathbf{s}_j\|$ or, for that matter, on $\mathbf{s}_i - \mathbf{s}_j$. Hence an MRF on a finite spatial lattice is generally *not stationary*, even when its conditional-probability specification is homogeneous.

Care must be taken with parameterizing the spatial-dependence matrix $\mathbf{C}$, as we now demonstrate. It is not difficult to show that $\{\text{corr}(Y(\mathbf{s}_i), Y(\mathbf{s}_j) \mid \{y(\mathbf{s}_k) : k \neq i, j\})\}^2 = c_{ij}c_{ji}$, which implies that $0 \leq c_{ij}c_{ji} \leq 1$. Recall the parameterization, $\mathbf{C} = \phi\mathbf{H}$, where $\mathbf{H} \equiv (h_{ij})$ is a given $n \times n$ matrix whose diagonal elements are all zero. Let $\mathbf{A} \equiv (a_{ij})$ denote the $n \times n$ spatial neighborhood matrix such that $a_{ij} = 1$ if $\mathbf{s}_j \in N(\mathbf{s}_i)$; otherwise, $a_{ij} = 0$. Then the number of neighbors of $\mathbf{s}_i$ is $|N(\mathbf{s}_i)| = \sum_{j=1}^n a_{ij}$. Also, write $\mathbf{M} = \tau^2\mathbf{\Delta}$, where $\mathbf{\Delta}$ is a diagonal matrix whose diagonal entries are known and positive. Cressie and Kapat (2008) give special cases of this class of CAR models. The simplest model to write down is the *homogeneous CAR (HCAR)* model with $\mathbf{\Delta} = \mathbf{I}$ and $\mathbf{H} = \mathbf{A}$.

The *weighted CAR (WCAR)* model with $\mathbf{\Delta} = \text{diag}(|N(\mathbf{s}_1)|^{-1}, \dots, |N(\mathbf{s}_n)|^{-1})$ and $h_{ij} = a_{ij}|N(\mathbf{s}_i)|^{-1}$ was proposed by Besag, York, and Mollié (1991). The WCAR model has the property that the square of the partial

correlation between $Y(\mathbf{s}_i)$ and $Y(\mathbf{s}_j)$ is $\phi^2 |N(\mathbf{s}_i)|^{-1} |N(\mathbf{s}_j)|^{-1}$, for $\mathbf{s}_j \in N(\mathbf{s}_i)$. Hence, $\phi^2 \in (0, |N(\mathbf{s}_i)| |N(\mathbf{s}_j)|)$, illustrating that the spatial-dependence parameter ϕ *cannot* be interpreted in the same manner as one would interpret a spatial correlation between 0 and 1.

The following model (called the ACAR model below),

$$\Delta = \text{diag}(|N(\mathbf{s}_1)|^{-1}, \dots, |N(\mathbf{s}_n)|^{-1}); h_{ij} = a_{ij} |N(\mathbf{s}_j)|^{1/2} |N(\mathbf{s}_i)|^{-1/2}, \quad (4.130)$$

has partial correlation,

$$\text{corr}(Y(\mathbf{s}_i), Y(\mathbf{s}_j) | \{Y(\mathbf{s}_k) : k \neq i, j\}) = \phi.$$

This, and a more general result for multivariate CAR models, was given in Sain and Cressie (2007). We call this model the *autocorrelated CAR* (ACAR) model. For further models, see the Bibliographic Notes in Section 4.5, which include a class of diagonally dominant CAR models due to Pettitt, Weir, and Hart (2002).

Finally, an attractive property of the CAR model is that there is an analytical expression for its joint distribution. In general, the normalizing constant of an MRF, $\sum_{\mathbf{t} \in \Omega} \exp(Q(\mathbf{t}))$ (or $\int_{\Omega} \exp(Q(\mathbf{t})) d\mathbf{t}$) given by the left-hand side of (4.118), is analytically intractable. The CAR model given by (4.120) is an exception. Its negpotential function given by (4.107) simplifies, and

$$\int_{\Omega} \exp(Q(\mathbf{t})) d\mathbf{t} = (2\pi)^{n/2} \left(\prod_{i=1}^n \tau_i^2 \right)^{1/2} |\mathbf{I} - \mathbf{C}|^{-1/2}. \quad (4.131)$$

However, for n large, computation of the determinant, $|\mathbf{I} - \mathbf{C}|$, in (4.131) can be problematic. Further discussion of this aspect of the CAR model can be found in Cressie, Perrin, and Thomas-Agnan (2005).

4.2.6 Hierarchical Modeling on a Spatial Lattice: Small Area Estimation

Thus far, the lattice process has been defined on a regular or irregular spatial lattice, $D_s = \{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ in $\mathbb{R}^d$. In fact, it is possible to define MRFs (and SAR models, SMA models, and so forth) without knowing exactly the coordinates $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ of the process $\{Y(\mathbf{s}_i) : i = 1, \dots, n\}$; a crucial component of the spatial-dependence model is which elements of the matrix $\mathbf{H}$ in Section 4.2.5 (or $\mathbf{W}$ or $\mathbf{E}$ in Section 4.2.7 or 4.1.6, respectively) are zero. In the case of the CAR model (a MRF), nonzero elements of $\mathbf{H}$ determine the neighborhood sets, $\{N(\mathbf{s}_i) : i = 1, \dots, n\}$, upon which the theory of joint-to-conditional (Section 4.2.2) and conditional-to-joint (Section 4.2.3) probabilities is based. Generally, for any MRF, the only spatial information one needs to specify is which subsets of D_s constitute the neighborhood sets.

Let $A_1, \dots, A_n$ be disjoint sub-areas (henceforth called *small areas*) of $\mathbb{R}^d$ that are spatially proximate and $D_s = \cup_{i=1}^n A_i$; let $\mathbf{s}_1, \dots, \mathbf{s}_n$ denote a well

defined feature of the respective small areas, such as geometric centroid, political capital city, and so forth. Then $D_s \equiv \{s_1, \dots, s_n\}$ defines a spatial lattice, and neighborhoods can be defined using geographical properties; for example, the neighborhood of the i th small area could be defined to be those small areas with which it shares a common border:

$$N(s_i) \equiv \{s_j : A_j \text{ borders } A_i\}. \quad (4.132)$$

That is, spatial relationships can be defined in ways other than the Euclidean distances between $\{s_i : i = 1, \dots, n\}$. Indeed, Figure 4.10 shows a dependence relationship between nodes of a lattice that may or may not have a spatial context.

In this subsection, we shall use the problem of *small area estimation* to illustrate spatial modeling of data that have been aggregated up to small areas. Associated with each small area, A_i , is a datum $Z(s_i)$, but the datum itself is assumed to be an average (or a transformation of an average).

Example: Doctor-Prescription Amounts per Consultation

Cressie, Perrin, and Thomas-Agnan (2005) consider the average doctor-prescription amounts per consultation in cantons of the Midi-Pyrénées in southwest France; Kang, Liu, and Cressie (2009) take logs of the data and analyze them using an HM with a Gaussian data model and a Gaussian process model.

There were 268 cantons in the Midi-Pyrénées where average doctor-prescription amounts (per consultation) were reported for the period January 1, 1999–December 31, 1999. Let s_{ij} denote the location where the j th prescription was written within canton A_i ; $j = 1, \dots, M_i$, $i = 1, \dots, 268$. Denote those prescription amounts by $\{P_{ij} : j = 1, \dots, M_i; i = 1, \dots, 268\}$. As is typical with health-related data, aggregation (here over cantons) is carried out to preserve doctor and patient confidentiality. That is, the individual-location and individual-prescription data $\{(s_{ij}, P_{ij})\}$ are unavailable. Instead, we only know that in canton A_i , there were M_i prescriptions written, and the average prescription amount was,

$$P_i \equiv \sum_{j=1}^{M_i} P_{ij} / M_i, \quad i = 1, \dots, 268.$$

Furthermore, we have geographic information, such as a shape file of the cantons' boundaries and the locations of their centroids, $\{s_i : i = 1, \dots, 268\}$. Then we define the spatial lattice to be $D_s \equiv \{s_1, \dots, s_{268}\}$, with concomitant spatial (attribute) data $\{P_1, \dots, P_{268}\}$.

Kang, Liu, and Cressie (2009), *hereafter KLC*, find that a log transformation of the data gives an appropriate scale on which to fit CAR models (Section 4.2.5). They write $Z(s_i) = k \cdot \log(P_i)$, where k is an unspecified constant to preserve data confidentiality. They specify the data model as

$$Z(s_i) | Y(s_i), \sigma_\epsilon^2 \sim \text{ind. Gau}(Y(s_i), \sigma_\epsilon^2 / M_i), \quad i = 1, \dots, 268.$$

and the process model as

$$\mathbf{Y}|\boldsymbol{\beta}, \tau^2, \phi \sim \text{Gau}(\mathbf{X}\boldsymbol{\beta}, \tau^2(\mathbf{I} - \phi\mathbf{H})^{-1}\boldsymbol{\Delta}),$$

where $\mathbf{Y} \equiv (Y(\mathbf{s}_1), \dots, Y(\mathbf{s}_{268}))'$; $\mathbf{X}$ is a 268×3 matrix made up of three covariates (1, percentage of patients ≥ 70 years, N-S coordinate of centroid) for each of the 268 cantons; $\boldsymbol{\beta}$ is a three-dimensional vector of regression coefficients; $\boldsymbol{\Delta} \equiv \text{diag}(M_1^{-1}, \dots, M_{268}^{-1})$; $\mathbf{H}$ is a 268×268 matrix expressing the spatial dependence:

$$h_{ij} = \begin{cases} (M_j/M_i)^{1/2}, & \mathbf{s}_j \in N(\mathbf{s}_i), \\ 0, & \text{elsewhere;} \end{cases}$$

$N(\mathbf{s}_i)$ is the (neighborhood) set of all cantons within 30 km of canton i ; τ^2 is the process-variance parameter; and ϕ is the parameter expressing the amount of spatial dependence in the process, chosen so that the matrix inverse exists. This choice of CAR model has the same partial-correlation properties as the ACAR model given by (4.130); Cressie, Perrin, and Thomas-Agnan (2005) give a justification for it and call it the *spatial rates* model, which is discussed below, near (4.141).

The analysis of KLC involves fitting an EHM and a BHM and comparing them; for the EHM, σ_ϵ^2 is estimated via the method-of-moments, and $\boldsymbol{\beta}$, τ^2 , and ϕ are estimated via maximum likelihood estimation. Figure 4.11 shows the 268 cantons and their centroids in the Midi-Pyrénées region, with a choropleth map of the data (the average prescription amounts $\{P_i\}$) superimposed. The centroids are the same as those shown in Figure 4.9. There is a good deal of variability in the data, which KLC conclude is mostly due to the measurement error in the data model: on the log scale, $\hat{\sigma}_\epsilon^2 = 292.9$, which is to be compared to $\hat{\tau}^2 = 77.47$.

Figure 4.12 shows a choropleth map of the (estimated) posterior means:

$$E(g^{-1}(Y(\mathbf{s}_i))|\mathbf{Z}, \hat{\boldsymbol{\beta}}, \hat{\sigma}_\epsilon^2, \hat{\tau}^2, \hat{\phi}), \quad i = 1, \dots, 268,$$

where recall that $g(x) \propto \log x$, and $\mathbf{Z} \equiv (Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_{268}))'$ are the data on the log scale. The maps properly account for bias correction through log-Gaussian relationships (e.g., Cressie, 1993, p. 135). The degree of smoothing in these predicted doctor-prescription amounts (posterior means) is remarkable, and the smoothed map follows roughly the political boundaries of the eight Départements in the region, even though that information was *not* used in the HM.

Accompanying the map shown in Figure 4.12 is a map showing the variability (or uncertainty) associated with the predicted values. In Figure 4.13, we show a choropleth map of:

$$\{\text{var}(g^{-1}(Y(\mathbf{s}_i))|\mathbf{Z}, \hat{\boldsymbol{\beta}}, \hat{\sigma}_\epsilon^2, \hat{\tau}^2, \hat{\phi})\}^{1/2}, \quad i = 1, \dots, 268.$$

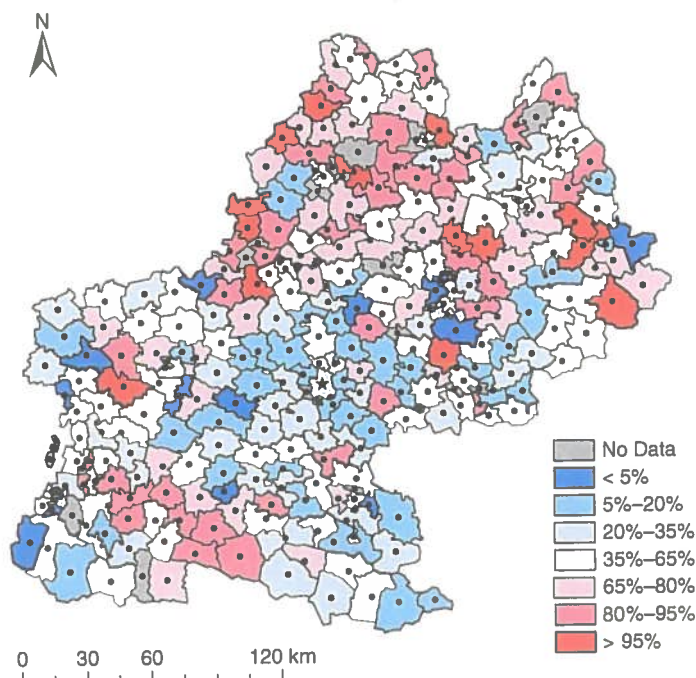


Figure 4.11 Choropleth map showing doctor-prescription amounts per consultation in the cantons of the Midi-Pyrénées (France). The “star” denotes the canton containing Toulouse. Percentiles used for shading are obtained from $\{P_i\}$.

On the original scale, predictions showing large average doctor-prescription amounts are accompanied by large variability (as expected). KLC also analyze the data using a BHM (with a prior on τ^2 and ϕ , and with $\hat{\beta}$ and $\hat{\sigma}_\varepsilon^2$ fixed). A sensitivity study to compare results of the EHM and BHM is carried out using the parameter model:

$$\tau^2 \sim U(0, 380), \quad \phi \sim U(0, \phi_{\max}),$$

where 380 is a value that is larger than the total estimated variation, and $\phi_{\max}$ is defined below equation (4.128). They find that the posterior means are largely unchanged, but the posterior standard deviations associated with the BHM are typically larger than those associated with the EHM (as expected). Consequently, the EHM can lead to inferences that are too liberal.

Disease Mapping

There is a literature closely related to small area estimation, known as *disease mapping*. Consider, as an example, the number of deaths caused by prostate cancer in the $n = 88$ counties of Ohio during the 10-year period, 1995–2004. Each individual (male) who died typically had a place of residence, which we denote as s_{ij} for the j th individual in A_i ; $j = 1, \dots, Z(s_i)$, $i = 1, \dots, n$.

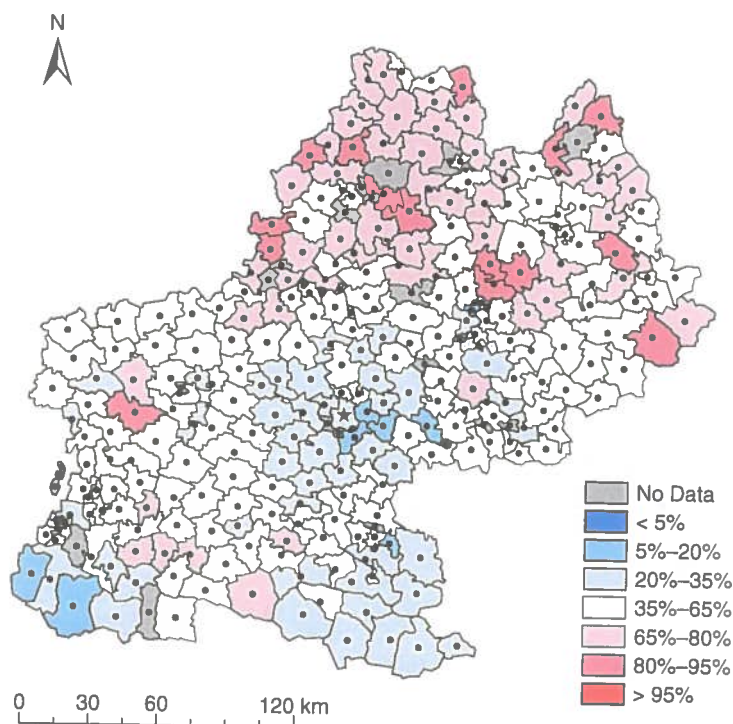


Figure 4.12 Choropleth map showing posterior mean of doctor-prescription amounts per consultation, in the cantons of the Midi-Pyrénées, based on the data in Figure 4.11. The “star” denotes the canton containing Toulouse. Percentiles used for shading are the same as in Figure 4.11.

where $\{Z(s_i) : i = 1, \dots, n\}$ are the *number* of deaths in the respective counties (small areas). A dominant question in disease mapping is whether the locations, $\{s_{ij}\}$, exhibit spatial dependence, specifically clustering. The locations can be viewed as a realization of a spatial point process and could be tested for complete spatial randomness (Section 4.3.2). However, the data $\{s_{ij}\}$ are typically unavailable, usually for confidentiality reasons, and they are aggregated to small areas and released to the public. Nevertheless, the question of clustering, now of small areas, is still relevant.

Assume for the moment that a given demographic group, males age 65–74, are considered in the study of prostate-cancer deaths introduced above. (Combining several demographic groups will be considered later.) For reasons of confidentiality or even ease of analysis, often only the disease counts $\{Z(s_i) : i = 1, \dots, n\}$ in the small areas $\{A_i : i = 1, \dots, n\}$ are available. Further discussion of this aggregation to counts is given in Section 4.3.4. A concomitant variable $\{N_i : i = 1, \dots, n\}$, the total numbers of *person* (males, age 65–74) *years at risk*, should also be available (typically from census data). Then the disease *rate* in the i th small area is defined as

$$R(s_i) \equiv Z(s_i)/N_i. \quad (4.133)$$

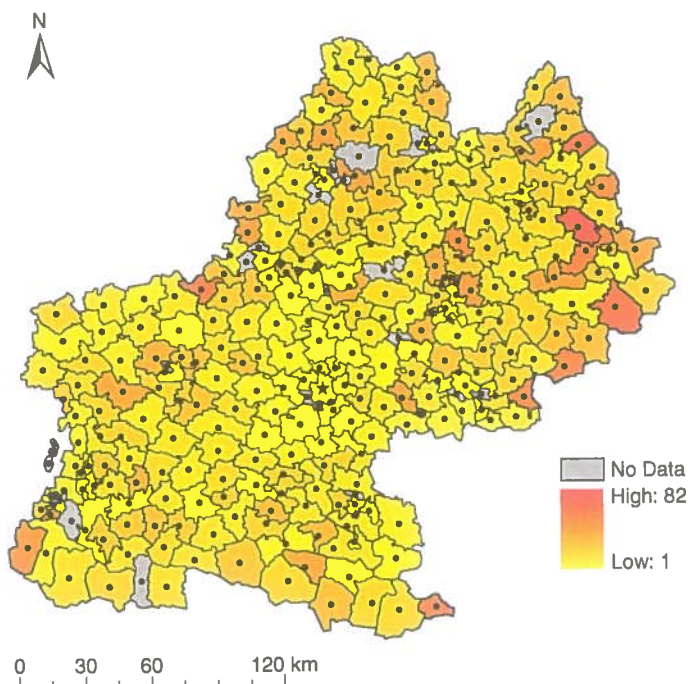


Figure 4.13 Choropleth map showing posterior standard error of doctor-prescription amounts per consultation, in the cantons of the Midi-Pyrénées, based on the data in Figure 4.11. The “star” denotes the canton containing Toulouse.

Under classical independence assumptions, one could model

$$Z(s_i) | \{p(s_j)\} \sim \text{ind. Bin}(N_i, p(s_i)), \quad i = 1, \dots, n, \quad (4.134)$$

where *Bin* denotes the binomial distribution, and $\{p(s_j): j = 1, \dots, n\}$ are the *disease risks* (i.e., the probabilities of getting the disease) in the small areas. Then, for $i = 1, \dots, n$, we obtain

$$E(R(s_i) | \{p(s_j)\}) = p(s_i) \quad (4.135)$$

and

$$\text{var}(R(s_i) | \{p(s_j)\}) = p(s_i)(1 - p(s_i))/N_i, \quad i = 1, \dots, n.$$

Consequently, small areas with smaller numbers of individuals at risk have observed rates that are more variable.

It is often the case that $\{p(s_i)\}$ are all small and $\{N_i\}$ are all large, which is perfect for a Poisson approximation to the binomial distribution in (4.134). For

a generic binomial random variable, $\text{Bin}(N, p)$, suppose $N \rightarrow \infty$ and $p \rightarrow 0$ such that $Np \rightarrow \mu$; then in distribution, $\text{Bin}(N, p) \rightarrow \text{Poi}(\mu)$.

The expected number of deaths in small area A_i is

$$N_i p(s_i) = (N_i \bar{p})(p(s_i)/\bar{p}) \equiv E_i \lambda(s_i),$$

where $\bar{p} \equiv \sum_{j=1}^n p(s_j)/n$ is the average risk and $E_i \equiv N_i \bar{p}$ is a *baseline* expected number of deaths. Then we can interpret $\{\lambda(s_i)\}$ as *relative risks*,

$$\lambda(s_i) = \frac{p(s_i)}{(E_i/N_i)}, \quad (4.136)$$

which is the risk divided by the “expected” risk for the i th small area; $i = 1, \dots, n$. These calculations motivate replacing (4.133) with the assumption

$$Z(s_i) | \{\lambda(s_j)\} \sim \text{ind. Poi}(E_i \lambda(s_i)), \quad i = 1, \dots, n. \quad (4.137)$$

Often, $\bar{p}$ and hence $\{E_i\}$ are unknown. A commonly used estimate of $\{E_i\}$ is

$$E_i \equiv N_i \left(\sum_{j=1}^n Z(s_j) / \sum_{j=1}^n N_j \right), \quad i = 1, \dots, n. \quad (4.138)$$

A quantity often considered by epidemiologists is the *standardized mortality ratio (SMR)*, which is defined as

$$\text{SMR}_i \equiv Z(s_i)/E_i, \quad i = 1, \dots, n. \quad (4.139)$$

Consequently, if we assume (4.137), then

$$E(\text{SMR}_i | \{\lambda(s_j)\}) = \lambda(s_i) \quad \text{and} \quad \text{var}(\text{SMR}_i | \{\lambda(s_j)\}) = \lambda(s_i)/E_i. \quad (4.140)$$

If $\{E_i\}$ have to be estimated [say using (4.138)], the equalities in (4.140) become approximations. Hence, up to these approximations, the SMR is an unbiased estimator of the relative risk with variability inversely proportional to the baseline expected number of deaths (equally, it is inversely proportional to the number of person years at risk). That is, for areas with a smaller E_i , the standardized mortality ratio, SMR_i , is a less precise estimator of its relative risk.

When the population in $D_s = \cup_{i=1}^n A_i$ is made up of several demographic groups or other obvious strata (e.g., age ranges, in the case of prostate cancer), then the calculation of the expected number of deaths E_i in the small area A_i is a little more involved: Define q_k to be the proportion of the at-risk population that are in group k ; $k = 1, \dots, K$, where K is the number of demographic groups. Then $\sum_{k=1}^K q_k = 1$. Consider the small area A_i . Because we have K

groups, the number of person-years-at-risk (N_i) for A_i can be decomposed into numbers of person-years-at-risk in each group ($\{N_{ik}\}$):

$$N_i = \sum_{k=1}^K N_{ik}.$$

Then the *baseline* expected number of deaths is defined by

$$E_i \equiv \sum_{k=1}^K N_{ik} q_k, \quad i = 1, \dots, n.$$

Often, $\{q_k: k = 1, \dots, K\}$, and hence $\{E_i\}$, are unknown. A commonly used (internally standardized) estimate of q_k is

$$\hat{q}_k = \sum_{j=1}^n Z(s_j)_k / \sum_{j=1}^n N_{jk}, \quad k = 1, \dots, K,$$

where $Z(s_j)_k$ is the observed number of deaths in small area A_j that occurred in the k th demographic group. Finally, the (internally standardized) estimate of the baseline expected number of deaths is defined by

$$E_i \equiv \sum_{k=1}^K N_{ik} \hat{q}_k, \quad i = 1, \dots, n.$$

Much of the recent literature on disease mapping has involved fitting an HM to the data in order to “borrow strength” from the areas that are more precise (larger E_i); the approach is sometimes called *shrinkage*, or *super-population modeling*. For example, see Lawson and Cressie (2000) and the text by Lawson (2001) for a review of that literature.

The spatial dependence is typically incorporated into the process model through the logarithm of the true rates:

Data model: Conditional on $\{\lambda(s_i)\}$ and for $i = 1, \dots, n$,

$$Z(s_i) | \lambda(s_i) \sim \text{Poi}(E_i \lambda(s_i)).$$

Process model: Conditional on β_C and θ and for $\mathbf{Y} \equiv (\log \lambda(s_1), \dots, \log \lambda(s_n))'$,

$$\mathbf{Y} | \beta, \theta_C \sim \text{Gau}(\mathbf{X}\beta, \Sigma(\theta_C)),$$

where

$$\Sigma(\theta_C) \equiv \sigma^2 \mathbf{D} + \tau^2 (\mathbf{I} - \phi \mathbf{H})^{-1} \Delta;$$

$\mathbf{D}$ and Δ are known $n \times n$ diagonal matrices; $\mathbf{H}$ is a known $n \times n$ matrix with zeros down the diagonal; and the lattice-model covariance parameters are $\theta_C \equiv (\sigma^2, \tau^2, \phi)'$, chosen so that the matrix inverse exists.

The first term in $\Sigma(\theta_C)$ is a known (apart from the parameter, σ^2) diagonal matrix that accounts for inhomogeneities, such as extra-Poisson variation. The second term is the CAR model with spatial-dependence parameter ϕ and variance heterogeneity expressed through $\tau^2\Delta$, where Δ is a known diagonal matrix.

The model of Besag, York, and Mollié (1991) uses $\mathbf{D} = \mathbf{I}$ and $\phi = \phi_{\max}$, resulting in $\tau^2(\mathbf{I} - \phi\mathbf{H})^{-1}\Delta$ being only nonnegative-definite. That is, the lattice-model parameters are $\theta_L = (\sigma^2, \tau^2)'$. Notice that, even when $\sigma^2 = 0$, it is still possible to obtain a proper posterior distribution through formal application of Bayes' Theorem. It reflects the often-seen, slowly decreasing spatial correlation as a function of distance between small areas. In contrast, the model of Cressie (1992) puts $\sigma^2 = 0$, and hence the lattice-model parameters are $\theta_C = (\tau^2, \phi)'$; this latter model allows inference on the strength of the spatial-dependence parameter in the interval $(\phi_{\min}, \phi_{\max})$, the parameter space for ϕ in the model (4.128). Cressie and Hartfield (1996) use a similar model for regular two-dimensional plots in an agricultural field trial; they estimated a well defined, positive-definite CAR matrix $\mathbf{C}$. Wright, Stern, and Cressie (2003) leave all three parameters, σ^2 , τ^2 , and ϕ in the process model, although the preponderance of choice in the disease-mapping literature has been to specify $\phi = \phi_{\max}$.

Cressie and Chan (1989) and Cressie (1992) make the following choices of $\Delta \equiv \text{diag}(\Delta_1, \dots, \Delta_n)$ and $\mathbf{H} \equiv (h_{ij})$, relating them to expected counts, $\{E_i\}$: For $i = 1, \dots, n$,

$$\begin{aligned} \text{A1 :} \quad & \Delta_i = 1/E_i, \\ \text{A2 :} \quad & h_{ij} = \begin{cases} (E_j/E_i)^{1/2}, & \mathbf{s}_j \in N(\mathbf{s}_i) \\ 0, & \text{elsewhere.} \end{cases} \end{aligned}$$

Importantly, the symmetry conditions, $\Delta_j h_{ij} = \Delta_i h_{ji}$, for all i, j , hold. This process model (like the data model given just above) results in smaller variances for areas with larger expected number of people at risk. Cressie, Perrin, and Thomas-Agnan (2005, Appendix A) give a justification of this choice of A1, A2 and called it the *spatial rates model*.

Besag, York, and Mollié (1991) made a different choice for Δ and $\mathbf{H}$. For $i = 1, \dots, n$,

$$\begin{aligned} \text{B1 :} \quad & \Delta_i = 1/|N(\mathbf{s}_i)|, \\ \text{B2 :} \quad & h_{ij} = \begin{cases} 1/|N(\mathbf{s}_i)|, & \mathbf{s}_j \in N(\mathbf{s}_i) \\ 0, & \text{elsewhere.} \end{cases} \end{aligned}$$

The symmetry conditions, $\Delta_j h_{ij} = \Delta_i h_{ji}$, for all i, j , hold. Recall that in Section 4.2.5, we called this a WCAR model.

One final way to compare the two models is to recall the result given in Section 4.2.5,

$$[\text{corr}(Y(s_i), Y(s_j) | \{Y(s_k) : k \neq i, j\})]^2 = \phi^2 h_{ij} h_{ji}. \quad (4.141)$$

In Section 4.2.5, we saw that for the WCAR (i.e., choose B1 and B2) model, the square of the partial correlation in (4.141) is $\phi^2 / |N(s_i)| |N(s_j)|$, making ϕ difficult to interpret. For the spatial-rates model (i.e., choose A1 and A2), the square of the partial correlation in (4.141) is ϕ^2 , which is invariant to the neighborhood structure. Consequently, ϕ is easily interpretable as a spatial-dependence parameter in $[-1, 1]$. The ACAR model defined in (4.130) has a similar structure to the spatial-rates model, where $N(s_i)$ replaces E_i in the definition of $\mathbf{\Delta}$ and $\mathbf{H}$. Both models have the property that ϕ can be interpreted as a partial correlation coefficient.

The HM just above becomes a BHM model by including a parameter model. We now give an example for a process model where $\mathbf{D} = \text{diag}(1/E_1, \dots, 1/E_n)$, and $\mathbf{\Delta}$ and $\mathbf{H}$ are given in A1 and A2:

Parameter model: The joint distribution of the parameters is

$$[\boldsymbol{\beta}, \sigma^2, \tau^2, \phi] = [\boldsymbol{\beta}] [\sigma^2] [\tau^2] [\phi],$$

where $\boldsymbol{\beta} \sim U((-\infty, \infty)^p)$; $[\sigma^2] \propto \exp(-b/\sigma^2)$, $[\tau^2] \propto \exp(-b/\tau^2)$, for b a small number (equal to 0.01); and $\phi \sim U(0, \phi_{\max})$.

Wright, Stern, and Cressie (2003) used this in a BHM for studying male lip cancer in Scotland. Implementation of the BHM is almost always through MCMC; see examples given in Gilks, Richardson, and Spiegelhalter (1996).

Change-of-Support in Small Areas: Ecological Bias

The *ecological fallacy* in its simplest form states that relationships seen at aggregated (here over space) levels could be very different from those seen at the individual level. Robinson (1950) coined the term and illustrated it with 1930s data on individuals in the United States, where part of the dataset contained the binary variables “foreign born” and “literate.” At the individual level, these two variables showed a negative correlation of -0.11 ; however, at the state level (the lower 48 states in the United States), the percent foreign born and the percent literate had a correlation of $+0.53$. The Geography literature has referred to the way inferences vary according to the level of aggregation as the *modifiable areal unit problem (MAUP)*, and it is simply Simpson’s Paradox in another guise (Chapter 1). In spatial statistics, the phenomenon is often called *change-of-support (COS)*; see Section 4.1.3. It is a difficult problem because typically one seeks explanation of an individual-level relationship, but all one has is aggregated-level data (due to data-collection methods or due to

confidentiality requirements). Much has been written about the problem, and we refer the interested reader to the special issue of the *Journal of the Royal Statistical Society, Series A*, Vol. 164, Part 1 (2001).

The problem is inherently a statistical one; a perfect correlation of ± 1 at the individual level implies a perfect correlation of ± 1 , respectively, at all aggregated levels, but once the individual correlation lies *strictly between* -1 and 1 , anything could happen to the correlation at the aggregated level. Our recommended approach to this vexing but fascinating problem is to use statistical modeling to resolve it; whenever possible, build statistical models at the individual level and look at their consequences under the operation of aggregation. Specifically, we recommend fitting the aggregated *individual-level model* to the *aggregated data*. While logically defensible, it hinges on the requirement that there is enough “knowledge” about the problem to build a good individual-level statistical model! But, if that knowledge is induced from aggregated data, there is a serious risk of being caught in the fallacy.

4.2.7 Simultaneously Specified Spatial Dependence

Whittle (1954) defined simultaneously specified spatial processes on the countable integer lattice in $\mathbb{R}^2$. Here we consider the finite irregular lattice; that is, $D_s \equiv \{\mathbf{s}_1, \dots, \mathbf{s}_n\}$. Let $\mathbf{v} \equiv (\nu(\mathbf{s}_1), \dots, \nu(\mathbf{s}_n))'$ denote a joint Gaussian distribution with zero mean and independent components. That is,

$$\mathbf{v} \sim \text{Gau}(\mathbf{0}, \mathbf{\Lambda}), \quad (4.142)$$

where $\mathbf{\Lambda} \equiv \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$. Analogous to (4.84), which recall is a first-order autoregressive model in $\mathbb{R}^1$, we define the *simultaneous autoregressive (SAR)* model in $\mathbb{R}^d$:

$$Y(\mathbf{s}_i) = \mu(\mathbf{s}_i) + \sum_{j=1}^n b_{ij}(Y(\mathbf{s}_j) - \mu(\mathbf{s}_j)) + \nu(\mathbf{s}_i), \quad (4.143)$$

where $b_{ii} = 0$; $i = 1, \dots, n$, and $(\mu(\mathbf{s}_1), \dots, \mu(\mathbf{s}_n))' \equiv \boldsymbol{\mu}$ is an n -dimensional vector representing the mean of $\mathbf{Y}$. Notice that (4.143) requires no ordering of the locations $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$. A special case of (4.143), for which the locations do have a type of ordering, is called a *unilateral process*. For example, D_s is a regular square lattice in $\mathbb{R}^2$ and b_{ij} is non-zero only if $\mathbf{s}_j$ is “southwest” of $\mathbf{s}_i$ (e.g., Tjostheim, 1983).

In terms of vectors and matrices, (4.143) can be written equivalently as

$$\mathbf{Y} - \boldsymbol{\mu} = \mathbf{B}(\mathbf{Y} - \boldsymbol{\mu}) + \mathbf{v};$$

that is,

$$\mathbf{Y} = \boldsymbol{\mu} + (\mathbf{I} - \mathbf{B})^{-1} \mathbf{v}, \quad (4.144)$$

where $\mathbf{B}$ has zeros down the diagonal, and it is assumed that the SAR coefficients, $\{b_{ij}\}$ are chosen such that $(\mathbf{I} - \mathbf{B})$ is invertible. Notice that $\mathbf{B}$ is not necessarily symmetric, and $(\mathbf{I} - \mathbf{B})$ is not necessarily positive-definite.

While $\mathbf{B}$ is typically sparse, $(\mathbf{I} - \mathbf{B})^{-1}$ is not. Nevertheless, the right-hand side of (4.144) expresses the spatial dependence in the SAR model as a spatial moving average of the elements of $\mathbf{v}$, which are *iid*.

If we write $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, then (4.144) can be written as the general linear model,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}, \quad (4.145)$$

where now $\boldsymbol{\delta}$ is a spatial Gaussian process that satisfies $(\mathbf{I} - \mathbf{B})\boldsymbol{\delta} = \mathbf{v}$, and $\mathbf{v} \sim \text{Gau}(\mathbf{0}, \boldsymbol{\Lambda})$. Hence,

$$\boldsymbol{\delta} \sim \text{Gau}(\mathbf{0}, (\mathbf{I} - \mathbf{B})^{-1} \boldsymbol{\Lambda} (\mathbf{I} - \mathbf{B}')^{-1}). \quad (4.146)$$

This representation of a SAR model should be contrasted with the representation of a CAR model given by (4.127). From (4.146), the spatial dependence is captured by $\mathbf{B}$. If $\mathbf{B} = \mathbf{0}$, the $n \times n$ matrix of zeros, then $\boldsymbol{\delta} \sim \text{Gau}(\mathbf{0}, \text{diag}(\sigma_1^2, \dots, \sigma_n^2))$, which is the joint distribution of independent components, $\delta(s_i) \sim \text{Gau}(0, \sigma_i^2)$, $i = 1, \dots, n$. In the CAR given by (4.127), we obtain a similar result if $\mathbf{C} = \mathbf{0}$, but the interpretation of *nonzero* $\mathbf{C}$ in (4.127) is totally different to the interpretation of *nonzero* $\mathbf{B}$ in (4.145).

The area of research known as *spatial econometrics* (e.g., Paelinck and Klaassen, 1979; Anselin, 1988) views the matrix $\mathbf{B}$ as a type of spatial backward-shift operator; compare this with the temporal backward-shift operator found in (3.15). For $\boldsymbol{\mu} = \mathbf{0}$, the SAR model can be written as $\mathbf{Y} = \mathbf{B}\mathbf{Y} + \mathbf{v}$, which is analogous to the first-order autoregression in time series, namely $Y_t = BY_t + W_t$, where $BY_t \equiv Y_{t-1}$.

Every SAR model with a sparse $\mathbf{B}$ results in a CAR model with a sparse $\mathbf{C}$. Define $\tilde{\mathbf{Y}} \equiv \boldsymbol{\Lambda}^{-1/2} \mathbf{Y}$, where $\boldsymbol{\Lambda}$ is given in (4.142) and $\mathbf{Y}$ satisfies (4.144). Then

$$\text{var}(\tilde{\mathbf{Y}}) = (\mathbf{I} - \tilde{\mathbf{B}})^{-1} (\mathbf{I} - \tilde{\mathbf{B}}')^{-1}, \quad (4.147)$$

where $\tilde{\mathbf{B}} \equiv \boldsymbol{\Lambda}^{-1/2} \mathbf{B} \boldsymbol{\Lambda}^{1/2}$. Because $\boldsymbol{\Lambda}$ is a diagonal matrix, $\mathbf{B}$ is sparse if and only if $\tilde{\mathbf{B}}$ is sparse. From (4.147), we have

$$(\text{var}(\tilde{\mathbf{Y}}))^{-1} = (\mathbf{I} - \tilde{\mathbf{B}}' - \tilde{\mathbf{B}} + \tilde{\mathbf{B}}'\tilde{\mathbf{B}}) \equiv \mathbf{M}^{-1}(\mathbf{I} - \mathbf{C}),$$

where $\mathbf{M}$ is an $n \times n$ diagonal matrix, $\mathbf{C}$ is an $n \times n$ matrix with zeros down the diagonal, and the formulas for $\mathbf{M}$ and $\mathbf{C}$ can be derived straightforwardly. It is easy to see that if $\tilde{\mathbf{B}}$ is sparse, so too is $\mathbf{C}$, albeit less sparse than $\tilde{\mathbf{B}}$. See Besag (1974) and Cressie (1993, Section 6.3), for comparisons between the SAR model and the CAR model.

There is one curiosity of the SAR model that illustrates once again the difficulty in generalizing from $s \in \mathbb{R}^1$ to $s \in \mathbb{R}^d$. A simple calculation (e.g.,

Cressie, 1993, p. 409) shows that the autoregressive “errors” $\{\nu(s_i)\}$ in (4.143) are correlated with $\{Y(s_i)\}$; specifically, $\text{cov}(\nu, \mathbf{Y}) = \Lambda(\mathbf{I} - \mathbf{B}')^{-1}$. That is, the SAR model has the property that the “noise” ν correlates with the “signal” $\mathbf{Y}$, making the autoregression in (4.143) difficult to interpret.

A simple example of an HM involving the SAR model is given as follows. For illustrative purposes, suppose $\mathbf{B} = \rho\mathbf{W}$, where ρ is a real-valued parameter and $\mathbf{W}$ is known and has zeros down the diagonal. Furthermore, suppose that $\Lambda = \sigma^2\mathbf{I}$.

Data model: Conditional on σ_ϵ^2 , and for $i = 1, \dots, n$,

$$Z(s_i) | Y(s_i), \sigma_\epsilon^2 \sim \text{ind. Gau}(Y(s_i), \sigma_\epsilon^2).$$

Process model: Conditional on β , σ^2 , and ρ ,

$$\mathbf{Y} | \beta, \sigma^2, \rho \sim \text{Gau}(\mathbf{X}\beta, \sigma^2[(\mathbf{I} - \rho\mathbf{W}')(\mathbf{I} - \rho\mathbf{W})]^{-1}),$$

where $\mathbf{W}$ is a known $n \times n$ matrix with zeros down the diagonal, and the parameter space of ρ ensures that the matrix inverse exists.

A BHM has a parameter (prior) model that specifies the distribution of (β', σ^2, ρ) . Because of the Gaussian and additive structure of the HM, the model for the data $\mathbf{Z}$ could be represented additively in terms of the marginal distribution of $\mathbf{Z}$. That is,

$$\mathbf{Z} = \mathbf{X}\beta + \delta + \epsilon,$$

where $\epsilon \equiv (\epsilon(s_1), \dots, \epsilon(s_n))'$ are the measurement errors, often $\epsilon \sim \text{Gau}(\mathbf{0}, \sigma_\epsilon^2\mathbf{I})$, and δ and ϵ are independent. Hence, the HM can be represented equivalently as a linear mixed model for “noisy” data $\mathbf{Z}$, with $E(\mathbf{Z}) = \mathbf{X}\beta$ and $\text{var}(\mathbf{Z}) = \sigma^2[(\mathbf{I} - \rho\mathbf{W}')(\mathbf{I} - \rho\mathbf{W})]^{-1} + \sigma_\epsilon^2\mathbf{I}$. Inference is needed for β , σ_ϵ^2 , σ^2 , ρ , δ , and hence the “signal” $\mathbf{Y}$. In an EHM, β , σ_ϵ^2 , σ^2 , and ρ are estimated, whereas in a BHM, their uncertainty is captured by a parameter model. Obviously, analogous conclusions hold if a CAR model like (4.98) replaces the SAR model in the HM; see Section 4.2.6.

4.2.8 Other Topics

The SMA and SARMA Models

In Section 4.1, where we discussed geostatistical processes, we presented the spatial moving average (or process convolution) models (4.67). The discretized version could be viewed as a lattice model. By analogy to the (mean-zero) SAR model, $(\mathbf{I} - \rho\mathbf{W})\mathbf{Y} = \nu$, we write the lattice *spatial moving average (SMA) model* as

$$\mathbf{Y} = (\mathbf{I} + \psi\mathbf{E})\nu, \quad (4.148)$$

where $\mathbf{E}$ is an $n \times n$ matrix with zeros down the diagonal, and $\mathbf{v} \equiv (v(s_1), \dots, v(s_n))'$ has mean zero and $\text{var}(\mathbf{v}) = \mathbf{\Lambda}$, a diagonal $n \times n$ matrix. Notice that when $\psi = 0$, $\mathbf{Y} = \mathbf{v}$, and hence the process $\mathbf{Y}$ exhibits no spatial dependence. From (4.148),

$$E(\mathbf{Y}) = \mathbf{0} \quad \text{and} \quad \text{var}(\mathbf{Y}) = (\mathbf{I} + \psi \mathbf{E}) \mathbf{\Lambda} (\mathbf{I} + \psi \mathbf{E}').$$

Haining (1978) gives a detailed discussion of this model.

A general linear model based on SMA dependence is obtained, if we write

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}, \quad (4.149)$$

where now $\boldsymbol{\delta} = (\mathbf{I} + \psi \mathbf{E})\mathbf{v}$, and $\mathbf{v} \sim \text{Gau}(\mathbf{0}, \mathbf{\Lambda})$. Therefore,

$$\mathbf{Y} \sim \text{Gau}(\mathbf{X}\boldsymbol{\beta}, (\mathbf{I} + \psi \mathbf{E}) \mathbf{\Lambda} (\mathbf{I} + \psi \mathbf{E}')). \quad (4.150)$$

Notice that the spatial dependence is captured through (sparse) $\text{var}(\mathbf{Y})$, and hence (4.150) is closer in type to a (discretized) geostatistical model.

A *spatial autoregressive moving average (SARMA) model* for $\mathbf{Y}$ is defined as follows:

$$(\mathbf{I} - \rho \mathbf{W})\mathbf{Y} = (\mathbf{I} + \psi \mathbf{E})\mathbf{v}; \quad (4.151)$$

the spatial dependence in the SARMA model can be expressed through,

$$\mathbf{Y} = (\mathbf{I} - \rho \mathbf{W})^{-1} (\mathbf{I} + \psi \mathbf{E})\mathbf{v} \equiv \mathbf{F}\mathbf{v},$$

which, in fact, is a spatial moving average of the independent elements of $\mathbf{v}$. It is easy to see that $E(\mathbf{Y}) = \mathbf{0}$, and that

$$\text{var}(\mathbf{Y}) = \{(\mathbf{I} - \rho \mathbf{W})^{-1} (\mathbf{I} + \psi \mathbf{E})\} \mathbf{\Lambda} \{(\mathbf{I} - \rho \mathbf{W})^{-1} (\mathbf{I} + \psi \mathbf{E})\}'. \quad (4.152)$$

The general linear model based on SARMA dependence is sometimes called a *SARMAX model*. Define

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}, \quad (4.153)$$

where the error term $\boldsymbol{\delta}$ follows the SARMA model. That is, $(\mathbf{I} - \rho \mathbf{W})\boldsymbol{\delta} = (\mathbf{I} + \psi \mathbf{E})\mathbf{v}$, where $\mathbf{v} \sim \text{Gau}(\mathbf{0}, \mathbf{\Lambda})$, and recall that $\mathbf{\Lambda} \equiv \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$. Hence,

$$\mathbf{Y} \sim \text{Gau}(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma}_Y),$$

where $\boldsymbol{\Sigma}_Y$ is given by $\text{var}(\mathbf{Y})$ in (4.152). Spatio-temporal versions of these models can be found in Section 6.4.2.

Recall that our goal in this section is to present lattice models for $\mathbf{Y}$ that incorporate spatial dependence. When the SARMAX model (4.153) is used as a

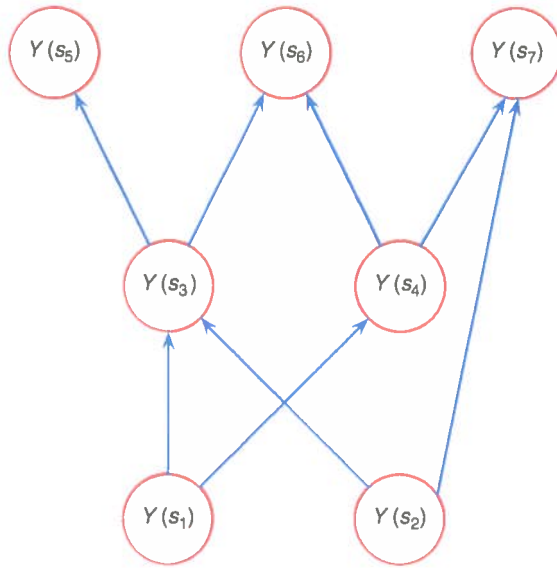


Figure 4.14 Directed graph with seven vertices, showing the (directed) edges between the vertices.

process model in an HM, which includes additive measurement error $\boldsymbol{\varepsilon}$, data $\mathbf{Z}$ can be represented additively in terms of its marginal distribution,

$$\mathbf{Z} = \mathbf{Y} + \boldsymbol{\varepsilon} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta} + \boldsymbol{\varepsilon},$$

where $\boldsymbol{\varepsilon} \equiv (\varepsilon(s_1), \dots, \varepsilon(s_n))'$, often $\boldsymbol{\varepsilon} \sim \text{Gau}(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{I})$, and $\boldsymbol{\delta}$ and $\boldsymbol{\varepsilon}$ are independent. To simplify discussion, assume that $\sigma_1^2 = \dots = \sigma_n^2 = \sigma^2$. This gives a linear mixed model, as in Section 4.2.7, and inference is needed for $\boldsymbol{\beta}$, σ_ε^2 , σ^2 , ρ , ψ , $\boldsymbol{\delta}$, and hence the “signal” $\mathbf{Y}$. In an EHM, $\boldsymbol{\beta}$, σ_ε^2 , σ^2 , ρ , and ψ are estimated, whereas in a BHM, their uncertainty is captured by a parameter model.

The Partially Ordered Markov Model (POMM)

Another approach taken to modeling spatial dependence in lattice data has been to impose some sort of directionality or ordering on the integer lattice, and to define analogues to (4.84), (4.90), or (4.91) based on that ordering. Abend, Harley, and Kanal (1965) did this on an integer lattice with simple directional dependence that they called a *Markov mesh model*; Cressie and Liu (2001) developed this further in the case of a binary process. Goutsias (1989) started with processes that satisfy a spatial generalization of (4.91), and he showed that this is equivalent to a spatial generalization of (4.90).

Cressie and Davidson (1998) define *partially ordered Markov models (POMMs)* on an acyclic directed graph of spatial locations, $\{\mathbf{s}_i : i = 1, \dots, n\}$,

through conditional probabilities,

$$[Y(s_i)|Y(pa(s_i))], \quad i = 1, \dots, n, \quad (4.154)$$

where $pa(s_i)$ is the set of “parents” of s_i , defined by the directed graph. They note that this is precisely the class of processes that Whittle (1963) was looking for in his attempt to generalize the one-dimensional Markov property to higher dimensions.

A simple example of a directed graph is shown in Figure 4.14; the nodes are at $D_s = \{s_1, \dots, s_7\}$. The set of parents, $pa(s_i)$, is the set of all nodes s_j with a directed edge from s_j to s_i . Further properties of such graphs are given by Lauritzen (1996).

Cressie and Davidson (1998) prove two main results. First, they prove

$$[Y] = [Y(D_0)] \prod_{s \in D_s \setminus D_0} [Y(s)|Y(pa(s))], \quad (4.155)$$

where $D_0 \subset D_s$ are all locations that have no parents (i.e., “minimum” elements in the partial ordering implied by the acyclic directed graph). From (4.155) we see that the joint distribution of POMMs can be written directly as a product of the conditional distributions (4.154), *without* the need to evaluate a normalizing constant. Of course, a (partial) ordering is needed on D_s in order to define (4.154), and so POMMs appear to be a rather restrictive class of spatial models for lattice data.

When taken out of its spatial context, we have seen in Section 2.4 that the acyclic directed graph represents precisely the type of relationships one sees in an HM. Suppose that the data model involves $[Z|Y, \theta_D]$, the process model involves $[Y|\theta_P]$, and the parameter model involves $[\theta]$, where $\theta = \{\theta_D, \theta_P\}$. Then the directed graph that captures the dependence structure in the HM is given by Figure 2.2b. Consequently, the joint distribution $[Z, Y, \theta_D, \theta_P]$ can be obtained from (4.155), and it is

$$[Z, Y, \theta_D, \theta_P] = [Z|Y, \theta_D][Y|\theta_P][\theta_D, \theta_P].$$

Cressie and Davidson’s second result shows that POMMs are in fact MRFs, and hence they are not as restrictive as it might first seem. Define the *closure of the parents* of s_i as, $\overline{pa}(s_i) \equiv pa(s_i) \cup \{s_i\}$. Then the *family* of s_i is defined as

$$fa(s_i) \equiv \bigcup \{\overline{pa}(s_j) : s_j \in \overline{pa}(s_i)\}; \quad (4.156)$$

furthermore, remove s_i from the family and define $fa^*(s_i) \equiv fa(s_i) \setminus \{s_i\}$. They show that a POMM’s conditional probabilities are given by

$$[Y(s_i)|Y_{-i}] = [Y(s_i)|Y(fa^*(s_i))], \quad i = 1, \dots, n. \quad (4.157)$$

That is, $\mathbf{Y}$ is an MRF with $N(s_i) = fa^*(s_i)$; $i = 1, \dots, n$.

Moreover, the conditional probabilities (4.157) that define the MRF can be calculated easily from (4.154). Define, for $i = 1, \dots, n$,

$$R(Y(s_i); \mathbf{Y}(fa^*(s_i))) \equiv \prod [Y(s_j) | \mathbf{Y}(pa(s_j))], \quad (4.158)$$

where the product in this expression is over $\{s_j \in D_s : s_i \in \overline{pa}(s_j)\}$. Then, from Cressie and Davidson (1998), the conditional probabilities are

$$[Y(s_i) | \mathbf{Y}_{-i}] = R(Y(s_i); \mathbf{Y}(fa^*(s_i))) / \sum_{t \in \Omega_i} R(t; \mathbf{Y}(fa^*(s_i))),$$

where recall that Ω_i is the support of $Y(s_i)$; $i = 1, \dots, n$. In summary, POMMs are defined by the set of parents $\{pa(s_i) : i = 1, \dots, n\}$, which then yield an MRF with neighborhoods and conditional probabilities defined by $\{fa^*(s_i) : i = 1, \dots, n\}$. These neighborhoods are equivalently obtained by forming the moral graph from the acyclic directed graph (i.e., forming edges between all parents of each node and then making all directed edges undirected); see Lauritzen (1996, p. 47). In fact, the MRF defined by the (undirected) graph in Figure 4.10 is the moral graph obtained from the directed graph in Figure 4.14.

While the POMMs look like a curiosity from a purely spatial point of view, when the time dimension is added it becomes sensible to order spatio-temporal locations according to dynamic relations in a statistical model; see Figures 6.20 and 6.21 in Chapter 6.

4.3 SPATIAL POINT PROCESSES

A spatial point process is a stochastic process governing the location of events (equivalently, points) $\{s_i\}$ in some set $D_s \subset \mathbb{R}^d$, where the number of such events in D_s is also random (e.g., Diggle, 2003). In the simplest case, the randomness is in the spatial locations and not in any attribute associated with the locations. For example, consider the locations of longleaf pines in 4 ha of a natural forest in Thomas county, Georgia, in 1979 (Platt, Evans, and Rathbun, 1988; Cressie, 1993; Rathbun and Cressie, 1994). Figure 4.15 shows the locations of adult and subadult trees. Trees develop for numerous reasons, due to seed dispersion, nutrients, water, light, competition, fire, animal foraging, and so forth. It is often reasonable to assume that their locations are the result of a superposition of many chance occurrences, and hence they can be modeled as a spatial point process. The proper definition of point processes requires some theoretical background, a brief description of which is given below.

Only *simple* spatial point processes in $\mathbb{R}^d$ (i.e., almost surely, either no event or a single event occurs at any point) will be considered. We characterize the spatial point process Z through subsets $A \subset D_s \subset \mathbb{R}^d$, which are Lebesgue measurable (i.e., subsets of D_s for which d -dimensional volume is defined).

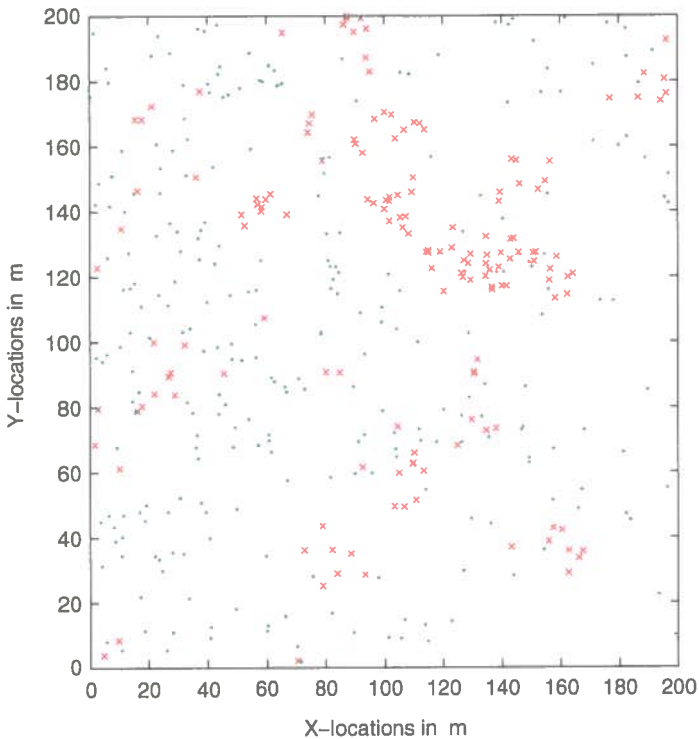


Figure 4.15 Spatial locations of longleaf pine trees in a 4-ha study area in the Wade Tract, Thomas County, Georgia, USA; the $m_A = 271$ adult longleaf pine trees are shown as green dots, and the $m_S = 159$ subadult longleaf pine trees are shown as red crosses.

Let $Z(A)$ denote the number of events in $A \subset D_s$; then $Z(\cdot)$ is a stochastic (counting) process that is defined on the set of Lebesgue measurable subsets of D_s . Henceforth, assume that D_s is *bounded* and that $Z(A)$ is (almost surely) *finite* for all $A \subset D_s$. Then $\{Z(A) : A \subset D_s\}$ characterizes the point process (e.g., Cressie, 1993, p. 619). We shall refer to the point process as $Z(\cdot)$ or Z , interchangeably.

An important, but simple, example is the *Poisson point process* Z , for which

$$Z(A)|\lambda^o \sim \text{Poi}(\lambda^o|A|), \quad A \subset D_s, \quad (4.159)$$

where $\lambda^o > 0$ is a parameter of the Poisson point process, and recall that $|A|$ is the d -dimensional volume of A . More details on this spatial point process are given in Section 4.3.1.

One fundamental property of a spatial point process is the expected number of events in a given region A (i.e., $E(Z(A))$). This can be calculated in terms of an intensity function $\lambda(s)$ (with units of “per d -dimensional volume”) that

is a measure of the *potential* for an event to appear at any location $\mathbf{s} \in D_s$. Let ds denote a small region located at $\mathbf{s}$ with volume $|ds|$. Then the *first-order intensity* function of the point process $Z(\cdot)$ is defined as

$$\lambda(\mathbf{s}) \equiv \lim_{|ds| \rightarrow 0} E(Z(ds))/|ds|, \quad \mathbf{s} \in D_s, \quad (4.160)$$

provided the limit exists. Hence,

$$E(Z(A)) = \int_A \lambda(\mathbf{s}) ds, \quad A \subset D_s. \quad (4.161)$$

For the Poisson point process, characterized by (4.159), we obtain

$$\lambda(\mathbf{s}) \equiv \lambda^0 > 0, \quad \mathbf{s} \in D_s.$$

That is, the *potential* for an event to appear anywhere in D_s is *constant*.

An obvious nonparametric estimator of $\lambda(\cdot)$, based on $m = Z(D_s)$ event locations $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_m\}$ in D_s , is (Diggle, 1985)

$$\hat{\lambda}(\mathbf{s}) \equiv \frac{1}{p_b(\mathbf{s})} \sum_{i=1}^m \kappa_b(\mathbf{s} - \mathbf{s}_i), \quad \mathbf{s} \in D_s, \quad (4.162)$$

where $\kappa_b(\cdot)$ is a probability density (kernel) function symmetric about the origin, $b > 0$ is the *bandwidth* that determines the amount of smoothing, and $p_b(\mathbf{s}) \equiv \int_{D_s} \kappa_b(\mathbf{s} - \mathbf{x}) d\mathbf{x}$ is an edge correction. For example, in $\mathbb{R}^2$, consider the simple choice, $\kappa_b(\mathbf{x}) = \rho_b(x_1)\rho_b(x_2)$, where $\mathbf{x} = (x_1, x_2)'$ and $\rho_b(\cdot)$ is the Epanechnikov kernel,

$$\rho_b(x) = (3/4)b^{-1}\{1 - (x/b)^2\}I(|x| \leq b). \quad (4.163)$$

Choice of b in (4.162) is based on the usual trade-off between bias and variance. Diggle recommends using $b = 0.68m^{-1/5}$ in (4.162) and (4.163), for D_s a unit square, but this is only a guideline.

Based on the locations of the $m_A = 271$ adult pine trees in the 4-ha of forest, depicted in Figure 4.15, we obtain $\hat{\lambda}(\cdot)$ using (4.162) and $b = 30$ m. The result is shown in Figure 4.16a, which indicates some inhomogeneity of intensity over D_s . In comparison, the $\hat{\lambda}(\cdot)$ associated with the $m_S = 159$ subadults and the same $b = 30$ m, shown in Figure 4.16b, indicates an intensity surface that is even more inhomogeneous and with a localized peak.

Knowledge of $\lambda(\cdot)$ does not completely characterize the spatial behavior of a point process; there are higher-order moment measures that express complex spatial interactions (e.g., Cressie, 1993, p. 622). For example, spatial statistics based on the *second-order intensity* function (or second-order product density),

$$\lambda_2(\mathbf{s}, \mathbf{x}) \equiv \lim_{\substack{|ds| \rightarrow 0 \\ |dx| \rightarrow 0}} E(Z(ds)Z(dx))/(|ds||dx|), \quad \mathbf{s}, \mathbf{x} \in D_s, \quad (4.164)$$

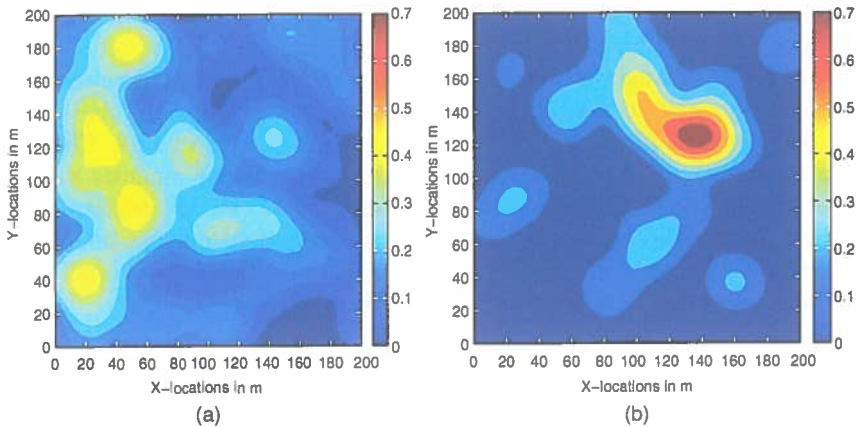


Figure 4.16 (a) Contour plot of estimated first-order intensity of $m_A = 271$ adult longleaf pine trees in the 4-ha study area in the Wade Tract; bandwidth $b = 30$ m. (b) Contour plot of estimated first-order intensity of $m_S = 159$ subadult longleaf pine trees in the same study area as (a); bandwidth $b = 30$ m.

are commonplace. Further discussion of $\lambda_2(\cdot, \cdot)$ is deferred until Section 4.3.2.

The study of spatial point processes has developed greatly in the last 15 years. It has been realized that many of the techniques in multivariate statistics have strong analogies in point processes. For example, the intensity function is like an unnormalized density function. Spatial dependence present in higher-order intensity functions is analogous to structure in multivariate density functions. Indeed, the theory for spatial point processes given in Section 4.3.3 shows that the spatial variability can be partitioned into $\Pr(Z(D_S) = m)$ and a conditional density function $f(s_1, \dots, s_m | m)$ that is proportional to the m th order intensity $\lambda_m(s_1, \dots, s_m)$. Therefore, many of the parametric models used in multivariate statistics have an analogue in spatial point processes (e.g., Møller and Waagepetersen, 2004; Illian et al., 2008).

4.3.1 The Poisson Point Process and the Cox Process

The canonical spatial point process that exhibits no spatial dependence (and is sometimes referred to as a *completely spatially random* process, or *CSR*) is the *homogeneous Poisson point process*. Here, $Z(A_1)$ and $Z(A_2)$ are independent whenever $A_1 \cap A_2 = \emptyset$, and $\lambda(s) \equiv \lambda^0 > 0$ in (4.160), for any $s \in D_S$. As a consequence (e.g., Rogers, 1974, pp. 13–14), $Z(A) \sim \text{Poi}(\lambda^0 |A|)$, which is the distributional result given in (4.159).

If we keep independence of $Z(\cdot)$ for non-overlapping sets, but choose a $\lambda(\cdot)$ in (4.160) that varies over D_S , the result is an *inhomogeneous Poisson point process*. Then,

$$Z(A) \sim \text{Poi}\left(\int_A \lambda(\mathbf{x}) d\mathbf{x}\right), \quad A \subset D_S. \quad (4.165)$$

The inhomogeneous Poisson point process is characterized by its first-order intensity $\lambda(\cdot)$; see, for example, Cressie (1993, Section 8.5.1).

If (4.165) is thought of as a data model in an HM, then an obvious process model is obtained by making $\lambda(\cdot)$ a spatial stochastic process. This is known as a Cox process (Cox, 1955); Møller and Waagepetersen (2004, Chapter 5) give a comprehensive discussion of the Cox process, which includes as special cases the Neyman–Scott process, the Thomas process, the Matérn Cluster process, the Shot-Noise Cox process, the Poisson-Gamma process, and the *Log-Gaussian Cox process* (LGCP). The latter process is often used in applications and it is given below in the form of an HM.

Data model: Conditional on $\lambda(\cdot)$, the point process $Z(\cdot)$ is an inhomogeneous Poisson point process with intensity $\lambda(\cdot)$.

Process model: Conditional on β and $C_Y(\cdot, \cdot)$, $Y(\cdot) \equiv \log \lambda(\cdot)$ is a Gaussian process with the following properties:

$$E(Y(\mathbf{s})) = \mathbf{x}(\mathbf{s})' \beta \quad \text{and} \quad C_Y(\mathbf{s}, \mathbf{x}) \equiv \text{cov}(Y(\mathbf{s}), Y(\mathbf{x})).$$

Suppose the data model is written generically as $[Z|Y]$ and the process model as $[Y]$. The Cox process is actually defined via marginalization, namely $[Z] = \int [Z|Y][Y]dY$. It is only relatively recently (e.g., Møller, Syversveen, and Waagepetersen, 1998) that the full power of an HM approach has been realized through the *joint* distribution, $[Z, Y] = [Z|Y][Y]$.

An optimal spatial predictor of $\lambda(\mathbf{s}_0)$ is $\lambda^*(\mathbf{s}_0)$, the posterior expectation of $\lambda(\mathbf{s}_0)$ given the point-process data (i.e., given the point pattern). The data are the number of events $Z(D_s)$ in D_s and the locations $\mathbf{s}_1, \dots, \mathbf{s}_{Z(D_s)}$ of those events. Hence, for the HM specified above, we have

$$\lambda^*(\mathbf{s}_0) = E(\exp(Y(\mathbf{s}_0)) | Z(D_s), \{\mathbf{s}_1, \dots, \mathbf{s}_{Z(D_s)}\}, \beta, C_Y(\cdot, \cdot)), \quad (4.166)$$

which is obtained from Bayes' Theorem. Recall that

$$\begin{aligned} & [Y(\cdot) | Z(D_s), \{\mathbf{s}_1, \dots, \mathbf{s}_{Z(D_s)}\}, \beta, C_Y(\cdot, \cdot)] \\ & \propto [Z(D_s), \{\mathbf{s}_1, \dots, \mathbf{s}_{Z(D_s)}\} | \exp(Y(\cdot))] [Y(\cdot) | \beta, C_Y(\cdot, \cdot)]. \end{aligned} \quad (4.167)$$

The second term in (4.167) is just the distribution of the Gaussian process Y . The first term is a distribution represented by the joint “density” function, $f((\mathbf{s}_1, \dots, \mathbf{s}_m), m)$, of the inhomogeneous Poisson point process (e.g., Cressie, 1993, p. 660):

$$\exp\left(-\int_{D_s} \lambda(\mathbf{x}) d\mathbf{x}\right) \times \begin{cases} 1, & m = 0, \\ \prod_{i=1}^m \lambda(\mathbf{s}_i) / m!, & m = 1, 2, \dots, \end{cases} \quad (4.168)$$

where $\lambda(\mathbf{s}) \equiv \exp(Y(\mathbf{s}))$; $\mathbf{s} \in D_s$. It will be seen in Section 4.3.3 that (4.168) can be used to simulate an inhomogeneous Poisson point process by first

drawing $Z(D_s)(=m)$ from a $Poi(\int_{D_s} \lambda(\mathbf{x})d\mathbf{x})$ distribution and then drawing $s_1, \dots, s_m$ iid from the density,

$$h(\mathbf{s}) \equiv \lambda(\mathbf{s}) / (\int_{D_s} \lambda(\mathbf{x}) d\mathbf{x}), \quad \mathbf{s} \in D_s. \quad (4.169)$$

The homogeneous Poisson point process plays a role analogous to that of a white-noise geostatistical process in Section 4.1; that is, the intensity function of events is constant, there is independence between counts in nonoverlapping regions, and (hence) the relative locations of events show neither positive spatial dependence (clustering) nor negative spatial dependence (regularity). The inhomogeneous Poisson point process represents a small generalization; it is analogous to adding a spatially varying trend term to a zero-mean white-noise process in Section 4.1. Consequently, the Poisson point process cannot be used to model spatial statistical dependence, such as that found in higher-order moment (i.e., intensity) measures. For example in the next subsection, the pair-correlation function (4.170) is seen to capture departures from “Poissonness.”

The next generalization introduces dependence through mixing of an independent process, namely the Cox process. On D_s , the density of the LGCP (with respect to a unit-intensity Poisson process) is proportional to the expected value of the expression (4.168), where recall that $\lambda(\cdot) = \exp(Y(\cdot))$ and the expectation is with respect to the Gaussian process Y . This is not generally known in closed form; however, it is not needed when taking an HM approach.

4.3.2 Spatial Statistical Dependence, Including the K Function

A spatial point process' second-order intensity function $\lambda_2(\cdot, \cdot)$, defined by (4.164), plays the analogous role to the covariance function for a geostatistical process (Section 4.1); we now consider some properties of λ_2 . If $Z(ds)$ and $Z(d\mathbf{x})$ are statistically dependent, then $\lambda_2(\mathbf{s}, \mathbf{x})$ cannot be expressed simply as $\lambda(\mathbf{s})\lambda(\mathbf{x})$. Thus, $\lambda_2(\mathbf{s}, \mathbf{x})$ can be used as a summary measure of spatial dependence in a spatial point process, and *the pair-correlation function*,

$$g(\mathbf{s}, \mathbf{x}) \equiv \lambda_2(\mathbf{s}, \mathbf{x}) / (\lambda(\mathbf{s})\lambda(\mathbf{x})), \quad (4.170)$$

is often used (Stoyan and Stoyan, 1994, Section 14.4; Baddeley et al., 2000). For the inhomogeneous Poisson process, $\lambda_2(\mathbf{s}, \mathbf{x}) = \lambda(\mathbf{s})\lambda(\mathbf{x})$; hence, in this case, $g(\mathbf{s}, \mathbf{x}) = 1$.

Now suppose that (4.170) can be written as $g(\mathbf{s}, \mathbf{x}) = g^*(\mathbf{s} - \mathbf{x})$, which is a type of second-order stationarity *after weighting* by $\lambda(\cdot)$. We see below that g^* leads to a more general definition of the K function.

Suppose that $Z(\cdot)$ is stationary and isotropic (i.e., invariant under translation and rotation); then

$$\lambda(\mathbf{s}) \equiv \lambda^o \quad (4.171)$$

and

$$\lambda_2(\mathbf{s}, \mathbf{x}) \equiv \lambda_2^o(\|\mathbf{s} - \mathbf{x}\|), \quad \mathbf{s}, \mathbf{x} \in D_s. \quad (4.172)$$

Furthermore, the pair-correlation function defined by (4.170) becomes,

$$g^o(h) \equiv \lambda_2^o(h)/(\lambda^o)^2, \quad h > 0.$$

A plot of $g^o(h)$ versus h (e.g., Stoyan and Stoyan, 1994, Figure 94) shows ranges of h where $g^o(h) < 1$, $= 1$, or > 1 . Small values of $g^o(h)$ occur if events at distance h -apart are rare; large values occur if events at distance h -apart appear frequently; a homogeneous Poisson point process has $g^o(h) \equiv 1$. In practice, $g^o(\cdot)$ is estimated by kernel-based methods (e.g., Illian et al., 2008, Section 4.3.3).

The K Function

For spatial point processes that are stationary and isotropic, there is another measure of spatial dependence, called the *K function* (Bartlett, 1964; Ripley, 1976). We now show a direct relationship between the second-order intensity function and the *K function*. Let $a(\mathbf{0}, dh)$ be an annulus of width dh on the sphere centered at $\mathbf{0}$ with radius h , and recall that $d\mathbf{0}$ is an infinitesimal region located at $\mathbf{0}$. Note that the surface area of a d -dimensional sphere of radius h is $d\pi^{d/2}h^{d-1}/\Gamma(1 + (d/2))$. Then, for $|d\mathbf{0}|$ and dh small, we obtain

$$\begin{aligned} E(Z(a(\mathbf{0}, dh))|Z(d\mathbf{0}) > 0) &= \frac{\Pr(Z(a(\mathbf{0}, dh)) > 0, Z(d\mathbf{0}) > 0)}{\Pr(Z(d\mathbf{0}) > 0)} \\ &= \frac{\lambda_2^o(h)\{d\pi^{d/2}h^{d-1}/\Gamma(1 + (d/2))\}dh}{\lambda^o}. \end{aligned}$$

Suppose $B(\mathbf{0}, h)$ is a d -dimensional sphere of radius h centered at $\mathbf{0}$. Then, consider

$$\begin{aligned} K(h) &\equiv (\lambda^o)^{-1} E(Z(B(\mathbf{0}, h))|Z(d\mathbf{0}) > 0) \\ &= (\lambda^o)^{-1} \int_0^h E(Z(a(\mathbf{0}, du))|Z(d\mathbf{0}) > 0) \\ &= (\lambda^o)^{-2} \{d\pi^{d/2}/\Gamma(1 + (d/2))\} \int_0^h u^{d-1} \lambda_2^o(u) du. \quad (4.173) \end{aligned}$$

Recall that the spatial point process is stationary and isotropic; hence, the quantity $E(Z(B(\mathbf{0}, h))|Z(d\mathbf{0}) > 0)$ is the expected number of extra events within distance h of an arbitrary event. Finally, the definition of the *homogeneous K function* is (Ripley, 1976)

$$K(h) \equiv (\lambda^o)^{-1} E \left(\frac{\text{number of extra events within}}{\text{distance } h \text{ of an arbitrary event}} \right), \quad h \geq 0. \quad (4.174)$$

Consequently, $K(\cdot)$ and $\lambda_2^o(\cdot)$ are related through the equation,

$$K(h) = (\lambda^o)^{-2} \{d\pi^{d/2} / \Gamma(1 + (d/2))\} \int_0^h u^{d-1} \lambda_2^o(u) du, \quad h \geq 0. \quad (4.175)$$

It is the K function that is famously used as a measure of spatial dependence in (stationary and isotropic) point processes. From (4.175), the second-order intensity is related directly to the K function by

$$\lambda_2^o(h) = \frac{(\lambda^o)^2 \Gamma(1 + (d/2))}{d\pi^{d/2}} \cdot \frac{K^{(1)}(h)}{h^{d-1}}, \quad h \geq 0, \quad (4.176)$$

where $K^{(1)}(h) \equiv dK(h)/dh$.

Also, from (4.175), the K function of the homogeneous Poisson point process in $\mathbb{R}^d$ is of the special form

$$K(h) = \pi^{d/2} h^d / \Gamma(1 + (d/2)), \quad h \geq 0. \quad (4.177)$$

In $\mathbb{R}^2$, (4.177) is given by

$$K(h) = \pi h^2, \quad h \geq 0. \quad (4.178)$$

Because the homogeneous Poisson point process represents a spatial point process with no spatial dependence, clustering or regularity can be detected more easily by considering the function

$$L(h) \equiv \{K(h)\Gamma(1 + (d/2))/\pi^{d/2}\}^{1/d} - h, \quad h \geq 0. \quad (4.179)$$

Ranges of h where the L function given by (4.179) is positive (negative) indicate clustering (regularity).

Estimation of the K function can be achieved by collecting and averaging pairs of events that are at distance h -apart or less from each other. Recall that D_s is a bounded region in $\mathbb{R}^d$. Then based on data $Z(D_s) = m$ and $\{s_1, \dots, s_m\}$, an obvious (but biased) estimator of $K(h)$ is

$$(\widehat{\lambda}^o)^{-1} \sum_{i=1}^m \sum_{\substack{j=1 \\ i \neq j}}^m I(\|s_i - s_j\| \leq h) / m,$$

where $\widehat{\lambda}_0 \equiv m/|D_s|$.

The bias in this estimator is caused by the randomness in $m = Z(D_s)$ and by *edge effects*; when an event in D_s is near its edge, the sphere of radius h centered at the event will not be wholly within D_s for larger h . Hence, there may be events outside D_s that should contribute to the estimator above; they

do not because they are unobserved. One way around this is to delete such events from the estimator.

Let d_j denote the distance from events s_j to the nearest edge of D_s ; $j = 1, \dots, m$. Then for $h > 0$, define the estimator

$$\widehat{K}(h) \equiv (|D_s|/m) \sum_{\substack{i=1 \\ i \neq j}}^m \sum_{j=1}^m I(\|s_i - s_j\| \leq h, d_j > h) / \sum_{j=1}^m I(d_j > h). \quad (4.180)$$

Other edge-corrected estimators are discussed by Cressie (1993, pp. 615–618). An estimator $\widehat{L}(\cdot)$ of the L function can be defined by substituting $\widehat{K}(\cdot)$ for $K(\cdot)$ in (4.179). In $\mathbb{R}^2$,

$$\widehat{L}(h) = (\widehat{K}(h)/\pi)^{1/2} - h, \quad h > 0. \quad (4.181)$$

Figures 4.17a and 4.17b show $\widehat{K}(\cdot)$ and $\widehat{L}(\cdot)$, respectively, for the subadult trees shown in Figure 4.15. In Figure 4.17b, we have superimposed 95% pointwise confidence limits. These are obtained from $\{\widehat{L}^{(\ell)}(\cdot): \ell = 1, \dots, 1000\}$, where each is estimated from 1000 simulations of CSR (Section 4.3.1) over the same region D_s and conditioned on there being 159 events. Clearly, from Figure 4.17, there is strong evidence of clustering at all scales.

From Figure 4.16a, we saw some heterogeneity in the first-order intensity function of adult trees. The K function also indicates mild clustering of adult trees at spatial scales less than 40 m; see Figure 4.18a, where the estimated L function is above the 95% confidence limits for $h < 40$ m. In order to calibrate this result, we show in Figure 4.18b the L function obtained from one simulated realization from CSR, conditioned on there being 271 events (the same as the number of adult trees). It appears that the adult trees are “settling down to a life of Poissonness”; contrast this to the subadult trees, which have yet to experience all the random attrition that an older tree would.

There is an *inhomogeneous* version of the K function based on $g^*(s - \mathbf{x})$ defined just below (4.170) that might be more appropriate for the longleaf pine trees. From (4.175), $K(h)$ is related to $g^o(h) = \lambda_2^o(h)/(\lambda^o)^2$ through integration. This motivates the definition

$$K^*(h) \equiv \int_{B(\mathbf{0}, h)} g^*(\mathbf{u}) d\mathbf{u}.$$

Since $g(s, \mathbf{x}) = g^*(s - \mathbf{x})$, it is true that for *any* set $A \subset \mathbb{R}^d$ we obtain

$$K^*(h) = \frac{1}{|A|} E \left\{ \sum_{s, \mathbf{x}}^* \frac{I(\|s - \mathbf{x}\| \leq h)}{\lambda(s)\lambda(\mathbf{x})} \right\},$$

where $\sum_{s, \mathbf{x}}^*$ denotes summation over all events $s \in Z \cap A$, $\mathbf{x} \in Z$, and such that $s \neq \mathbf{x}$. Kernel estimation of $K^*(\cdot)$ is discussed by Illian et al. (2008, Section 4.3.3); Guan (2009) gives a nonparametric variance estimator for $K^*(\cdot)$;

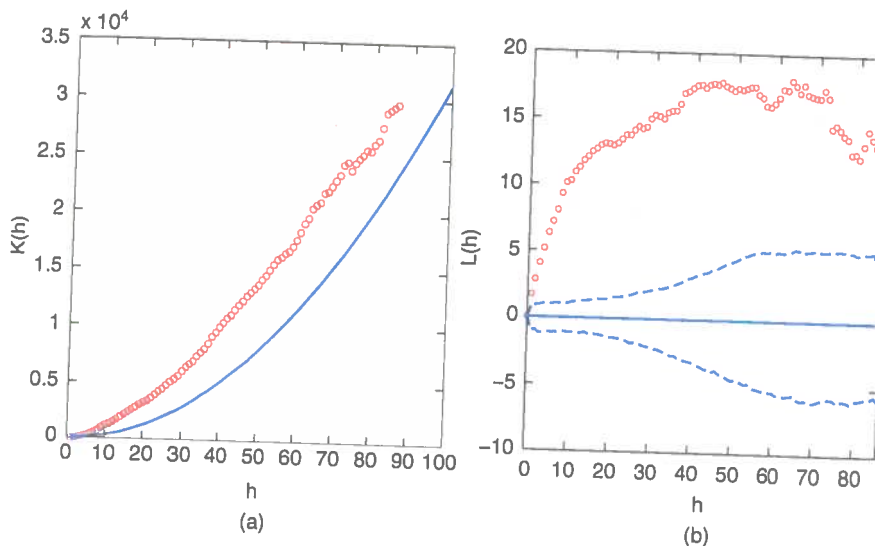


Figure 4.17 (a) Estimated K function for the $m_S = 159$ subadult longleaf pine trees (red circles) in the 4-ha study area in the Wade Tract; the theoretical K function for CSR (solid blue line) is superimposed. (b) Estimated L function for the $m_S = 159$ subadult longleaf pine trees (red circles) obtained from (a); the 95% pointwise confidence limits for L values based on 1000 CSR realizations (dashed blue lines) are superimposed.

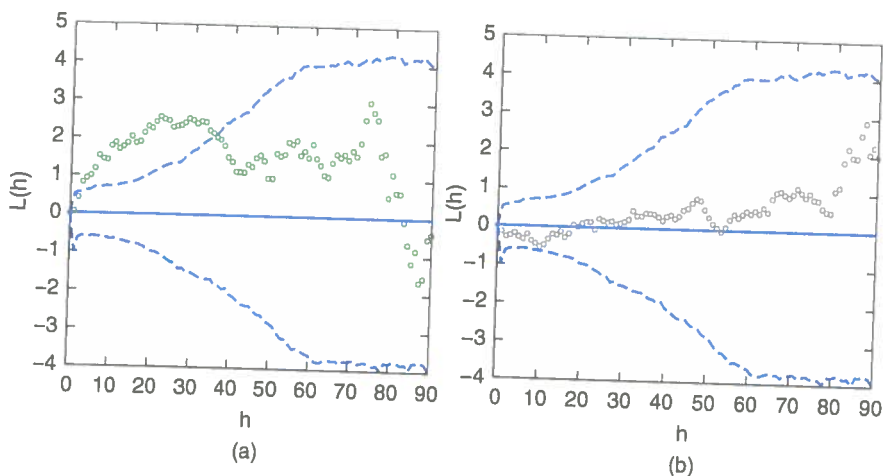


Figure 4.18 (a) Estimated L function for the $m_A = 271$ adult longleaf pine trees (green circles) in the 4-ha study area in the Wade Tract; the 95% confidence limits for L values based on 1000 CSR realizations (dashed blue lines) are superimposed. (b) Estimated L function for $m_A = 271$ locations generated from a single realization of a CSR process (black circles); the same 95% confidence limits for L values, as shown in (a) (dashed blue lines), are superimposed.

and Henrys and Brown (2009) present statistical inference for comparing two spatial point processes based on their respective $K^*(\cdot)$.

Dependence in the Log-Gaussian Cox Process

There are different ways that spatial statistical dependence can be modeled when the data are event locations in a spatial domain $D_s \subset \mathbb{R}^d$. Assuming an HM like that in Section 4.3.1 (e.g., with a LGCP), spatial dependence is present through a hidden geostatistical process that models the (log) intensity function. Then conditional on the intensity function $\lambda(\cdot)$, the events of $Z(\cdot)$ occur independently such that the number of events in non-overlapping subsets are independent Poisson. To emphasize further that the Poisson process has no spatial statistical dependence, it can be seen (e.g., Cressie, 1993, p. 651; Section 4.3.3) that conditional on $Z(D_s) = m$, the events $s_1, \dots, s_m$ are *independent* and identically distributed according to the density,

$$h(\mathbf{s}) = \lambda(\mathbf{s})I(\mathbf{s} \in D_s) / (\int_{D_s} \lambda(\mathbf{x}) d\mathbf{x}), \quad \mathbf{s} \in \mathbb{R}^d.$$

We have already mentioned that, marginally, the point process $Z(\cdot)$ is known as a *Cox process*, which inherits spatial dependence from its so-called directing measure $\lambda(\cdot)$. We shall now calculate the second-order intensity function of the LGCP $Z(\cdot)$, where recall that $\lambda(\cdot)$ is a log Gaussian process.

$$\begin{aligned} E(Z(ds)Z(d\mathbf{x})) &= E\{E(Z(ds)Z(d\mathbf{x})|\lambda(\cdot))\} \\ &= E\{\lambda(ds)\lambda(d\mathbf{x})\} \\ &= E(e^{Y(s)} \cdot e^{Y(\mathbf{x})})|ds| |d\mathbf{x}|, \end{aligned} \quad (4.182)$$

where $\lambda(\cdot) \equiv \exp\{Y(\cdot)\}$. Since $Y(\cdot)$ is a Gaussian process, then using well known results for the lognormal distribution (e.g., Aitchison and Brown, 1957) we see that for the LGCP, the first-order intensity function defined by (4.151) is $E(e^{Y(s)}) = \exp\{\mu_Y(s) + C_Y(s, s)/2\}$, and the second-order intensity function defined by (4.164) is

$$\lambda_2(\mathbf{s}, \mathbf{x}) = \exp\{\mu_Y(s) + \mu_Y(\mathbf{x}) + C_Y(s, s)/2 + C_Y(\mathbf{x}, \mathbf{x})/2 + C_Y(s, \mathbf{x})\}. \quad (4.183)$$

Only when $C_Y(s, \mathbf{x}) = 0$, can we write $\lambda_2(\mathbf{s}, \mathbf{x}) = \lambda(s)\lambda(\mathbf{x})$. Hence, *conditionally* on $\lambda(\cdot)$, $Z(\cdot)$ has *no* spatial statistical dependence, but *marginally* $Z(\cdot)$ is an LGCP with spatial statistical dependence characterized by (4.183). The $g(\mathbf{s}, \mathbf{x}) = \exp\{C_Y(s, \mathbf{x})\}$, which is a function of $\mathbf{s} - \mathbf{x}$ if C_Y is stationary. More generally, spatial statistical dependence in the process model for $\lambda(\cdot)$ leads to spatial dependence in the Cox process $Z(\cdot)$.

4.3.3 Distribution Theory for Spatial Point Processes

Recall we assumed that $Z(\cdot)$ is simple (i.e., with probability 1, at any spatial location, only single events can occur). Then a spatial point process $Z(\cdot)$ can be characterized by a sequence of nonnegative, permutation-invariant intensity functions, $\{\lambda_m(\mathbf{s}_1, \dots, \mathbf{s}_m): m = 1, 2, \dots\}$, such that the conditional density function is expressed as

$$\begin{aligned} f(\mathbf{s}_1, \dots, \mathbf{s}_m | m) d\mathbf{s}_1 \dots d\mathbf{s}_m \\ \equiv \Pr(Z(d\mathbf{s}_1) = 1, \dots, Z(d\mathbf{s}_m) = 1 | Z(D_s) = m) \\ = \frac{\lambda_m(\mathbf{s}_1, \dots, \mathbf{s}_m) d\mathbf{s}_1 \dots d\mathbf{s}_m}{\int_{D_s} \dots \int_{D_s} \lambda_m(\mathbf{x}_1, \dots, \mathbf{x}_m) d\mathbf{x}_1 \dots d\mathbf{x}_m}, m = 1, 2, \dots, \end{aligned} \quad (4.184)$$

and the count distribution in D_s is given by

$$\begin{aligned} \Pr(Z(D_s) = m) &\equiv p_m \\ &\propto \begin{cases} e^{-|D_s|}, & m = 0, \\ (e^{-|D_s|}/m!) \int_{D_s} \dots \int_{D_s} \lambda_m(\mathbf{x}_1, \dots, \mathbf{x}_m) d\mathbf{x}_1 \dots d\mathbf{x}_m, & m = 1, 2, \dots, \end{cases} \end{aligned} \quad (4.185)$$

where $\sum_{m=0}^{\infty} p_m = 1$ (Daley and Vere-Jones, 1988, p. 129).

For example, because the (inhomogeneous) Poisson point process has independent counts in disjoint regions, for that model, (4.184) becomes

$$f(\mathbf{s}_1, \dots, \mathbf{s}_m | m) = \prod_{i=1}^m \{\lambda(\mathbf{s}_i) / \int_{D_s} \lambda(\mathbf{x}) d\mathbf{x}\},$$

and from (4.185),

$$p_m = \exp(-\int_{D_s} \lambda(\mathbf{x}) d\mathbf{x}) \{ \int_{D_s} \lambda(\mathbf{x}) d\mathbf{x} \}^m / m!, \quad m = 0, 1, \dots,$$

which is the probability mass function of a Poisson distribution. Consequently, the simulation of the Poisson point process is straightforward: Generate $Z(D_s) \sim \text{Poi}(\int_{D_s} \lambda(\mathbf{x}) d\mathbf{x})$ and, conditional on $Z(D_s) = m$, generate $\mathbf{s}_1, \dots, \mathbf{s}_m$ iid according to the density, $\lambda(\cdot) / \int_{D_s} \lambda(\mathbf{x}) d\mathbf{x}$. In the special case of a homogeneous Poisson point process, this latter density is just the uniform density, $1/|D_s|$.

The likelihood for a simple spatial point process is the Radon–Nikodym derivative of the point-process measure with respect to the unit-intensity Poisson-process measure. That is,

$$\ell(\theta) = e^{-|D_s|} \lambda_m(\mathbf{s}_1, \dots, \mathbf{s}_m; \theta), \quad (4.186)$$

where λ_m is on the right-hand side of (4.184) and has been written to emphasize any unknown parameters θ . The simplicity of (4.186) belies its difficulty to use in practice. Typically, it is only known up to a normalizing constant (that is a function of θ). Geyer (1999) discusses optimization of (4.186). Alternatively, Jensen and Møller (1991) and Baddeley and Turner (2000) optimize a composite likelihood based on the Papangelou conditional intensity, while Guan (2006) bases his composite likelihoods on pairwise loglikelihoods.

In a classical Bayesian setting, there is a prior distribution on θ . Based on this prior and (4.186), the posterior distribution $[\theta|Z]$ is available through an MCMC algorithm (Møller et al., 2006).

Now, the m th intensity function λ_m in (4.185) can be factored uniquely into its Gibbsian form (e.g., Daley and Vere-Jones, 1988, p. 129):

$$\begin{aligned} \lambda_m(s_1, \dots, s_m) \propto \exp\left\{\sum_i g_i(s_i) + \sum_{i < j} g_{ij}(s_i, s_j) + \dots \right. \\ \left. + g_{12\dots m}(s_1, s_2, \dots, s_m)\right\}, \end{aligned} \quad (4.187)$$

where it is important to remember that the proportionality constant here and in (4.184) is chosen so that $\sum_{m=0}^{\infty} p_m = 1$. For example, suppose $g_i(s) \equiv \log \lambda^o$; $g_{ij}(s, u) \equiv (\log \gamma) I(\|s - u\| \leq \rho)$; and all higher-order g -functions are zero, where $\theta \equiv (\lambda^o, \gamma, \rho)'$ are parameters. Then

$$\lambda_m(s_1, \dots, s_m) = (\lambda^o)^m \gamma^{Y_m(\rho)} / c(\lambda^o, \gamma, \rho), \quad (4.188)$$

where $Y_m(\rho)$ is the number of pairs of event locations in $\{s_1, \dots, s_m\}$ whose distances are less than or equal to ρ , and $c(\lambda^o, \gamma, \rho)$ is a normalizing constant to guarantee that $\sum_{m=0}^{\infty} p_m = 1$. This spatial point process is known as the *Strauss process*; its parameter space is $\{(\lambda^o, \gamma, \rho) : c(\lambda^o, \gamma, \rho) < \infty\} = (0, \infty) \times [0, 1] \times [0, \infty)$; see Kelly and Ripley (1976). For the Strauss process, the likelihood is

$$\ell(\lambda^o, \gamma, \rho) = e^{-|D_s|} (\lambda^o)^m \gamma^{Y_m(\rho)} / c(\lambda^o, \gamma, \rho).$$

An infinite-order interaction model is considered by Widom and Rowlinson (1970).

Notice that the decomposition (4.187) is obtained by direct analogy with the expression for the negpotential $Q(\cdot)$ in terms of the G -functions for lattice processes; see (4.107).

Markov Point Processes

A *Markov point process* is obtained when the representation (4.187) of λ_m is truncated at a fixed number of terms, for all $m = 1, 2, \dots$. These have also been

called *Gibbs processes*, because they were originally introduced into physics by Gibbs (1902); they have a long history in Statistical Mechanics.

An example of a Markov point process is when, for all i and j , $g_i(\mathbf{s})$ is $\xi(\mathbf{s}; \boldsymbol{\beta})$, $g_{i,j}(\mathbf{s}, \mathbf{u}) = \psi(\|\mathbf{s} - \mathbf{u}\|; \boldsymbol{\theta})$, and all the other g -functions are zero; here, $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ denote unknown parameters. Large-scale variation in Markov point processes can be captured through covariates $\mathbf{x}(\cdot)$, by putting $\xi(\cdot, \boldsymbol{\beta}) = \mathbf{x}(\cdot)' \boldsymbol{\beta}$. When $\xi(\cdot; \boldsymbol{\beta})$ is a constant, we obtain the *pair-potential Markov point process*, and a special case of this is the Strauss process given by (4.188), where $\xi(\mathbf{s}; \boldsymbol{\beta}) = \log(\lambda^0)$; $\mathbf{s} \in D_s$. In that case,

$$\psi(\|\mathbf{s} - \mathbf{u}\|; \boldsymbol{\theta}) = (\log \gamma) I(\|\mathbf{s} - \mathbf{u}\| \leq \rho). \quad (4.189)$$

Finally, a Markov point process with general $\xi(\mathbf{s}; \boldsymbol{\beta})$ and pair-potential given by (4.189) with $\rho = 0$, corresponds to no spatial dependence. This is an inhomogeneous Poisson point process with intensity function,

$$\lambda(\mathbf{s}) = \exp(\xi(\mathbf{s}; \boldsymbol{\beta})), \quad \mathbf{s} \in D_s.$$

4.3.4 Disease Mapping from Event Locations

In Section 4.2.6 we saw examples of HMs on lattices, in particular a discussion of disease mapping. The data were small-area counts, and inference was on (hidden) relative risks. After introducing an HM for disease-event locations, we shall see that the HM in that section can be derived straightforwardly.

An HM for Disease-Event Locations

To keep it simple, we assume that individuals at risk come from a homogeneous population (e.g., a single sex-age stratum, like males, age 65–74 years). The spatial locations of these *individuals at risk* are modeled as “events” of an inhomogeneous Poisson point process $Z^0(\cdot)$, with first-order intensity function $\delta(\cdot)$. Generally, $\delta(\cdot)$ is known, obtained from independent demographic studies. From the random locations of individuals at risk, there are some who have the disease being studied. We model their locations randomly, via a “thinning function,”

$$T(\mathbf{s}) = \begin{cases} 1, & \text{if an individual at } \mathbf{s} \text{ has the disease,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.190)$$

For noninfectious diseases (e.g., cancer), we can assume that for $\mathbf{s}_1 \neq \mathbf{s}_2$, $T(\mathbf{s}_1)$ and $T(\mathbf{s}_2)$ are independent. The risk function, defined in terms of probability, is

$$p(\mathbf{s}) \equiv E(T(\mathbf{s})) = \Pr(\text{individual at } \mathbf{s} \text{ has the disease}), \quad \mathbf{s} \in D_s. \quad (4.191)$$

This is a primary process of interest, and epidemiologists want to estimate it and map it.

Define a new spatial point process,

$$Z(ds) \equiv T(s)Z^0(ds), \quad s \in D_s, \quad (4.192)$$

where recall that ds is an infinitesimal region located at s . Then by a well known result for thinned Poisson point processes (e.g., Cressie, 1993, p. 690), $Z(\cdot)$ is also a Poisson point process with (first-order) intensity function, $p(\cdot)\delta(\cdot)$. Since $\delta(\cdot)$ is assumed known from demographic studies, $p(\cdot)$ is the source of randomness in the process-model part of the following HM.

Data model: Conditional on $p(\cdot)$ (and $\delta(\cdot)$), the point process $Z(\cdot)$ is an inhomogeneous Poisson point process with intensity function $p(\cdot)\delta(\cdot)$.

Process model: Conditional on β and $C_v(\cdot, \cdot)$,

$$p(\cdot) \equiv \exp\{\mathbf{x}(\cdot)'\beta + v(\cdot)\},$$

where $v(\cdot)$ is a zero-mean Gaussian process with covariance function $C_v(\cdot, \cdot)$, and $\mathbf{x}(\cdot)$ is a vector of covariates.

Data from the observed point pattern is $Z(D_s) = m$ and $\{s_1, \dots, s_m\}$, from which we wish to make inference on β and $v(\cdot)$ (and hence on $p(\cdot)$). In particular, there may be interest in the null hypothesis that certain β_i are equal to zero. Furthermore, a *risk map*, $\{p(s) : s \in D_s\}$, could be inferred by computing the posterior distribution,

$$[p(\cdot)|Z(D_s), \{s_1, \dots, s_{Z(D_s)}\}].$$

Of course, $\mathbf{x}(\cdot)'\beta + v(\cdot)$ has to be negative to ensure $p(\cdot)$ is a probability. If this is problematic, a logistic transformation could be used in the process model instead.

Aggregation of Events to Small Areas

Now suppose that we have the region $D_s = \bigcup_{j=1}^J A_j$, where $\{A_j : j = 1, \dots, J\}$ are disjoint small areas. Then $\mathbf{Z} \equiv (Z(A_1), \dots, Z(A_J))'$ represent the number of disease cases in the J small areas. Moreover, conditional on $p(\cdot)$, the data model above implies that

$$Z(A_j) \sim \text{ind. Poi}\left(\int_{A_j} p(\mathbf{u})\delta(\mathbf{u}) d\mathbf{u}\right), \quad j = 1, \dots, J. \quad (4.193)$$

The mean number of individuals at risk in the j -th small area is $N_j \equiv \int_{A_j} \delta(\mathbf{u}) d\mathbf{u}$, $j = 1, \dots, J$. Now make the assumption that $p(\cdot)$ is (approximately) constant within each of the $\{A_j\}$. That is, for $s \in A_j$, write $p(s) \equiv p_j$; $j = 1, \dots, J$. Consequently,

$$Z(A_j) \sim \text{Poi}(N_j p_j), \quad (4.194)$$

and an analogous development of (4.133) through (4.136) yields the relative risks,

$$\lambda_j \equiv \frac{p_j}{(E_j/N_j)}, \quad (4.195)$$

where recall that $E_j \equiv N_j \bar{p}$ is the baseline expected number of disease incidences; $j = 1, \dots, J$, and $\bar{p} \equiv \sum_{j=1}^J p_j/J$ is the average risk. That is, under aggregation of the data model and the process model into small areas, $\{A_j : j = 1, \dots, J\}$, the previous HM has an aggregated form.

Data model: Conditional on $\{\lambda_j\}$ (and $\{E_j\}$), and for $j = 1, \dots, J$,
 $Z(A_j) | \lambda_j \sim \text{ind. Poi}(E_j \lambda_j)$.

Process model: Conditional on β , θ_C , and for $\mathbf{Y} \equiv (\log \lambda_1, \dots, \log \lambda_J)'$,
 $\mathbf{Y} | \beta_C, \theta \sim \text{Gau}(\mathbf{X}\beta, \Sigma(\theta_C))$,

where $\Sigma(\theta_C)$ is a $J \times J$ positive-definite matrix that depends on covariance parameters θ_C .

For example, for the HM following (4.136)–(4.140) in Section 4.2.6, $\Sigma(\theta_C)$ was chosen to be a MRF plus a component representing extra, non-spatial variation. However, it is not generally known how to disaggregate such a model so that it corresponds to a well-defined spatial point process.

If the piecewise-constant assumption for $p(\cdot)$ is dropped, the data model is still

$$Z(A_j) \sim \text{Poi}(N_j p_j), \quad (4.196)$$

but where now p_j is the weighted average,

$$p_j = \int_{A_j} p(\mathbf{u}) \delta(\mathbf{u}) d\mathbf{u} / \int_{A_j} \delta(\mathbf{x}) d\mathbf{x}. \quad (4.197)$$

Now this requires knowledge of $\delta(\cdot)$ at the point level; unfortunately, it may only be known at the small-area level, where it has been assumed that demographic studies yield $\{N_j = \int_{A_j} \delta(\mathbf{x}) d\mathbf{x} : j = 1, \dots, J\}$. Hence, keeping the piecewise-constant assumption for $p(\cdot)$ can be viewed as a simplifying assumption that allows the statistical analysis to move forward. A Bayesian way to deal with $\delta(\cdot)$ unknown would be to put a prior on it that respects the constraints, $N_j = \int_{A_j} \delta(\mathbf{x}) d\mathbf{x}$; $j = 1, \dots, J$.

Aggregation of the point-level process model given in this section requires careful calibration. Noting that $N_j p_j = E_j \lambda_j$, $j = 1, \dots, J$, then making inference on the relative risks, $\{\lambda_j\}$, is equivalent to making inference on the risks, $\{p_j\}$. Intuitively, areas that have historically had more individuals at risk would have a risk value that is more certain because there would be a lot more

past disease outcomes to assess. For this reason, Cressie (1992) proposed a heteroskedastic model for $\{\lambda_j\}$, such that $\text{var}(\lambda_j)$ is inversely related to E_j , $j = 1, \dots, J$. Define $Y_j \equiv \log \lambda_j$; $j = 1, \dots, J$. Then, assuming a CAR model for $\mathbf{Y}$ (Section 4.2.5), the heteroskedasticity is captured through

$$\text{var}(Y_j | \mathbf{Y}_{-j}) = \tau^2 / E_j, \quad (4.198)$$

and through the spatial-rates CAR model (Cressie, Perrin, and Thomas-Agnan, 2005),

$$\Sigma(\theta_C) = \tau^2 (\mathbf{I} - \phi \mathbf{H})^{-1} \Delta, \quad (4.199)$$

where $\theta_C = (\tau^2, \phi)'$,

$$\Delta \equiv \text{diag}(1/E_1, \dots, 1/E_J), \quad \text{and}$$

$$(\mathbf{H})_{ij} = (E_j^{1/2} / E_i^{1/2}) I(A_j \text{ is a neighbor of } A_i).$$

There is a curious lacuna between the spatial point process models used in disease mapping from event locations, and the Markov random field models used for (aggregated) small area counts. Aggregation of a point-level log-Gaussian Cox process defined on D_s , does *not* yield an easily interpretable (i.e., “sparse”) MRF defined on $\{A_j\}$ and, in general, aggregation of a “sparse” MRF does not yield a “sparse” aggregated MRF. Therefore, in spite of a tradition in the disease-mapping literature to use MRFs (particularly CAR models), perhaps the spatial process models of choice should be geostatistical-type models, which are able to maintain “sparseness” under aggregation?

4.3.5 Other Topics

Marked Point Processes

Recall the example shown in Figure 4.15 of longleaf pine tree locations in 4-ha of a natural forest. The trees have varying size, and if their *diameters at breast height (DBH)* are measured, along with their location, then a spatial statistical model should account for tree size through an additional process (e.g., Rathbun and Cressie, 1994). In fact, Figure 4.15 shows the trees stratified according to whether they are considered adults or subadults, a stratification that is defined by their DBH.

Any accompanying measurements associated with the events of the spatial point process are considered a sampling from a process that is called the *mark process*, and we denote it by $M(\cdot)$. A marked spatial point process on D_s consists of the initial spatial point process, $Z(\cdot)$. Then, conditional on the realization of $Z(\cdot)$, namely $Z(D_s)$, $\mathbf{s}_1, \dots, \mathbf{s}_{Z(D_s)}$, there is a subsequent (geostatistical or lattice) model for the mark process, $M(\cdot)$:

$$[M(\mathbf{s}_1), \dots, M(\mathbf{s}_{Z(D_s)}) | Z(\cdot)].$$

For example, see Rathbun and Cressie (1994) for a spatial statistical model (conditional on the initial point process) for DBHs of longleaf pine trees in a natural forest. For marked spatial point processes in an HM, the data model is partitioned into two levels, namely the top level for the spatial point process $Z(\cdot)$ and the next level for the mark process $M(\cdot)$, conditional on $Z(\cdot)$.

An example of an HM for a marked spatial point process can be easily developed, as follows.

Data model: Conditional on $\lambda(\cdot)$, the point process $Z(\cdot)$ is an inhomogeneous Poisson process with intensity $\lambda(\cdot)$.

Conditional on $Z(D_s) = m$, $\mathbf{s}_1, \dots, \mathbf{s}_m$, and σ_ε^2 ,

$$M(\mathbf{s}_i) | L(\mathbf{s}_i), \sigma_\varepsilon^2 \sim \text{ind. Gau}(L(\mathbf{s}_i), \sigma_\varepsilon^2)$$

Process model: Conditional on β_Y and $C_Y(\cdot, \cdot)$, the log intensity function, $Y(\cdot) \equiv \log \lambda(\cdot)$, is a Gaussian process with the following properties:

$$E(Y(\mathbf{s})) = \mathbf{x}_Y(\mathbf{s})' \beta_Y,$$

$$\text{cov}(Y(\mathbf{s}), Y(\mathbf{x})) \equiv C_Y(\mathbf{s}, \mathbf{x}).$$

Conditional on $Y(\cdot)$, $L(\cdot)$, is a Gaussian process whose first moment depends on covariates $\mathbf{x}_L(\cdot)$ and possibly $Y(\cdot)$, and whose second moments exhibit spatial statistical dependence.

Further properties of marked spatial point processes can be found, for example, in Cressie (1993, Section 8.7), Baddeley and van Lieshout (1995), Schlather (2001), and Schlather, Ribeiro, and Diggle (2004).

Models for Location Error

In much of spatial statistics, it is assumed that spatial locations, $\mathbf{s}_1, \dots, \mathbf{s}_m$, are known precisely. In reality, *both* attribute and location are subject to measurement errors. The locational information often comes from global positioning systems (GPSs), which are commonplace in modern society. Each comes with a declaration of its locational precision; generally, the greater the precision of the GPS, the higher its price!

Suppose that $Z^0(\cdot)$ is a point process on $D_s^0 \subset \mathbb{R}^d$, which is characterized by its number of events, $Z^0(D_s^0) = m$, and the event locations, $\{\mathbf{s}_1^0, \dots, \mathbf{s}_m^0\}$. These might be locations of trees in a natural forest D_s , such as shown in Figure 4.15. If the trees in the forest are mapped using a GPS, then the resulting point pattern of trees is in D_s , where D_s is equal to D_s^0 augmented by a buffer zone to account for every tree's location in the forest. Consequently, if we model the point pattern as being generated by a point process $Z(\cdot)$, then $Z(D_s) = Z^0(D_s^0)$, in distribution. A reasonable model for the event locations of $Z(\cdot)$ is

$$\mathbf{s}_i = \mathbf{s}_i^0 + \varepsilon_i, \quad i = 1, \dots, m, \quad (4.200)$$

for $\epsilon_1, \dots, \epsilon_m$ iid random vectors in $\mathbb{R}^d$ that represent location error (e.g., the GPS's measurement error). The model (4.200) for location error was called the *feature-positioning model* by Cressie and Kornak (2003). Dobrushin (1963) showed that if the location-error variances tend to infinity, then under regularity conditions, $Z(\cdot)$ converges to a homogeneous Poisson point process no matter what the underlying point process $Z^0(\cdot)$ is.

The feature-positioning model can be expressed as an HM; an example is given below.

Data model: Conditional on $Z^0(\cdot)$ and R_ϵ , the point process $Z(\cdot)$ is given by $Z(D_s) = Z^0(D_s^0)$, and

$$\mathbf{s}_i = \mathbf{s}_i^0 + \epsilon_i, \quad i = 1, \dots, m,$$

where $\{\mathbf{s}_i^0\}$ are the event locations of Z^0 in D_s^0 , $\{\epsilon_i\}$ are iid random d -dimensional locations uniform on the ball, $b(\mathbf{0}, R_\epsilon) \equiv \{\mathbf{x}: \|\mathbf{x}\| \leq R_\epsilon\}$, and $D_s = \bigcup_{\mathbf{s} \in D_s^0} b(\mathbf{s}, R_\epsilon)$.

Process model: Conditional on β and θ , $Z^0(\cdot)$ is a Markov point process with regression parameters β in $\xi(\mathbf{s}; \beta) = \mathbf{x}(\mathbf{s})'\beta$ and small-scale spatial dependence parameters θ in (4.189).

At the level of the process model, we could alternatively express spatial dependence in the point process $Z^0(\cdot)$ as follows.

Process model 1: Conditional on $\lambda(\cdot)$, $Z^0(\cdot)$ is an inhomogeneous Poisson point process with intensity $\lambda(\cdot)$.

Process model 2: Conditional on β and $C_Y(\cdot, \cdot)$, $Y(\cdot) \equiv \log \lambda(\cdot)$ is a Gaussian process with the following properties:

$$E(Y(\mathbf{s})) = \mathbf{x}(\mathbf{s})'\beta \quad \text{and} \quad C_Y(\mathbf{s}, \mathbf{x}) \equiv \text{cov}(Y(\mathbf{s}), Y(\mathbf{x})).$$

Diggle (1993, pp. 97–99) provides some results that relate to estimation of parameters of the hidden spatial point process $Z^0(\cdot)$, from data $Z(D_s^0) = m$ and $\mathbf{s}_1, \dots, \mathbf{s}_m$. For a stationary, isotropic $Z^0(\cdot)$ and a radially symmetric distribution for the location errors $\{\epsilon_i\}$, the first-order intensity function for $Z(\cdot)$ is the same as that for $Z^0(\cdot)$. However, when $Z^0(\cdot)$ exhibits clustering, Diggle's general conclusion is that the K function for $Z(\cdot)$ is smaller than that for $Z^0(\cdot)$. That is, K -function estimators based on the data $Z(D_s^0) = m$ and $\mathbf{s}_1, \dots, \mathbf{s}_m$, generally underestimate the K function for the hidden, location-error-free process $Z^0(\cdot)$.

Fanshawe and Diggle (2011) extend the feature-positioning model of Cressie and Kornak (2003) by adopting a BHM for a marked spatial point process that

is based on (4.200). Their goal is optimal spatial prediction of the mark process.

Directed Markov Point Process (DMPP)

Recall from Section 4.2.8, the discussion of Partially Ordered Markov Models (POMMs), where a partial order (equivalently an acyclic directed graph) defines the spatial dependence in lattice data. This directional type of dependence in lattice processes has an analogue for spatial point processes. Intuitively, this is not surprising, if we view a point process as an ultra-fine-grid lattice process with pixel ds_j located at pixel center $\mathbf{s}_j$; $j = 1, \dots, J$, where $\bigcup_{j=1}^J ds_j = D_s$, and $Z(ds_j) = 1$ or 0, according to a spatially dependent binary process.

If a lattice process is a binary POMM on an ultra-fine grid, and hence it has a directed type of spatial dependence, we would expect it to approximate a spatial point process that also has a directed type of spatial dependence. Under regularity conditions, including $\max\{|ds_j|: j = 1, \dots, J\} \rightarrow 0$ as $J \rightarrow \infty$, the binary POMM should converge to a spatial point process; Cressie et al. (2000b) show this to be the case. They define *Directed Markov Point Processes (DMPPs)* as limits of binary POMMs.

Define the coordinate-wise *partial order* (notated as $\prec$) between two locations in $\mathbb{R}^d$ as

$$\mathbf{s} \prec \mathbf{x} \text{ if } s_1 \leq x_1 \text{ and } s_2 \leq x_2 \dots \text{ and } s_d \leq x_d,$$

where $\mathbf{s} \equiv (s_1, \dots, s_d)'$ and $\mathbf{x} \equiv (x_1, \dots, x_d)'$. Then the cone of $\mathbf{x} \in \mathbb{R}^d$ is defined as

$$\text{cone}(\mathbf{x}) \equiv \{\mathbf{s} \in D_s: \mathbf{s} \prec \mathbf{x} \text{ and } \mathbf{s} \neq \mathbf{x}\}.$$

Furthermore, let $\psi(\mathbf{s}, \mathbf{x})$ denote a spatial interaction function. Then Cressie et al. (2000b) show that a (homogeneous) DMPP can be written in terms of the joint “density” function as follows:

$$\begin{aligned} & f((\mathbf{s}_1, \dots, \mathbf{s}_m), m) \\ &= \exp\{-|D_s|\} \exp\left\{-\lambda \int_{D_s} \prod_{\mathbf{s}_j \in \text{cone}(\mathbf{x})} \psi(\mathbf{x}, \mathbf{s}_j) d\mathbf{x}\right\} \lambda^m \prod_{i=1}^m \prod_{\mathbf{s}_j \in \text{cone}(\mathbf{s}_i)} \psi(\mathbf{s}_i, \mathbf{s}_j). \end{aligned} \quad (4.20)$$

In fact, DMPPs can be defined more generally, in terms of a directed conditional intensity function, which characterizes the spatial dependence in the DMPP. More details of the definition and the properties of this type of point process can be found in Baddeley, Nair, and Cressie (2002).

4.4 RANDOM SETS

Recall from Chapter 1 our discussion of the object view (versus the field view) of spatial (and dynamical) phenomena. This book is largely about the field view, but in this section we shall discuss the role of random objects in an HM framework. Consider the data to be a collection of objects whose variability we describe using the theory of random sets (Matheron, 1971, 1975). For example, Cressie and Hulting (1992) used random sets to describe the evolution of breast cancer cells growing in a nutrient medium on a glass plate. The datum at a given time was an image of breast cancer cells on the plate.

Analogous to the theory of random variables, where variability is modeled as a real-valued measurable mapping from a probability space to a measure space on the real numbers, a *random set* in $\mathbb{R}^d$ is a measurable mapping from a probability space into a particular measure space, which we now describe. Define $\mathcal{F}$ to be the set of all closed subsets of $\mathbb{R}^d$ (including the empty set $\emptyset$). Let $\mathcal{K}$ denote the set of all compact sets, and let $\mathcal{K}' \equiv \mathcal{K} \setminus \emptyset$ be the set of all nonempty compact sets. For any $A \subset \mathbb{R}^d$, define

$$\mathcal{F}_A \equiv \{F \in \mathcal{F}: F \cap A \neq \emptyset\} \quad \text{and} \quad \mathcal{F}^A \equiv \{F \in \mathcal{F}: F \cap A = \emptyset\}. \quad (4.202)$$

For K compact and $G_1, \dots, G_n$ open, generate all sets of the form $\mathcal{F}^K \cap \mathcal{F}_{G_1} \cap \dots \cap \mathcal{F}_{G_n}$. Matheron (1975) has shown that this class of subsets of $\mathcal{F}$ is a *base* of a topology on $\mathcal{F}$ (which he called the *hit-or-miss topology*) and that the topological space is compact, Hausdorff, and separable. Furthermore, Matheron (1975, p. 28) proved that only hit-or-miss information, either on the set of all compact sets or on the set of all open sets, is needed to specify a random set; we return to this point later. Equipped with a topology on $\mathcal{F}$, one can now be rigorous about convergence of a sequence of closed sets. Furthermore, by taking countable unions and intersections of the open sets of the topological space $\mathcal{F}$, a σ -algebra $\sigma_{\mathcal{F}}$ on $\mathcal{F}$ can be generated.

A *random closed set* or RACS (which is often just called a *random set*) is defined as a measurable mapping Y from a probability space $(\Omega, \mathcal{A}, Q)$ into the measure space $(\mathcal{F}, \sigma_{\mathcal{F}})$. This is a very general definition of a random quantity, which includes the definitions of familiar random quantities, such as random variables, random vectors, point processes, and so on. Let $\text{Pr}(\cdot)$ be the law of Y , namely, the probability induced on $\sigma_{\mathcal{F}}$ by

$$\text{Pr}(\nu) \equiv Q(Y^{-1}(\nu)), \quad \nu \in \sigma_{\mathcal{F}}.$$

4.4.1 Hit-or-Miss Topology

The hit-or-miss topology is basic to the theory of random sets. It was chosen because it reflects the way image data in $\mathbb{R}^d$ are analyzed; that is, its roots are in practical applications. When studying an object Y (e.g., a work of art, whose authenticity is in question), some sort of systematic (nondestructive)

probing is needed, which leads to the use of structuring elements B (chosen independently of Y) to check whether “ B hits Y ” ($B \cap Y \neq \emptyset$) or “ B misses Y ” ($B \cap Y = \emptyset$).

It can be shown that all the interesting set transformations (dilation, erosion, opening, closing, convexification, etc.) of a RACS Y are themselves RACS (Matheron, 1975, p. 28). The analogue to this for a random variable Y is that a transformation $g(Y)$ is also a random variable; this is true provided $g(\cdot)$ is measurable. Therefore, for a RACS to be useful in an HM approach to modeling random objects, the development of a collection of random-set models is essential. How can the models be specified? What are the important events that make two random sets different? For a partial answer, we return to the hit-or-miss topology. If we can specify $\Pr(Y \in \mathcal{F}^K \cap \mathcal{F}_{G_1} \cap \dots \cap \mathcal{F}_{G_k})$ for all compact K , for all open $G_1, \dots, G_k$, and for all integers k , in a consistent way, then Y is well defined. Fortunately, a great reduction of test sets is possible.

Definition: For any $K \in \mathcal{K}$, define the *hitting function* T as

$$T(K) \equiv \Pr(Y \in \mathcal{F}_K) = \Pr\{Y \cap K \neq \emptyset\}. \quad (4.203)$$

Henceforth, we refer to K in $\Pr\{Y \cap K \neq \emptyset\}$ as a *test set*.

The hitting function T defined in (4.203) has the following properties (Matheron, 1975, p. 29):

1. $T(\emptyset) = 0$ and $0 \leq T \leq 1$.
2. T is increasing.
3. T satisfies the following recurrence relations. For any $k \geq 0$, let $S_k(B_0; B_1, \dots, B_k)$ denote the probability that Y misses B_0 but hits $B_1, \dots, B_k$. Then

$$\begin{aligned} S_0(B_0) &= 1 - T(B_0) \geq 0 \\ S_1(B_0; B_1) &= T(B_0 \cup B_1) - T(B_0) \geq 0 \\ &\vdots \\ S_k(B_0; B_1, \dots, B_k) &= S_{k-1}(B_0; B_1, \dots, B_{k-1}) \\ &\quad - S_{k-1}(B_0 \cup B_k; B_1, \dots, B_{k-1}) \geq 0. \end{aligned}$$

That is, T is a *Choquet capacity of infinite order* (Matheron, 1975, p. 30).

A powerful result, proved independently by Matheron (1971) and by Kendall (1974), is Choquet's Theorem in the context of random-set theory; it says that the converse of property 3. above is true. In other words, if a given T on $\mathcal{K}$ is a Choquet capacity of infinite order, there exists a *necessarily unique* P_T on $\sigma_{\mathcal{F}}$ such that

$$P_T(\mathcal{F}_K) = T(K), \quad \text{for all } K \in \mathcal{K}. \quad (4.204)$$

An immediate example of its use is when the RACS Y is a simple spatial point process (i.e., no more than one event at any location) in $\mathbb{R}^d$. In Section 4.3, we denoted a point process as $Z(\cdot)$, and $Z(A)$ is the number of points in $A \subset D_s \subset \mathbb{R}^d$. Then Choquet's Theorem says the point process is completely specified from

$$\begin{aligned} T(K) &= 1 - \Pr(Y \cap K = \emptyset) \\ &= 1 - \Pr(Z(K) = 0), \quad \text{for all } K \in \mathcal{K}, \end{aligned}$$

where recall that the RACS Y is the point process $Z(\cdot)$ in this example. This observation, that the point process is uniquely determined from $\{\Pr(Z(K) = 0) : K \in \mathcal{K}\}$, was made by Ripley (1976, p. 989). It is tempting to bracket Choquet's Theorem with the result that says that a random variable Y is well defined once the probabilities of events $\{Y \leq y\}$, for all $y \in (-\infty, \infty)$, are specified in a consistent way. But the two results, while being similar, are not exactly analogous. In fact, Choquet's Theorem for a random variable, where the RACS Y is a one-point set in $\mathbb{R}^1$, involves "test sets," $\{[a, b] : -\infty < a \leq b < \infty\}$; a little more work is needed to modify the necessary test sets down to $\{(-\infty, y] : -\infty < y < \infty\}$. It is in this domain where results (and hence random-set models) are scarce, namely finding ways to reduce the hitting-function test sets down from the full complement $\mathcal{K}$.

There are other ways to model uncertainty in objects and patterns. The theory of random patterns (i.e., Pattern Analysis) due to Grenander (1996) has proved amenable to incorporation into an HM (Grenander and Miller, 2007).

4.4.2 Hierarchical Models for Objects Based on Random Sets

There are a number of ways that random-set models of objects can be used in an HM. The *observed* RACS is denoted as Z , which we consider to be a "noisy" version of the hidden RACS Y . For the purpose of illustration, suppose that the true process Y is the union of compact sets. Then the *observed* boundaries of the individual compact sets could be modeled through independent Brownian Bridges superimposed on the respective boundary loci of the compact sets. [A Brownian Bridge with mean zero is a Gaussian process that has the property that it starts at 0 and always returns to 0 on a finite interval (e.g., Glasserman, 2004).] Then Z is the union of the sets defined by the *observed* boundaries.

Given the hierarchical nature of the model, it is straightforward to simulate from the joint distribution. Formally,

$$[Z, Y] = [Z|Y][Y];$$

first simulate Y from $[Y]$ and then Z from $[Z|Y]$. The posterior distribution is proportional to the joint distribution; that is,

$$[Y|Z] \propto [Z, Y] = [Z|Y][Y],$$

but the proportionality constant, which is $1/[Z]$, can be difficult to obtain.

In principle, a Metropolis or Metropolis–Hastings algorithm could be implemented to obtain a simulation from $[Y|Z]$; see Section 2.3.2. This approach to simulating from the posterior distribution is quite general and could be used for making inference on a hidden object Y , given an observed object Z .

Sets can also appear in spatial statistics as supports of density or intensity functions. Recall the inhomogeneous Poisson point process with intensity $\{\lambda(s) : s \in D_s\}$. In the example of trees in a natural forest, there may be a region $B \subset D_s$ where soil conditions are conducive to growth, but not so outside B ; that is, $\lambda(s) \equiv 0$ for $s \in D_s \setminus B$. Based on data, $Z(D_s) = m$ and $\{s_1, \dots, s_m\}$ on the point process $Z(\cdot)$, we may wish to make inference on the support B . Define $Y(s) \equiv I(s \in B) = 1$ if $s \in B$, and zero otherwise. Then the following HM is well defined for making inference on the indicator function $Y(\cdot)$ (equivalently, on B).

Data model: Conditional on $B_0 \subset D_s$ and $\mu_0(\cdot)$, the point process $Z_1(\cdot)$ is an inhomogeneous Poisson point process with intensity $\lambda_1(s) \equiv \mu_0(s)I(s \in B_0)$; $s \in D_s$.

Process model 1: Conditional on β and $C_Y(\cdot, \cdot)$, $Y_0(\cdot) \equiv \log \mu_0(\cdot)$ is a Gaussian process on B_0 with the following properties:
 $E(Y_0(\cdot)) = \mathbf{x}(\cdot)' \beta$, and we write $C_Y(\mathbf{u}, \mathbf{v}) \equiv \text{cov}(Y_0(\mathbf{u}), Y_0(\mathbf{v}))$.

Process model 2: Conditional on random-set model parameters θ_R , B_0 is a RACS defined by its hitting function,

$$T(K; \theta_R) \equiv \Pr(K \cap B_0 \neq \emptyset), \quad K \in \mathcal{K}.$$

Figure 4.20a shows a simulation of the HM on the unit square: First, B_0 is a (discretized) Boolean model (Section 4.4.3); a realization is obtained by superimposing Figure 4.19a onto a fine spatial grid, resulting in the set of black pixels shown in Figure 4.19b. Second, a realization of $\mu_0(\cdot) \equiv \exp(Y_0(\cdot))$, a log-Gaussian process, is obtained (see Figure 4.19c). Third, $\lambda_1(s) \equiv \mu_0(s)I(s \in B_0)$ is shown in Figure 4.19d. Finally, a realization of the spatial point process $Z_1(\cdot)$ is shown in Figure 4.20a.

4.4.3 The Boolean Model

The random set simulated in Figure 4.19b is called a *Boolean model*, which is obtained by centering independent realizations of a compact random set S in $\mathbb{R}^d$ at each point of a realization $\{\mathbf{u}_i\}$ of a homogeneous Poisson point process Z_0 in $\mathbb{R}^d$, and taking the union of all the centered compact sets. That is,

$$B = \bigcup_i (S_i \oplus \mathbf{u}_i), \quad (4.205)$$

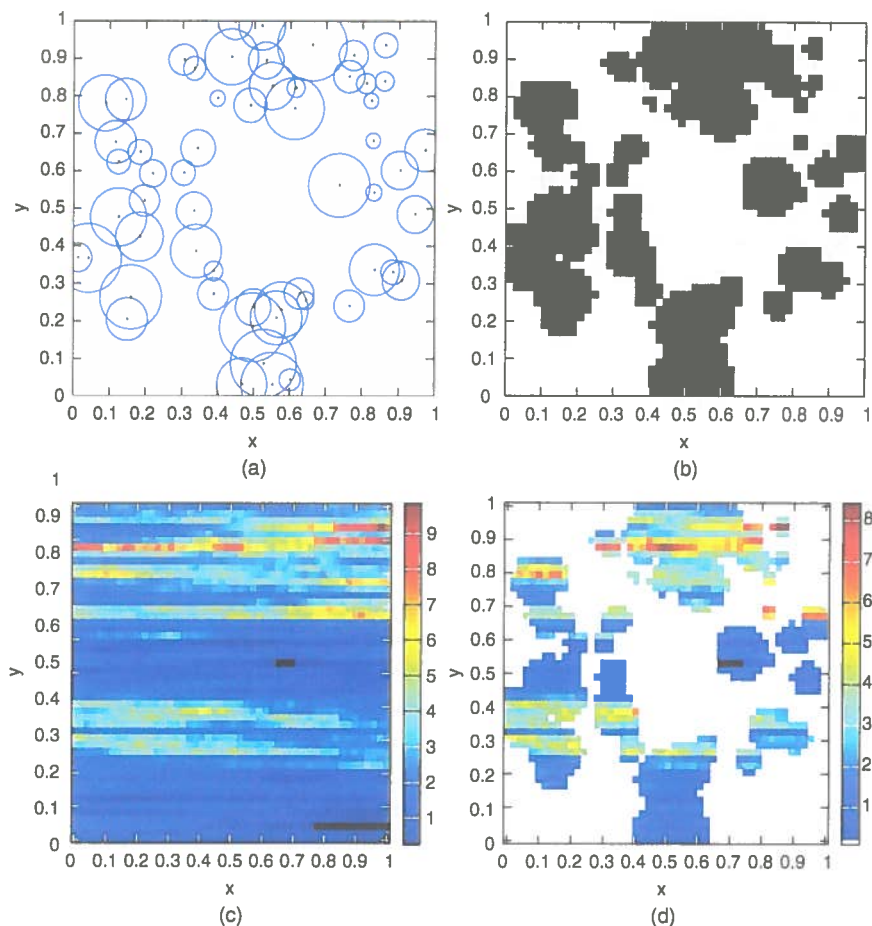


Figure 4.19 (a) Randomly generated disk centers and disk radii in the unit square D_s . A realization of the Boolean model is the union of the disks centered around the $Z_0(D_s) = 57$ foci locations shown. The 57 foci are from the CSR point process with intensity $\lambda^o = 50$, and the 57 radii are a random sample from the uniform distribution, $U[.02, 1]$. (b) Discretized Boolean model B_0 on a regular 50×50 grid of pixels on the unit square; its area $|B_0| = .3980$. A pixel is declared to be contained in the (discretized) Boolean model if the pixel center is contained in at least one of the disks shown in (a). (c) Log-Gaussian process discretized onto the 50×50 grid. The Gaussian process defining it is stationary and has mean 0, variance 1, and covariance function $C(\cdot)$ given by Figure 6.3, considered as a two-dimensional spatial covariance function. (d) Intensity function defined by the log-Gaussian process on the Boolean model, and defined to be zero elsewhere.

where $S \oplus \mathbf{u} \equiv \{\mathbf{s} + \mathbf{u} : \mathbf{s} \in S\}$; the random sets $\{S_i\}$ are independent RACS identically distributed as S ; and $\{\mathbf{u}_i\}$ are the events of Z_0 . Thus, there are two sources of randomness in the model, namely the homogeneous Poisson point process characterized by its intensity, λ^o , and the probability measure associated with the compact random set, S .

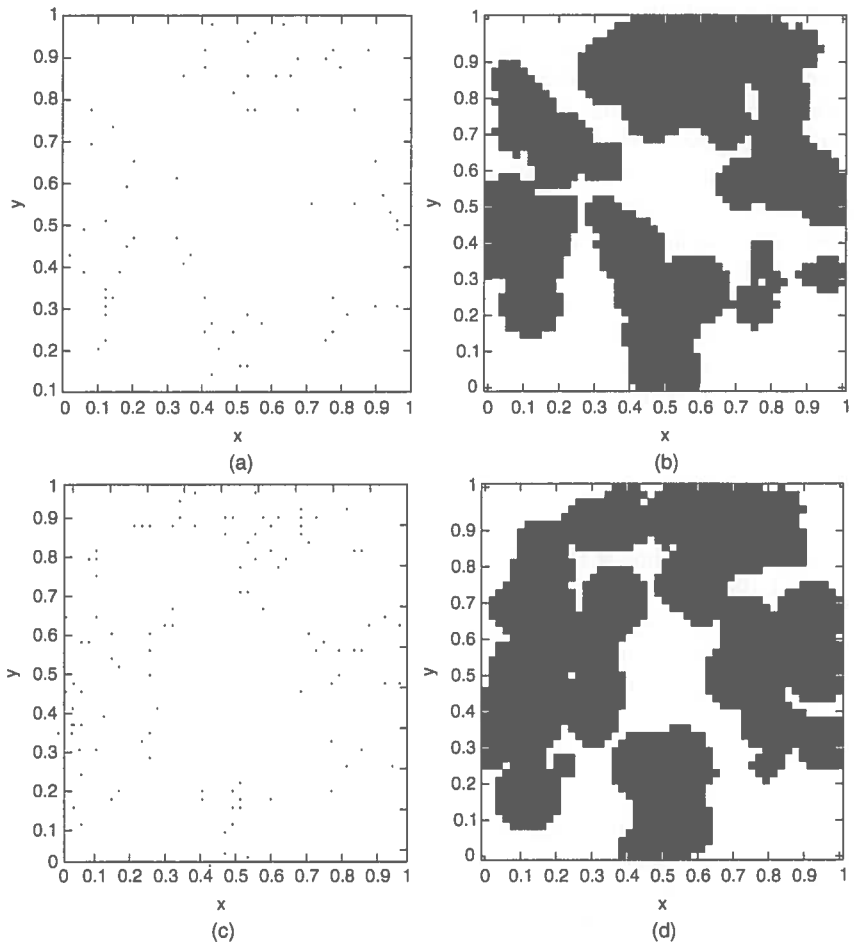


Figure 4.20 (a) Randomly generated foci of growth in the unit square, generated according to an inhomogeneous Poisson process with intensity function given by Figure 4.19d; there are $Z_1(D_S) = 70$ foci. (b) The (discretized) Boolean model B_1 is obtained the same way as in Figure 4.19b, using the same $U[.02, .1]$ distribution to generate the random radii; its area is $|B_1| = .5420$, indicating growth. (c) Using the same log-Gaussian process as in Figure 4.19c, the intensity function restricted like in Figure 4.19d yields randomly generated foci with $Z_2(D_S) = 114$. (d) The (discretized) Boolean model B_2 is obtained the same way as Figure 4.19b, using the same $U[.02, .1]$ distribution to generate the random radii; its area is $|B_2| = .6092$, indicating growth.

The hitting function of B given by (4.205) can be calculated. From the random set S , define $\check{S} \oplus K \equiv \{-s + v : s \in S \text{ and } v \in K\}$. Then the hitting function of B is (e.g., Cressie, 1993, p. 754)

$$T(K) = 1 - \exp\{-\lambda^0 E(|\check{S} \oplus K|)\}, \quad K \in \mathcal{K}, \quad (4.206)$$

where recall that $|A|$ is the d -dimensional volume of a measurable set A . The Boolean model is a very useful model for random sets because its hitting

function is available in closed form, and because it is formed through a simple operation (union) on compact sets.

Cressie and Hulting (1992) developed a set-valued autoregressive process to model tumor growth that is very much like the process model shown in select panels of Figures 4.19 and 4.20. *In vitro* breast-cancer cells on a glass slide at time 1 are randomly declared to be foci of growth with local growth rates that are also considered random. The local growth is modeled as a disk of random radius, and the *union* of all disks centered at the foci makes up the set of all cancer cells at time 2.

In what follows, we show how to simulate a process model for a sequence of random sets $\{B_0, B_1, B_2, \dots\}$ that exhibit “random growth.” Let Z_1 denote the inhomogeneous spatial Poisson point process whose events represent the foci of growth; a realization is given in Figure 4.20a. Its first-order intensity function is, for $B_0 \subset D_s$ and $Y_0(\cdot)$ a Gaussian process,

$$\lambda_1(s) = \exp\{Y_0(s)\}I(s \in B_0), \quad s \in D_s. \quad (4.207)$$

For example, Figure 4.19b shows B_0 , Figure 4.19c shows $\exp\{Y_0(\cdot)\}$, and Figure 4.19d shows $\lambda_1(\cdot)$ given by (4.207). Then Figure 4.20a is a realization from Z_1 with $Z_1(D_s) = 70$ event locations, $\{\mathbf{u}_{1i} : i = 1, \dots, 70\}$. Now we can start the whole sequence of steps again to generate a set-valued autoregressive process. Figure 4.20b shows

$$B_1 \equiv \bigcup_i (S_{1i} \oplus \mathbf{u}_{1i}), \quad (4.208)$$

a generalized Boolean model whose foci $\{\mathbf{u}_{1i}\}$ are generated by the *inhomogeneous* Poisson point process Z_1 ; in (4.208), $\{S_{1i}\}$ are *iid* random disks with center $\mathbf{0}$ and random radius $U[.02, .1]$.

By replacing B_0 with B_1 , $Y_0(\cdot)$ with $Y_1(\cdot)$, and repeating, we obtain Z_2 and then B_2 in the same way B_1 was obtained in (4.208). Figure 4.20c shows Z_2 and Figure 4.20d shows B_2 . The model that was simulated shows growth: $|B_0| = .3980$, $|B_1| = .5420$, and $|B_2| = .6092$. It is also possible to model a receding tumor through low intensities $\exp\{Y_0(\cdot)\}$, $\exp\{Y_1(\cdot)\}$, etc., and small disk radii of $\{S_{1i}\}$, $\{S_{2i}\}$, etc. Cressie and Hulting (1992) carry out statistical inference on intensities and local growth rates for *in vitro* breast cancer cells observed 72 hours apart. This discussion around Figures 4.19 and 4.20 opens up the possibility of building dynamical spatio-temporal set processes, something we refer to briefly in Section 6.5. There is a growing literature in growth processes using random sets (e.g., Jónsdóttir, Schmiegel, and Jensen, 2008).

The Boolean model’s parameters can be estimated by discretizing its 0–1 values onto a fine grid and then using the method-of-moments on (4.206) for different test sets K (e.g., Cressie and Laslett, 1987). Writing down the Boolean model’s likelihood has always been difficult to achieve, except for the simplest cases, because of the overlapping nature of the sets $\{S_i \oplus \mathbf{u}_i : i = 1, 2, \dots\}$. The various combinations of possibilities of how an $s \in D_s$ might be in or out

of the Boolean model B has made the likelihood elusive. To deal with this, Møller and Helisova (2009) choose a model where there is dependence between the events $\{u_i\}$ and the sets $\{S_i\}$, they then define a conditional likelihood that is more conducive to MCMC analysis than the regular likelihood, and finally they use MCMC to (approximately) optimize the conditional likelihood.

Micheas and Wikle (2009) take a different approach and consider a random non-overlapping disk model to model objects from a hierarchical perspective. They model random disks and random foci in a multistage hierarchical Bayesian framework. The BHM approach facilitates inference without the need for tedious calculation of the hitting probabilities. Critically, their approach is based on modeling a point from the random set, instead of the set itself, which allows direct inference on the parameters of the model via MCMC. This also allows inference on *individual* random disks in the model. They also demonstrate how this methodology can be used for modeling the growth of objects.

4.5 BIBLIOGRAPHIC NOTES

A number of texts that cover spatial random processes have been mentioned in the introduction to this chapter. Review articles and other texts that give presentations of topics at various levels of difficulty include Yaglom (1957, 1962), Matérn (1960), Bartlett (1975), Cliff and Ord (1981), Ripley (1981), Upton and Fingleton (1985), Ripley (1988), Upton and Fingleton (1989), Cressie (1991, 1993), Possolo (1991), Bailey and Gatrell (1992), Hjort and Omre (1994), Gregoire et al. (1997), Griffith and Layne (1999), Moore (2001), Griffith (2003), Haining (2003), Møller (2003), Banerjee, Carlin, and Gelfand (2004), Waller and Gotway (2004), Longley et al. (2005), Schabenberger and Gotway (2005), Le and Zidek (2006), Bivand, Pebesma, and Gómez-Rubio (2008), Fortin and Dale (2009), LeSage and Pace (2009), Gaetan and Guyon (2010), and Gelfand et al. (2010, Part II).

This chapter has divided up naturally into four types of spatial processes. These are geostatistical processes, spatial lattice processes, spatial point processes, and random-set processes.

Geostatistical Processes

Matheron (1962, 1963) and Gandin (1963) wrote comprehensive accounts of statistics for spatial processes for audiences in mining and atmospheric sciences, respectively. Matheron coined the term “geostatistics,” and a comprehensive account of mining geostatistics appears in the books by David (1977), Journel and Huijbregts (1978), Clark (1979), and Chilés and Delfiner (1999). Rivoirard (2005) summarizes the concept and methods in a review chapter. Various other accounts of geostatistical processes, for different purposes, can be found in Yaglom (1962), Yadrenko (1983), Bras and Rodriguez-Iturbe (1985), Isaaks and Srivastata (1989), Brooker (1991), Christakos (1992), Goovaerts (1997), Kitanidis (1997), Müller (1998), Hohn (1999), Stein (1999), Webster and Oliver (2001), and Lantuéjoul (2002).

Ripley (1981), Christensen (1991), and Cressie (1991, 1993) present a statistician's perspective of the subject and draw on examples beyond Mining, particularly in Crop Science, Hydrology, and the Environmental Sciences. These references do not use hierarchical models, by and large. Banerjee, Carlin, and Gelfand (2004) do, but they do not always distinguish clearly between the measurement error and the microscale variation of the latent spatial process $Y(\cdot)$, occasionally resulting in spatial prediction of only the smooth component of $Y(\cdot)$. Le and Zidek (2006) and Diggle and Ribeiro (2007) also give a statistician's perspective, and their Bayesian presentations involve prediction of the hidden spatial process $Y(\cdot)$. Banerjee et al. (2008) and Finley et al. (2009) predict $Y(\cdot)$ through predictive processes defined on support points. Arab (2007) and Lindgren et al. (2011) use a spectral representation of $Y(\cdot)$ derived from a Galerkin expansion of a stochastic PDE.

Measures of Spatial Dependence

Fundamental to kriging and generally to calculating the predictive/posterior distribution is specification of the spatial covariance function. The variogram, along with the variance function, will often play the same role. Estimation of these important quantities in various scenarios has recently been considered by Stein, Chi, and Welty (2004), Zhang (2004), Stein (2005), Zhang and Zimmerman (2007), and Kaufman, Schervish, and Nychka (2008). An innovative approach to variogram estimation (and spatial prediction) based on block empirical likelihoods (which avoids specification of a joint likelihood from a parametric family) can be found in Nordman (2008a,b) and in Nordman and Caragea (2008).

Spatial dependence can also be summarized in the spectral domain, where estimation can likewise proceed (e.g., Fuentes, 2002, 2007; Im, Stein, and Zhu, 2007; Huang, Hsing, and Cressie, 2010). Other summaries of spatial dependence include the joint distribution and the spatial cumulative distribution function (Lahiri et al., 1999).

Covariance functions that are locally stationary have been considered by Silverman (1957) and by Sampson and Guttorp (1992). They transform spatial coordinates into a space where the covariance function is isotropic. Paciorek and Schervish (2006) consider process convolutions using inhomogeneous kernel functions. Global data have also fostered the need for nonstationary covariance models (Section 4.1.4; Cressie and Johannesson, 2008; Jun and Stein, 2008). The use of copulas to model spatial dependence is a more recent approach (e.g., Bárdossy and Li, 2008).

Proving that estimators of these measures of spatial dependence are consistent is complicated, requiring notions of a spatial domain that grows, along with samples that infill the domain. These issues are discussed by Ivanov and Leonenko (1989), Lahiri (1996), Lahiri et al. (1999), Stein (1999), and Park et al. (2009).

Spatial Prediction and Mapping

Kriging can be equated to spatial best linear unbiased prediction (Cressie, 1990). It is a special instance (i.e., it is linear) of optimal spatial prediction of

a hidden process $Y(\cdot)$ based on noisy, incomplete data $\mathbf{Z}$. This more general problem is solved by computing the posterior distribution $[g(Y(\cdot))|\mathbf{Z}, \hat{\theta}]$ for an EHM, or $[g(Y(\cdot))|\mathbf{Z}]$ for a BHM, where $g(Y(\cdot))$ is some functional of interest, like $Y(s_0)$, $Y(B)$, $\nabla Y(s_0)$ (Banerjee and Gelfand, 2006), or $\exp(Y(s_0))$ (e.g., Cressie, 2006). A point predictor is obtained by summarizing the posterior distribution by its mean, or median, or mode, and so on.

The idea of using process models with parameters that are themselves processes was presented by Wikle, Berliner, and Cressie (1998) in the spatio-temporal context and was considered later by Gelfand et al. (2003) in the spatial-only context. Non-Gaussian HMs were considered in Section 4.1.2, to which contributions by Christensen and Waagepetersen (2002), Zhang (2002, 2003), Gelfand, Kottas, and MacEachern (2005), Cressie (2006), Monestiez et al. (2006), Zhu, Gu, and Peterson (2007), Crainiceanu, Diggle, and Rowlingson (2008), Musio, Augustin, and von Wilpert (2008), and Berliner et al. (2008) should be added.

Optimal spatial prediction of extremes of $Y(\cdot)$ is a nonlinear prediction problem, for which kriging is not well suited, since it “fills in the valleys” and “cuts off the hills.” De Oliveira and Ecker (2002), Cooley, Nychka, and Naveau (2007), and Zhang, Craigmile, and Cressie (2008) use HMs for optimal spatial prediction of extremes.

Change-of-Support

In the climate and remote-sensing communities, the problem of change-of-support is usually referred to as *downscaling*, where inference is on $Y(s_0)$ at the point or small-grid-cell level, based on data $Z(B_1), \dots, Z(B_m)$. Upscaling is then concerned with inference on $Y(B)$ based on data $Z(s_1), \dots, Z(s_m)$. It is the latter problem that mining engineers are particularly interested in, since ore is mined and sent to the mill in units made up of a block B . Also, environmental remediation or management of an ecological resource is typically made at the block level. Ecologists use the term *scale* instead of support, in the context of pattern and scale (e.g., Levin, 1992).

Change-of-support is commonplace in spatial statistics. Openshaw and Taylor (1979) show how hard the problem is in general to solve. The downscaling problem has been considered from a geostatistical viewpoint by Kyriakidis (2004). A comprehensive review of the statistical problem of change-of-support has been written by Gotway and Young (2002), and Section 4.1.3 features other approaches.

Spatial Moving Average (SMA) and Multiresolution Statistical Models

The SMA models are easy to define and easy to simulate from. They also offer a convenient way to model complicated spatial dependence with reduced rank models; see the review by Wikle (2010a). The temptation is to let the moving average weights (equivalently, process-convolution kernels) be as general as possible, but then they typically lack physical interpretation and lead to non-sparse covariance matrices. Choosing which reduced-rank model to use can be

guided by the physical structure of the process. In the context of deterministic smoothing, Johannesson and Cressie (2004) give a method for choosing from among several families of spherical basis functions on the globe, based on a measure of *spatial coherence*. In the context of Fixed Rank Kriging, where the emphasis is on modeling a spatial covariance function, Shi and Cressie (2007) give a method for choosing among multiresolution wavelet basis functions. Beyond a general recommendation of multiresolutional structure, the question of basis-function choice is open. For the change-of-resolution models (Section 4.1.6), the random effects are “drawn upward” from fine resolution to coarse resolution through hierarchical representations (Kolaczyk, 1999; Huang and Cressie, 2001; Kolaczyk and Nowak, 2004; Huang, Cressie, and Gabrosek 2002; and Tzeng, Huang, and Cressie, 2005).

Multivariate Spatial Processes

There are classes of multivariate spatial models that assume symmetric dependence through $\text{cov}(Y_1(\mathbf{s}), Y_2(\mathbf{s} + \mathbf{h})) = \text{cov}(Y_2(\mathbf{s}), Y_1(\mathbf{s} + \mathbf{h}))$, most notably, the linear model of coregionalization, or LMC (e.g., Wackernagel, 1995; Webster, Atteia, and Dubois, 1994; Banerjee, Carlin, and Gelfand, 2004, Chapter 7). However, there is no *a priori* reason for this type of symmetry to hold. Ver Hoef and Cressie (1993) avoid making this assumption by defining and working with cross-variograms. Sain and Cressie (2007) and Sain, Furrer, and Cressie (2011) allow asymmetry parameters in their multivariate (lattice) spatial models and note an asymmetric multivariate spatial dependence in the data they analyze. Gneiting, Kleiber, and Schlather (2010) define Matérn cross-covariance functions that show considerable flexibility, including the possibility of asymmetry.

Generally, the goal is spatial prediction of the multivariate spatial process $\mathbf{Y}(\cdot)$, or of one component, say $Y_1(\cdot)$, based on multivariate (noisy) spatial data, $\{\mathbf{Z}(s_1), \dots, \mathbf{Z}(s_m)\}$. When the component $Y_1(\cdot)$ is predicted in an EHM setting, the associated methodology is called *cokriging*. References for multivariate spatial prediction include Myers (1982, 1992), Ver Hoef and Cressie (1993), Wackernagel (1995), Cressie and Wikle (1998b), and Huang et al. (2009). In a BHM setting, see the articles by Majumdar and Gelfand (2007) and by Finley et al. (2008).

Banerjee, Carlin, and Gelfand (2004) and Gelfand et al. (2004) make a comparison of the LMC defined in Section 4.1.5, with the conditional-probability approach in (4.83), which deserves revisiting, here for the bivariate case $K = 2$. Under the LMC, they report properties of the conditional-probability approach given by (4.83) that look unusual to them, which they interpret as casting doubt on the general applicability of (4.83). In fact, there are a couple of misconceptions that, once identified, clear up these unusual-looking properties. First, taking the conditional-probability approach requires modeling $[\mathbf{Y}_2|\mathbf{Y}_1] = [Y_2(s_1), \dots, Y_2(s_n)|\mathbf{Y}_1]$. Any consideration of the conditional behavior of a component of $\mathbf{Y}_2$, say $Y_2(s_i)$, is always referred to the conditional distribution $[Y_2(s_i)|\mathbf{Y}_1]$, *never* to $[Y_2(s_i)|Y_1(s_i)]$. Part of Gelfand et al.’s (2004) discussion

refers to the behaviors of $[Y_2(\mathbf{s}_1)|Y_1(\mathbf{s}_1)], \dots, [Y_2(\mathbf{s}_n)|Y_1(\mathbf{s}_n)]$. They point out, correctly, that trying to interpret these quantities in some joint manner makes no sense. However, (4.83) considers the joint behavior of $\{Y_2(\mathbf{s}_1), \dots, Y_2(\mathbf{s}_n)\}$, conditional on the *vector* $\mathbf{Y}_1$. The reason for Gelfand et al.'s misconception appears to come from observing that, in the LMC,

$$\mathbf{Y}_2|\mathbf{Y}_1 \sim \text{Gau}\left(\frac{a_{21}}{a_{11}}\mathbf{Y}_1, a_{22}^2\boldsymbol{\Sigma}_{2|1}\right),$$

where the (i,j) th entry of $\boldsymbol{\Sigma}_{2|1}$ is $\rho_2(\mathbf{s}_i - \mathbf{s}_j)$. They then notice that in equation (4.82), $\mathbf{W}_2 \equiv (W_2(\mathbf{s}_1), \dots, W_2(\mathbf{s}_n))' \sim \text{Gau}(\mathbf{0}, \boldsymbol{\Sigma}_{2|1})$. Finally, they put $\mathbf{W}_2(\mathbf{s})$ equal to the error in the regression equation; that is, they write

$$Y_2(\mathbf{s}_i) = \frac{a_{21}}{a_{11}}Y_1(\mathbf{s}_i) + a_{22}W_2(\mathbf{s}_i), \quad i = 1, \dots, n.$$

It is not clear to us, either algebraically or statistically, how this equation above is correct.

We conclude that the conditional-probability approach, when considered properly, has many modeling advantages. Even when $K \geq 3$, there are often nested relations that allow a decomposition of the joint probability (e.g., for $K = 3$, $[\mathbf{Y}_1, \mathbf{Y}_2, \mathbf{Y}_3] = [\mathbf{Y}_3|\mathbf{Y}_1, \mathbf{Y}_2][\mathbf{Y}_2|\mathbf{Y}_1][\mathbf{Y}_1]$).

Bayesian Hierarchical Geostatistical Model Computations

Computational issues in the Bayesian implementation of hierarchical spatial models are covered in the books by Banerjee, Carlin, and Gelfand (2004) and by Diggle and Ribeiro (2007), as well as in the review chapter by Wile (2010b). The dominant method used is a Markov chain Monte Carlo (MCMC) approach. In cases where parameters are random, which we argue are more practical, extra care must often be taken to balance computational considerations (e.g., convergence, efficiency) with the choice of the prior distribution on the parameters. For example, Gaussian-process priors typically include parameters associated with the variance, spatial dependence, and smoothness. In many Bayesian analyses in the literature, the smoothness parameter is assumed to be known (as a percentage of total variance), and attention is focussed on the variance and spatial-dependence parameters. For example, inverse gamma or discrete uniform priors are often chosen for the spatial-dependence parameter. In certain cases, one can use more "objective" priors for spatial parameters (e.g., Berger, De Oliveira, and Sansó, 2001; Paulo, 2005), but typically the variance parameters are given a relatively noninformative conjugate prior (e.g., inverse gamma). However, the use of noninformative priors for variance components can be problematic (see Browne and Draper, 2006; Gelman, 2006). It is increasingly common to reparameterize the spatial dependence parameters (e.g., see Zhang, 2004; Christensen, Roberts, and Skold, 2006; Palacios and Steel, 2006; Diggle and Ribeiro, 2007; and Yan et al., 2007). Experience suggests that it can be difficult for certain datasets to be fitted to these models without a great deal of careful sensitivity analyses.

Information Content

In Section 2.2, a brief discussion was given of the notion of information content. This relates to the properties of statistical inference as the data size increases; as it tends to infinity, is the uncertainty in the inference removed? For standard, *iid* error assumptions, the answer is generally “yes.” For the type of short-term temporal dependence typically seen in Chapter 3, the answer is again generally “yes.” However, in cases of infill sampling in a fixed domain, the situation is delicate (Lahiri, 1996; Lahiri, Lee, and Cressie, 2002; Lee and Lahiri, 2002; Bhattacharjya, Eidsvik, and Mukerji, 2010).

Model Selection

This book is almost silent on model selection. However, there has been some innovative work in the area of geostatistical model selection; see Hoeting et al. (2006) and Huang and Chen (2007). Likewise, Zhu and Carlin (2000) and Zhu, Huang, and Reyes (2010) consider selection of models for spatial lattice data.

Spatial Lattice Processes

Such processes are distinguished from geostatistical processes by the finite (or countably infinite) size of the spatial index set D_s . However, there are methodologies that the two spatial statistical modeling approaches (lattice and geostatistical) can share, particularly because in practice geostatistical processes are evaluated on a fine (regular) grid $\{s_1, \dots, s_n\}$. In Section 4.2, many of the lattice models are MRFs (Sections 4.2.2–4.2.5), which are based on a conditional-distribution specification of spatial dependence. This is different from the bivariate-moments specification of the geostatistical models in Section 4.1. Some comparison of the two will be given below in a separate set of bibliographic notes.

Not all lattice processes are MRFs, and in fact there is an area that has become known as Spatial Econometrics that is based around the simultaneous-spatial-dependence models such as SAR, SARMA, and SARMAX. These are specified in Sections 4.2.7 and 4.2.8. In that literature (e.g., Anselin, 1988; Arbia, 1989; Getis, Mur, and Zoller, 2004; LeSage and Pace, 2009), researchers have often written their models as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{B}\mathbf{Y} + \boldsymbol{\varepsilon},$$

by analogy to models in (temporal) econometrics. See Section 4.2.7 for definitions, and an explanation that $\mathbf{B}$ is now a “spatial-shift” operator in place of the usual time-series backward-shift operator. While the decomposition of variability above seems natural, it is not hard to see that $E(\mathbf{Y}) = (\mathbf{I} - \mathbf{B})^{-1}\mathbf{X}\boldsymbol{\beta}$, leading to a confounding of the covariates $\mathbf{X}$ and the spatial dependence in $(\mathbf{I} - \mathbf{B})^{-1}$. In general, for processes that include both CAR and SAR models with fixed effects (covariates), one must take care regarding potential confounding between fixed effects and random effects when making inference on the unknowns (Reich et al., 2006; Paciorek, 2010; Hodges and Reich, 2010).

An additive decomposition of variability does not generalize easily to non-Gaussian, nonlinear lattice models; a decomposition based on a sequence of conditional-probability models *does* generalize, which is a theme of this book.

Texts and monographs in the area of lattice processes include Cliff and Ord (1981), Ripley (1981, 1988), Anselin (1988), Griffith (1988), Haining (1990), Cressie (1991, 1993), Guyon (1995), Griffith and Layne (1999), Tiefelsdorf (2000), Fotheringham, Brunson, and Charlton (2002), Getis, Mur, and Zoller (2004), Rue and Held (2005), some chapters in Gamerman and Lopes (2006), Lloyd (2007), Le Sage and Pace (2009), and Gelfand et al. (2010, Part III). Those with an emphasis on epidemiology include Elliott et al. (2000), Lawson (2001), Waller and Gotway (2004), and Lawson (2009).

The Markov Random Field (MRF)

This is a very elegant way to incorporate dependence in *any* multivariate distribution, not just spatial dependence. The dependence is succinctly captured by an undirected graph (Sections 2.4 and 4.2.2), which actually shows the conditional *independencies* but does not show the type and strength of the dependencies.

There has been much written about the theoretical properties of MRFs, including their properties in the limit as the lattice extends to infinity (e.g., Spitzer, 1971; Prum and Fort, 1991; van den Berg, 1993; Guyon, 1995), and their properties as the lattice spacing tends to zero (e.g., Besag, Milne, and Zachary, 1982; Cressie et al., 2000b). However, the statistical-modeling implications of these properties were dispersed in hard-to-find places. The Hammersley–Clifford Theorem (Hammersley and Clifford, 1971) was in an unpublished Oxford University manuscript; Clifford (1990) gives some of the history surrounding the result, as well as the original proof. The articles by Brook (1964) and Moussouris (1974) are noteworthy, but the article that captured the past, the present (at the time the article was published), and in some sense the future (at the time of writing this), was Besag (1974). Its impact on spatial statistics has been enormous.

A downside to MRFs is that they do not aggregate. In other words, if $\{Y(s_1), Y(s_2), Y(s_3), \dots, Y(s_n)\}$ forms a sparse MRF (i.e., local neighborhood structure), then $\{Y(s_1) + Y(s_2), Y(s_3), \dots, Y(s_n)\}$ is typically *not* a sparse MRF. Furthermore, $\{Y(s_2), Y(s_3), \dots, Y(s_n)\}$ may not be a sparse MRF; that is, MRFs do not marginalize either. In applications of spatial statistics, aggregation and marginalization are commonplace. See Rue, Steinsland, and Erland (2004), Rue and Held (2005), and Rue, Martino, and Chopin (2009) for further discussion. Statistical image analysis often uses MRFs, and they are well suited to this application because the pixels are fixed and the data are usually complete. [For a generic review of statistical image analysis, see, for example, the encyclopedia entry by Cressie and Davidson (1998b).]

The fundamental problem for modelers using MRFs is to know when their conditional-probability models are specified coherently (i.e., give rise to a joint distribution). This is discussed at length in Section 4.2.3. The conditions are

easy to check for CAR (i.e., Gaussian) models, but as spatial statistical models evolve toward nonlinear, non-Gaussian models to capture more real-world phenomena, conditions to ensure coherent MRF models will be essential (Kaiser and Cressie, 2000; Lee, Kaiser, and Cressie, 2001).

Finally, implementation of MRFs in HMs will almost always be done by numerical statistical approximations (e.g., rejection sampling, Laplace approximation, importance sampling, slice sampling, Markov chain Monte Carlo, etc., or some combination of these). The MRF generally has a normalizing constant that cannot be obtained analytically and, when found in an HM, the HM's normalizing constant is likewise unavailable. A review of the considerable literature on statistical computing for MRFs can be found in Rue and Held (2005), Martino and Rue (2009), and Rue, Martino, and Chopin (2009).

CAR Models

The conditionally autoregressive (CAR) models are more correctly called auto Gaussian models (in the terminology of Besag's auto spatial models). Cressie (1993) called them conditionally specified Gaussian (CG) models and contrasted them with simultaneously specified Gaussian (SG). The "CAR" terminology has remained, but it should be remembered that these models are specifically for *Gaussian* spatial data.

Besag (1974, 1981) had noticed that if one assumes a CAR model on a regular lattice in $\mathbb{R}^2$, where $\text{var}(\mathbf{Y}) = \tau^2(\mathbf{I} - \phi\mathbf{A})^{-1}$ and $\mathbf{A}$ is the nearest-neighbor adjacency matrix, then $\text{corr}(Y(s_1, s_2), Y(s_1 + 1, s_2))$ tends to 1 very slowly as $\phi \rightarrow \phi_{\max}$. That is, to obtain substantial spatial dependence, ϕ has to be *very close* to $\phi_{\max}$; Besag then recommended that $\phi = \phi_{\max}$, which results in a CAR model that is *not* defined in n -dimensional space, but it is defined on some lower-dimensional subspace. This has become known as the *intrinsic autoregressive* (IAR) model (e.g., Banerjee, Carlin, and Gelfand, 2004, Section 3.3). The time-series analogue to "CAR versus IAR" is a stationary autoregressive process versus a random walk, where it has long been recognized that the two processes' behaviors are fundamentally different (spawning a large literature on *unit-root tests*). The spatial-modeling literature has generally concentrated on the IAR model without any determination of whether a proper CAR or an IAR model is appropriate. Besag's concern seems to have its roots more in the choice of the *nearest-neighbor* adjacency matrix $\mathbf{A}$ than in finding $\hat{\phi}$ very close to $\phi_{\max}$. Hrafnkellson and Cressie (2003) showed empirically that if a *range* parameter is introduced into the CAR model, ϕ may take reasonable values and still generate strong spatial correlation between neighbors.

A family of CAR models due to Pettitt, Weir, and Hart (2002), termed *diagonally dominant*, keeps $(\text{var}(\mathbf{Y}))^{-1}$ from being singular throughout the whole parameter space. In the notation of Section 4.2.5, we have

$$\text{var}(\mathbf{Y}) = (\mathbf{I} - \mathbf{C})^{-1}\mathbf{M},$$

where $\mathbf{C} = (c_{ij})$, $\mathbf{M} = \text{diag}(m_{11}, \dots, m_{nn})$, $c_{ii} = 0$,

$$c_{ij} = \frac{\kappa g_{ij}}{1 + |\kappa| \sum_{s_k \in N(s_i)} g_{ik}}, \quad i \neq j,$$

$$m_{ii} = \frac{1}{1 + |\kappa| \sum_{s_k \in N(s_i)} g_{ik}},$$

and $\mathbf{G} = (g_{ij})$ is a symmetric matrix whose diagonal entries are zero and whose nondiagonal entries are nonnegative. Notice the spatial-dependence parameter $\kappa \in (-\infty, \infty)$, and the CAR model is proper over the whole parameter space. Ferreira and De Oliveira (2007) modify this model slightly by specifying $\kappa \in [0, \infty)$.

Multivariate CAR Models

Mardia (1988) introduced a multivariate Markov random field (MRF) model for image processing. More recently, Billheimer et al. (1997), Kim, Sun, and Tsutakawa (2001), Pettitt, Weir, and Hart (2002), Carlin and Banerjee (2003), Gelfand and Vounatsou (2003), Jin, Carlin, and Banerjee (2005), Jin, Banerjee, and Carlin (2007), Sain and Cressie (2007), and Sain, Furrer, and Cressie (2011) have explored these multivariate MRF models and their role in Bayesian hierarchical modeling.

CAR Models versus Geostatistical Models

As we specified in Section 4.2.5, there is *always* a relationship between a geostatistical model and a CAR model defined on a finite number of small areas. It is their respective properties that we would like to exploit. For example, to take the inverse of a spatial-dependence matrix, we would like to model with a CAR model, but to derive properties of aggregated processes, we would like to model with a geostatistical model. An approach that has good success in limited settings (small to moderate n) is to approximate a geostatistical model with a CAR model, when needed (Rue and Tjelmeland, 2002; Rue, Steinsland, and Erland, 2004; Rue and Held, 2005, Chapter 5; Cressie and Verzelen, 2008; Rue, Martino, and Chopin, 2009). More direct comparisons between CAR models and geostatistical models can be found in Banerjee, Wall, and Carlin (2003), Hrafnkelsson and Cressie (2003), and Song, Fuentes, and Ghosh (2008).

Ecological Inference in a Spatial Setting

In spatial statistics, ecological inference is known as the change-of-support problem, and it is really a manifestation of Simpson's Paradox in a spatial setting where aggregation is commonplace. The book by King (1997) addresses the general problem; one would need to adapt these results to the spatial setting. In Section 4.2.6, the special issue of the *Journal of the Royal Statistical Society, Series A*, Vol. 164, Part 1 (2001) was mentioned as a resource for

problems with ecological bias. The following articles in that special issue are particularly illuminating: Best et al. (2001), Darby, Deo, and Doll (2001), and Wakefield and Salway (2001). Using a powerful model of aggregation from spatial point processes (Section 4.3) to lattice processes (Section 4.2), Best, Ickstadt, and Wolpert (2000) give an analysis of public health and exposure to automobile emissions that allows data measured at disparate resolutions to be combined consistently. The review paper by Gotway and Young (2002) also covers statistical change-of-support for lattice data.

Spatial Point Processes

Matérn (1960) discussed spatial point processes as part of his landmark publication in spatial statistics. In the area of ecology, Pielou (1977) developed descriptive methods for spatial point patterns. Ripley's (1981) book and Diggle's (1983, 2003) editions put spatial point processes on a more mathematical footing, which continued with the monographs by Cressie (1991, 1993), Stoyan and Stoyan (1994), van Lieshout (2000), Møller and Waagepetersen (2004), Illian et al. (2008), and Gelfand et al. (2010, Part IV). The review by Møller and Waagepetersen (2007) takes a modern distributional point of view to modeling and estimation of spatial point processes. A part of that work involves Log-Gaussian Cox Processes; Møller, Syversveen, and Waagepetersen (1998) discussed their marginal properties and, importantly, saw their potential use in an HM.

Intensities for Spatial Point Processes

The first-order intensity function has played a prominent role in the modeling and analysis of spatial point patterns. There is an attractive alternative, namely the Papangelou conditional intensity, which is an intensity function of the spatial point process Z , *conditional* on there being an event of Z at $s \in \mathbb{R}^d$. Baddeley et al. (2005) use it to develop a residual analysis for spatial point processes.

First- and higher-order intensities are often estimated using kernel estimation, which emphasizes the strong connection between spatial point processes in $\mathbb{R}^d$ and multivariate analysis of random spatial locations. Inference for such nonparametric estimators has been more recently based on resampling/subsampling rather than on asymptotic Gaussian distribution theory (e.g., Guan, Sherman, and Calvin, 2004; Guan and Loh, 2007; Guan and Sherman, 2007).

The recent literature on spatial-point-process modeling and inference is statistically rich (e.g., Møller and Waagepetersen, 2007; Guan, 2008; Waagepetersen and Guan, 2009). Inference in the form of hypothesis testing can be handled through Monte Carlo testing, provided that the spatial point process is straightforward to simulate from. The idea was developed by Hope (1968), and it has proved effective in a number of applications (e.g., Kulldorf, 1997; Priebe, Olson, and Healy, 1997; Kornak, Irwin, and Cressie, 2006).

Marked Spatial Point Processes

This represents a very attractive generalization of the spatial point process, by combining a stochastic model for attributes (mark process), along with a

stochastic model for event locations (point process). The mark process and the point process are typically dependent (e.g., Penttinen, Stoyan, and Henttonen, 1992; Henderson, Shimakura, and Gorst, 2002; Schlather, Ribeiro, and Diggle, 2004). Comas, Delicado, and Mateu (2010) consider marks that are modeled as random functions. One mark process that is a direct consequence of the spatial point process itself is the set of random Dirichlet (or Voronoi) tessellations associated with the random locations of the events. Let s_i be the i th event from among events, $\{s_1, s_2, \dots\}$ of the point process. Then the Dirichlet cell associated with s_i is $C_i \equiv \{\mathbf{x} : \|\mathbf{x} - s_i\| = \min_j \|\mathbf{x} - s_j\|\}$ (e.g., Cox and Isham, 1980, Chapter 6; Schoenberg, Barr, and Seo, 2008). For further connections to random sets, see Baddeley and Møller (1989).

Bayesian Inference for Spatial Point Processes

Much of the spatial-point-process literature is not concerned with hierarchical models. Rather, the spatial point process is characterized by $[Z|\theta]$ and a prior $[\theta]$ is put on θ (e.g., Heikkinen and Penttinen, 1999; Møller, 1999; Cressie and Lawson, 2000; Møller et al., 2006; Berthelsen and Møller, 2008). When the Cox process is viewed conditionally, a BHM approach to inference is natural (e.g., Kottas and Sansó, 2007; Møller and Waagepetersen, 2007). In principle, a BHM should allow change-of-support to be handled, provided that the process model is consistent through different levels of aggregation (e.g., Wolpert and Ickstadt, 1998).

Applications of Spatial Point Processes

Principal areas of application have been in forestry, earthquakes, rainfall modeling, astronomy, and epidemiology. See the review by Stoyan and Penttinen (2000) for spatial point process methods in forestry. Ogata et al. (1998) and Choi and Hall (2001) are two examples of articles that study earthquake point-process data. Point process models for rainfall were developed by Rodriguez-Iturbe and co-authors (e.g., Rodriguez-Iturbe, Cox, and Isham, 1988). In astronomy, there has been considerable development in astrostatistics, which includes spatial point processes (e.g., Babu and Feigelson, 1996). Epidemiology, particularly related to environmental factors, has a large spatial component; point process models are relevant when the disease locations are available (e.g., Diggle, 1993). Often epidemiologists are interested in testing for the presence of clusters; for example, see Wakefield, Kelsall, and Morris (2000) for a discussion of this methodology.

Random Set Processes

Matheron (1975) is a book that is path-breaking in the way one can take an object view of the world (Chapter 1) and model the uncertainty associated with that view with probability theory. Well known for his work in geostatistics, Matheron also established a group at the School of Mines in Fontainebleau, France, working on what he called Mathematical Morphology. Matheron (1975) contains the theory of random sets, interwoven with

definitions of set transforms (e.g., set addition, opening, closing, etc.) that are useful in image analysis and that he showed defined other random sets. This was all done in $\mathbb{R}^d$, and stereology (i.e., the measurement of an object in lower-dimensional spaces that allow the object to be reconstructed—with some uncertainty—in $\mathbb{R}^d$) was also considered. Kendall (1974) published a very similar theory of random sets at about the same time, but it was the French group who took it further through one of Matheron's early students, who published a book (Serra, 1982) linking Mathematical Morphology directly with image analysis being done in their laboratory. Cressie (1979) presented a central limit theorem for sums of random sets, and Molchanov (1993) presented limit theorems for unions of random closed sets, which is a beginning to establishing a statistical theory for random sets. More recent developments in random-set theory can be found in Molchanov (2005) and Nguyen (2006).

In the related area of Pattern Analysis, a strong statistical component where uncertainties are expressed in an HM and patterns are quantified and compared statistically was developed by a U.S. group at Brown University (e.g., Grenander and Miller, 2007). There are a number of other related areas, and these approaches are left for the interested reader to pursue: Stochastic Geometry (e.g., Stoyan, Kendall, and Mecke, 1987; Stoyan and Stoyan, 1994), Geometric Probability (e.g., Solomon, 1978), and Shape Analysis (e.g., Dryden and Mardia, 1998). There is interesting cross-over work that relates the field view of the world with the object view (Adler, 1981).

Stereology has developed from a theory based on probability to a statistical theory based on sampling (of objects). Here the randomness is derived from random sampling with probes, something that is important in medical testing where biopsies are taken and imaged. Early work concentrated on the first moment of the estimator, but that changed when it was realized that sampling theory, albeit for more complicated populations, could be used to develop (estimates of) variances of the sample-based statistics (Baddeley and Jensen, 2005).